

Hedging on the Frontier: Learning New Tasks with Few Samples

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Abstract

When a learner faces a new task with few samples, it must leverage any available side information. In practice, this often comes in the form of model evaluations on related tasks in public benchmarks. A key question then is how to model task relatedness such that it is both realistic and the benchmark evaluations lead to provable gains. Empirically, we observe that *weak monotonicity* is often approximately satisfied: if a model dominates another on many benchmarks, it also tends to outperform on the new task. We explore the statistical complexity of learning under (approximate) weak monotonicity, leveraging it within two learning paradigms: transfer learning and model selection aggregation. We show that not only can we prune the model class based on monotonicity, but we can also further adapt to the geometry of the available trade-offs by *hedging on the frontier*.

1. Introduction

Foundation models are designed to perform well across a vast range of downstream applications (Bommasani et al., 2021). But, in general, it is hard to determine whether one model is strictly more capable than another, or to know beforehand how suitable it is for a new task. Benchmarks can provide some signal by evaluating these models on a fairly narrow set of tasks (Liang et al., 2023; Chiang et al., 2024; Srivastava et al., 2023; Hardt, 2026). However, models are often ranked inconsistently across different benchmarks (Zhang and Hardt, 2024), which raises the issue of *transferability*: with the proliferation of models and benchmarks, can they be reliably used by a practitioner to solve a new target task with limited data?

Underlying the belief that benchmarks are useful is often the

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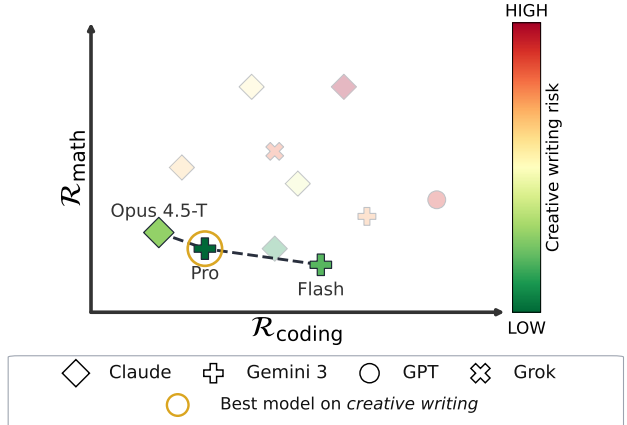


Figure 1. Pareto Frontier (Coding vs Math) with risk scores from a subset of Huggingface LMArena dataset (Chiang et al., 2024).¹

implicit assumption that performance transfers: a model that dominates another on a related benchmark task will likely outperform the other also in the target task. Indeed, this phenomenon has already been observed in the context of out-of-distribution (OOD) versus in-distribution generalization (Miller et al., 2021), although it fails to hold in most arguably more realistic scenarios (Sanyal et al., 2024; Salaudeen et al., 2025). Hence, instead of relating just two tasks, in this work, we formalize the weaker assumption of *weak monotonicity*, which relates a target task to multiple source tasks.² Weak monotonicity admits comparison across a *suite* of benchmarks: a more capable model that is consistently better on a whole set of benchmarks will also tend to be better on the target. It does not imply that there is a single dominating model; rather, there may be an entire *frontier* of potentially suitable models, and a need to be able to select from it using some limited target data. This is depicted in Fig. 1 using data from the Huggingface LMArena, where math and coding scores separately do not predict the best model for creative writing, but it is contained in the joint frontier. Importantly, a key feature is that the assumption of weak monotonicity only further weakens as the number of benchmark tasks increases. Moreover, we do not always require it to hold strictly: we quantify the degree to which it holds using the *modulus of monotonicity*, a quantity that appears in different variants in our theoretical results.

For the theoretical results, we assume throughout that the benchmark risks are known exactly. In transfer learning ter-

minology, this corresponds to the idealized setting of infinite source samples; we leave the finite source sample case to future work. We then ask the theoretical question: *What is the sample complexity of learning under accurate benchmark evaluations and (approximate) weak monotonicity?*

To answer this question, we distinguish between two classes of functions we may compare to. Firstly, we consider the *full* hypothesis class for which evaluations are available. This then turns into the question of (classical) *transfer learning* (also called supervised domain adaptation) (Ben-David et al., 2010), where the aim is to achieve small excess risk with respect to the full hypothesis class. And secondly, we may compare only to the dictionary of best models for each benchmark alone. This falls into the realm of *model selection aggregation* (Tsybakov, 2003) in a setting where the dictionary is not arbitrary in that the models are optimal for individual benchmarks. The goal is then to achieve small excess risk with respect to the best model in the dictionary. We introduce model selection aggregation in detail in Section 5.

In the existing transfer learning literature (or domain generalization), most approaches are based on assumptions on the relationship between source and target domains that typically fall into two categories: *Interpolation*-based approaches exploit proximity across source and target domains such as distributional similarity, shared features, or empirical performance correlations (Ben-David et al., 2010; Long et al., 2015; Mansour et al., 2008; 2021). On the other hand, *extrapolation*-based approaches can transfer beyond observed environments by learning invariant or causal representations (Arjovsky et al., 2019; Krueger et al., 2021; Peters et al., 2016; Schölkopf et al., 2021). Importantly, learning under weak monotonicity strictly generalizes previous settings in transfer learning, e.g., when the target distribution is a *mixture* of source distributions (Mansour et al., 2008; 2021), cf. Section 2.1. Both types of modeling assumptions are often too restrictive in practice and are not reflective of the actual shift between the tasks. For instance, causal methods often do not outperform simple baselines (Nastl and Hardt, 2024; Ahuja et al., 2021) and DRO style methods fail, e.g., on the prominent WILDS dataset (Koh et al., 2021). For modern foundation models, transfer is often framed as model selection from a *zoo* of pretrained candidates (Zhang et al., 2023; Dong et al., 2022), for example by ranking candidates via transferability scores (Nguyen et al., 2020; You et al., 2021) typically using access to model representations. However, for state-of-the-art models, we often only have black-box access to their evaluations (Lee et al., 2024), so that these methods cannot be used.

We assume weak monotonicity partly to address these shortcomings. The main insight from our work is that under monotonicity, statistically efficient algorithms should not only *prune* models that are dominated. When target data is

limited, they must also be selective over the frontier models, and a form of *hedging* can be facilitated by avoiding improper trade-offs on the benchmarks. We show that the effectiveness of hedging is determined by the geometry on the frontier induced by monotonicity.

In short, our contributions can be summarized as follows:

- We are the first to formally study the setting of learning a target task under (approximate) *weak monotonicity* (Section 2), motivated by empirical findings that approximate monotonicity holds in real benchmarking datasets, such as HELM and VHELM (Figs. 3 and 5 and Section 4.3).
- We introduce the Pareto covering number (Section 3), a complexity measure that adapts naturally to the geometry of the Pareto frontier while inherently encoding the monotonicity assumption. We further derive its limiting distribution; this result may be of independent interest.
- We prove sample complexity guarantees in terms of the Pareto covering number and modulus of monotonicity in the transfer learning setting (Section 4.1). We also present a parameter-free algorithm that enjoys a fall-back guarantee for monotonicity violations (Section 4.2).
- Under assumptions on the curvature of the Pareto front, we show that monotonicity enables fast rates in model selection aggregation, without requiring a Bernstein condition or strongly convex loss (Section 5).

Datasets. While the contributions of this paper are mainly theoretical, we also validate our assumptions and methods on two benchmark suites: HELM (Holistic Evaluation of Language Models) (Liang et al., 2023) and VHELM (Holistic Evaluation of Vision Language Models) (Lee et al., 2024). Both report evaluations of LLMs (respectively, VLMs) across a collection of scenarios and metrics.³

2. Problem Setting and Preliminaries

Let $\mathcal{F}$ be a model class and let $\mathcal{R}_Q : \mathcal{F} \rightarrow [0, 1]$ measure risk for a task of interest.⁴ Specifically, we let Q be a distribution on an arbitrary space $\mathcal{Z}$ and $\mathcal{R}_Q(f) = \mathbb{E}_{z \sim Q} \ell(f, z)$ for any function $f \in \mathcal{F}$, where the loss is bounded, $\ell(f, z) \in [0, 1]$. We aim to find a model with minimal target risk. Further, we assume full access to a set of K benchmarks $\mathbf{R} \equiv (\mathcal{R}_1, \dots, \mathcal{R}_K) : \mathcal{F} \rightarrow [0, 1]^K$, each of them minimized in $\mathcal{F}$ by f_k . A learning algorithm $\mathcal{A}$ observes n i.i.d. samples from Q , denoted $\{z_i\}_{i=1}^n \sim Q^{\otimes n}$,

²Risks are mapped to $[0, 1]$ using models rankings, where rankings are taken from the LMArena Overview leaderboard (snapshot taken from data available on Jan. 10th, 2026).

³We use *benchmark* and *source task* interchangeably.

⁴Data and code are available at <https://github.com/FedericoDiGennaro/Hedging-on-the-Frontier>.

⁵Table 1 in the appendix provides a full overview of notation.

and outputs a model $\hat{f} \equiv \mathcal{A}(\{z_i\}_{i=1}^n; \mathbf{R}, \mathcal{F})$ given knowledge of $\mathbf{R}$ and $\mathcal{F}$. In this paper, we analyze the excess risk $\mathcal{E}_Q(\hat{f}, \mathcal{H}) := \mathcal{R}_Q(\hat{f}) - \inf_{h \in \mathcal{H}} \mathcal{R}_Q(h)$ with respect to the following two function classes:

$$\mathcal{H} = \begin{cases} \mathcal{F} & \text{transfer learning (Section 4),} \\ \{f_k\}_{k=1}^K & \text{model selection aggregation (Section 5).} \end{cases}$$

We study the sample complexity of achieving ε -excess risk with high probability uniformly over Q in some class of distributions $\mathcal{Q}$; that is, the minimal sample size n to achieve, for some $\delta \in (0, 1)$,

$$\inf_{Q \in \mathcal{Q}} \mathbb{P}_{\{z_i\}_{i=1}^n \sim Q^{\otimes n}} \left(\mathcal{E}_Q(\hat{f}, \mathcal{H}) \leq \varepsilon \right) \geq 1 - \delta.$$

Without any relationship between $\mathbf{R}$ and $\mathcal{R}_Q$, the sample complexity is generally equivalent to a standard learner that (only) has access to data $\{z_i\}_{i=1}^n$ from Q , and the additional availability of the benchmarks does not provide any advantage. In the following, we propose a new assumption to model relationships between benchmarks and the new task.

2.1. Weak Monotonicity

To formalize our assumption, we use the following notation: for vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^K$ we write $\mathbf{v} \preceq \mathbf{w}$ if $v_k \leq w_k$ for all $k \in [K]$, $\mathbf{v} \prec \mathbf{w}$ if $\mathbf{v} \preceq \mathbf{w}$ and $\mathbf{v} \neq \mathbf{w}$, and $\mathbf{v} \ll \mathbf{w}$ if $v_k < w_k$ for all $k \in [K]$. To model the information that benchmark performance contains about the new task, define the *preorder* $\preceq_{\mathbf{R}}$ on the hypothesis space $\mathcal{F}$ as

$$f \preceq_{\mathbf{R}} f' \quad :\Longleftrightarrow \quad \mathbf{R}(f) \preceq \mathbf{R}(f').$$

This is only a preorder, because there can exist hypotheses f and f' that are *incomparable*, i.e., neither of the two comparisons $f \preceq_{\mathbf{R}} f'$ or $f \succeq_{\mathbf{R}} f'$ holds, and if $\mathbf{R}$ is not injective, $\preceq_{\mathbf{R}}$ is not antisymmetric. In an ideal setting, the benchmarks provide a reliable signal on the performance of the new task of interest. We formalize this intuition as $\mathcal{R}_Q$ preserving the preorder $\preceq_{\mathbf{R}}$.

Definition 1 (Monotonicity). We say that $\mathcal{R}_Q$ is (weakly) *monotonic* with respect to $\mathbf{R}$ if for all $f, f' \in \mathcal{F}$,

$$f \preceq_{\mathbf{R}} f' \quad \implies \quad \mathcal{R}_Q(f) \leq \mathcal{R}_Q(f'). \quad (\text{MON})$$

The sense in which this monotonicity is *weak* is that the stronger conditional $\mathbf{R}(f) \prec \mathbf{R}(f')$ does not necessarily imply $\mathcal{R}_Q(f) < \mathcal{R}_Q(f')$. Throughout the work, we use monotonicity and *weak* monotonicity interchangeably. Monotonicity is connected to *Pareto optimality*.

Definition 2 (Pareto optimality). A model $f \in \mathcal{F}$ is called Pareto optimal w.r.t. $\mathcal{F}$ and $\mathbf{R}$ if there is no other model $f' \in \mathcal{F}$ so that $\mathbf{R}(f') \prec \mathbf{R}(f)$. We denote the set of all Pareto optimal models in $\mathcal{F}$ w.r.t. $\mathbf{R}$ as $\text{Par}(\mathcal{F}, \mathbf{R}) \subseteq \mathcal{F}$.

Pareto optimality and monotonicity are related through the following lemma. In what follows, to avoid pathologies, we always implicitly assume that the set $\mathbf{R}(\mathcal{F}) \subseteq [0, 1]^K$ is compact, and $\mathcal{R}_Q$ attains its minimum on $\mathcal{F}$, i.e., there exists $f^* \in \mathcal{F}$ such that $\mathcal{R}_Q(f^*) = \inf_{f \in \mathcal{F}} \mathcal{R}_Q(f)$.

Lemma 1 (Monotonicity and Pareto optimality). *For every $f \in \mathcal{F}$ there exists an $f' \in \text{Par}(\mathcal{F}, \mathbf{R})$ with $f' \preceq_{\mathbf{R}} f$. Moreover, if $\mathcal{R}_Q$ satisfies (MON), then there exists an $f^* \in \arg \min_{f \in \mathcal{F}} \mathcal{R}_Q(f)$ that is Pareto optimal.*

Notice that not all minimizers need to be in the Pareto set, and the fact that the Pareto set contains a minimizer does not directly imply monotonicity. Beyond Lemma 1, we can also show that monotonicity implies that the target risk is not sensitive to changes in f that do not affect the benchmarks.

Lemma 2 (Monotonicity implies sufficiency). *Suppose $\mathcal{R}_Q$ is monotonic with respect to $\mathbf{R}$. Then, there exists a monotonic map $s : \mathbb{R}^K \rightarrow \mathbb{R}$ such that $\mathcal{R}_Q = s \circ \mathbf{R}$ on $\mathcal{F}$.*

Given Lemmas 1 and 2 (proven in Appendix G.1), monotonicity may seem to be a strong assumption. However, the assumption becomes weaker as the number of benchmarks K increases: in general, the larger K , the fewer models are comparable in terms of the preorder, and hence the easier it is for $\mathcal{R}_Q$ to be monotonic with respect to $\mathbf{R}$.

Relation to other transfer models. Assume that the benchmarks are risks over distributions P_k on $\mathcal{Z}$, that is, $\mathcal{R}_k(f) = \mathbb{E}_{z \sim P_k} \ell_k(f, z)$ for some $\ell_k(f, z) \in [0, 1]$. It is then easy to show that monotonicity holds if $\ell_k = \ell$ for all k and the target distribution is a mixture of the source distributions $Q \in \text{conv}(P_1, \dots, P_K)$ (see Lemma 5 in the appendix). Monotonicity is therefore at least as general as the assumptions made in classical multi-source domain adaptation with convex mixtures (Mansour et al., 2008; 2021; Hoffman et al., 2018) and group DRO (Sagawa et al., 2019). However, it is also strictly more general, as one can appreciate from Example 2 in the appendix. Moreover, as we argue in Section 4.1 and Appendix D, our approach can also yield tighter bounds even under the mixture assumption.

Another paradigm of modeling distribution shifts is based on invariances such as covariate or label shift (Shimodaira, 2000), and, more generally, shared (or invariant) causal structure (Peters et al., 2016; Rojas-Carulla et al., 2018; Arjovsky et al., 2019). But, both covariate and label shift are neither necessary nor sufficient for monotonicity (see Lemma 8). And the same holds for the mixtures of conditionals assumption in (Zhang et al., 2015).

2.2. Two Moduli of Monotonicity

In practice, monotonicity may hold only approximately. In this section we introduce two moduli of monotonicity to

quantify, for any target distribution, how much it violates monotonicity. The moduli are defined using the following notion of distance.

Definition 3 (Pareto distance). The *Pareto distance* on $\mathcal{F}$ with respect to $\mathbf{R}$ is the map $d_{\mathbf{R}} : \mathcal{F} \times \mathcal{F} \rightarrow \mathbb{R}$ defined as

$$d_{\mathbf{R}}(f, f') = \max_{k \in [K]} \{\mathcal{R}_k(f) - \mathcal{R}_k(f')\}.$$

In general, this distance is not symmetric and can be negative. The *Pareto ball* $\{f \in \mathcal{F} : d_{\mathbf{R}}(f_0, f) \leq t\}$ includes all functions that are not more than t better than f_0 across all objectives, see Fig. 2a. Another way to express the Pareto distance is $d_{\mathbf{R}}(f, f') = \min \{t \in \mathbb{R} : \mathbf{R}(f) - t\mathbf{1} \preceq \mathbf{R}(f')\}$ (see Lemma 3), where $\mathbf{1} = (1, \dots, 1)$ is the all-ones vector. In this formulation, we can interpret the Pareto distance as the smallest amount $t \in \mathbb{R}$ that the objectives $\mathcal{R}_k(f)$ need to improve so that it is no worse than $\mathcal{R}_k(f')$.

With the Pareto distance, we can define two ways to quantify the *degree* of monotonicity. Our results are expressed in terms of these moduli. They are closely related to the modulus of transfer from Hanneke and Kpotufe (2024).

Definition 4. The *upper modulus of monotonicity* of $\mathcal{R}_Q$ and $\mathbf{R}$ is the function $\bar{\omega}_Q : [0, 1] \rightarrow [0, 1]$,

$$\bar{\omega}_Q(t) = \sup_{f, f' \in \mathcal{F} : d_{\mathbf{R}}(f, f') \leq t} \{\mathcal{R}_Q(f) - \mathcal{R}_Q(f')\}.$$

Similarly, the *lower modulus of monotonicity* is defined as the function $\underline{\omega}_Q : [0, 1] \rightarrow [-1, 1]$ given by

$$\underline{\omega}_Q(t) = \inf_{f, f' \in \mathcal{F} : d_{\mathbf{R}}(f, f') \leq -t} \{\mathcal{R}_Q(f') - \mathcal{R}_Q(f)\},$$

where we set $\underline{\omega}_Q(t) = 1$ if the set from above is empty.

By definition, the upper modulus of monotonicity satisfies for any $t \geq 0$ and $f, f' \in \mathcal{F}$, that if $\mathcal{R}_k(f) \leq \mathcal{R}_k(f') + t$ for all k , then $\mathcal{R}_Q(f) \leq \mathcal{R}_Q(f') + \bar{\omega}_Q(t)$; it measures how much worse a function can be on the target when it is at most t worse on all source risks. On the other hand, the lower modulus satisfies that if $\mathcal{R}_k(f) \leq \mathcal{R}_k(f') - t$ for all k , then $\mathcal{R}_Q(f) \leq \mathcal{R}_Q(f') - \underline{\omega}_Q(t)$; that is, it measures the minimal gain on the target when all source risks are improved by at least t . The lower and upper moduli are both *non-decreasing* functions of t , and it holds that

$$\bar{\omega}_Q(0) = 0 \iff \underline{\omega}_Q(0) \geq 0 \iff (\text{MON}).$$

The results in Sections 4 and 5 show that a small upper modulus, or respectively a large lower modulus, when t is close to zero, lead to stronger generalization bounds. In particular, $\bar{\omega}_Q(0)$ upper bounds the “approximation error”, in the sense that the best model outside the Pareto set can have risk on $\mathcal{R}_Q$ that is at most $\bar{\omega}_Q(0)$ smaller than the best model inside the Pareto set. Further, it can be shown that, if

s from Lemma 2 is a weighted average, then $\bar{\omega}_Q(t) \leq t$ and $\underline{\omega}_Q(t) \geq t$ (see Lemma 4 in the Appendix).

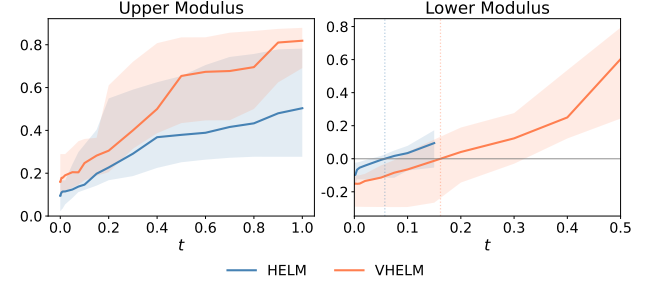


Figure 3. Median and quantiles of moduli (over all combinations of $K = 3$ sources and target Q) for both HELM and VHELM.⁵

The moduli of HELM and VHELM. We now compute the moduli on HELM and VHELM. Fig. 3 (left) reports, for a fixed K , the upper modulus $\bar{\omega}_Q$ from Definition 4, aggregated by taking the median across all combinations of K sources and a target task Q . From Fig. 3, we observe that the moduli are quite close to zero at $t = 0$, suggesting that the datasets are *almost* monotone. Further, the target performance decreases significantly with benchmark performance.

3. Pareto Covering Numbers

To capture the statistical complexity, we first introduce a notion of geometry on the Pareto front⁶

$$\mathfrak{R} := \{\mathbf{R}(f) \in [0, 1]^K : f \in \text{Par}(\mathcal{F}, \mathbf{R})\}.$$

First, we define the *Pareto distance* $d : \mathfrak{R} \times \mathfrak{R} \rightarrow \mathbb{R}$ on the front $\mathfrak{R}$ as the map $d(\mathbf{r}, \mathbf{r}') = d_{\mathbf{R}}(f, f')$, for $\mathbf{r} = \mathbf{R}(f), \mathbf{r}' = \mathbf{R}(f')$ using the Pareto distance $d_{\mathbf{R}}$ given in Definition 3. This distance is a quasi-metric *on the Pareto front*: it satisfies all the usual axioms of a metric except symmetry (see Lemma 3 and Fig. 2). We use it to define the notion of a t -Pareto cover and set.

Definition 5 (Pareto cover and set). Let $A \subset \mathfrak{R}$ be a set in the Pareto front. A subset $S \subset A$ is a t -Pareto cover of A if for all $\mathbf{r} \in A$, there is an $\mathbf{r}_0 \in S$ so that $d(\mathbf{r}_0, \mathbf{r}) \leq t$. The t -Pareto covering number $N_{\text{Par}}(t, A)$ is the minimal size of any t -Pareto cover of A . A set $\{f_1, \dots, f_N\} \subset \mathcal{F}$ is a t -Pareto set with respect to $\mathbf{R}$ in $\mathcal{F}$ if $\mathbf{R}(\{f_1, \dots, f_N\})$ is a t -Pareto cover of $\mathfrak{R}$. We write $N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}) := N_{\text{Par}}(t, \mathfrak{R})$.

Fig. 2a visualizes a Pareto ball; the part of $\mathfrak{R}$ covered by a single point, and Fig. 2b shows how a collection of Pareto balls forms a Pareto cover. A consequence of the definition

⁵For larger t , the set defining the lower modulus is empty in HELM and we cut off the plot there.

⁶We can pull the geometry back onto the Pareto set. To do so, identify $\mathfrak{R}$ with the quotient space $\text{Par}(\mathcal{F}, \mathbf{R})/\mathbf{R}$. Formally, $f \sim g$ are equivalent when $\mathbf{R}(f) = \mathbf{R}(g)$. The equivalence class containing $f \in \text{Par}(\mathcal{F}, \mathbf{R})$ is in bijection with $\mathbf{R}(f) \in \mathfrak{R}$.

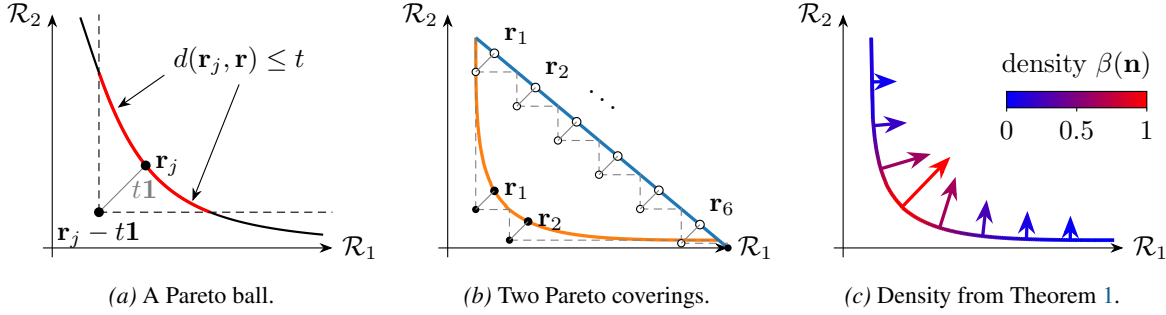


Figure 2. Using the Pareto quasi-metric from Definition 5 to cover the Pareto front. In Fig. 2a the area covered by a single point $\mathbf{r}_j$ (i.e., a Pareto ball). The number of points required adapts to the geometry of the front (2b). And in the limit, as $t \rightarrow 0$, the density of the covering β depends on the normal vector (2c), as specified in Theorem 1.

of a t -Pareto set $\{f_1, \dots, f_N\}$ and Lemma 1 is that for every $f \in \mathcal{F}$, there is some f_j so that $\mathbf{R}(f_j) \preceq \mathbf{R}(f) + t\mathbf{1}$ (or equivalently $d_{\mathbf{R}}(f_j, f) \leq t$), as visualized in Fig. 2. Also, a crude upper bound on the t -Pareto covering number is always $\log N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}) \leq (K-1) \log(1 + 1/t)$, as proved in Lemma 9 in Appendix C.

3.1. The Limiting Distribution

The next result quantifies the growth rate of the t -Pareto covering number as $t \rightarrow 0$. This rate depends on the curvature of the Pareto front. From a technical perspective, it is a main contribution of this work.

Assumption 1 (Nice Pareto front). Let $\mathfrak{R} \subset [0, 1]^K$ be a compact and smooth $(K-1)$ -dimensional submanifold (with boundary) that admits a smooth normal field $\mathbf{n} : \mathfrak{R} \rightarrow \mathbb{S}^{K-1}$ in its interior, where the unit normal vector at $\mathbf{r} \in \mathfrak{R}$, denoted by $\mathbf{n}(\mathbf{r}) = (n_1(\mathbf{r}), \dots, n_K(\mathbf{r}))$, is uniformly bounded away from zero; $n_k(\mathbf{r}) > \eta > 0$ for all $\mathbf{r} \in \mathfrak{R}$ and $k \in [K]$. We say such a Pareto front is *nice*.

Theorem 1. Let $\mathcal{H}$ denote the $(K-1)$ -dimensional Hausdorff measure and Assumption 1 hold. Then, there is a constant C_K that depends only on K so that, for any $A \subset \mathfrak{R}$ that has a boundary ∂A of $\mathcal{H}$ -measure zero, it holds that

$$\lim_{t \rightarrow 0} N_{\text{Par}}(t, A) \cdot t^{K-1} = \mu(A),$$

where μ is the following measure on the Pareto front $\mathfrak{R}$ with Borel measurable sets B :

$$\mu(B) = \int_B \beta(\mathbf{n}(\mathbf{r})) d\mathcal{H}(\mathbf{r}), \quad \beta(\mathbf{n}) = C_K \frac{\prod_{k \in [K]} n_k}{(\mathbf{1} \cdot \mathbf{n})^{K-1}}.$$

We provide further comments on the role of the assumptions at the start of the proof in Appendix F. It may initially be surprising that the density $d\mu/d\mathcal{H}$ is not uniform: the measure depends on the normal direction $\mathbf{n}(\mathbf{r})$ and puts more mass on parts of the Pareto front with larger β . In fact, a small β is closely related to a large gain-to-loss ratio for some pair of benchmarks; this is called an improper trade-off (Geoffrion, 1968) and is generally undesirable from a

multi-objective perspective. Hence, the limiting distribution avoids putting weight on (nearly) improper solutions, implying that the upcoming algorithms *hedge* using only proper trade-offs. The density depending on the normal vectors is visualized in Fig. 2c.

The measure μ also naturally induces a distribution over the Pareto set: Without loss of generality (by Lemma 2), consider the quotient space $\text{Par}(\mathcal{F}, \mathbf{R})/\mathbf{R}$. By pulling back μ from the Pareto front, we define the *prior distribution* $\pi = \mathbf{R}^* \mu / \mu(\mathfrak{R})$ over the quotient space with density

$$\frac{d\pi}{d(\mathbf{R}^* \mathcal{H})}(f) \propto \beta(\mathbf{n}(\mathbf{R}(f))), \quad f \in \text{Par}(\mathcal{F}, \mathbf{R})/\mathbf{R}. \quad (1)$$

As we will see, this enables a PAC-Bayesian perspective of the learning problem, with the prior naturally encoding where a learner should expect to find the solution of $\mathcal{R}_Q$.

3.2. Comparison with Other Coverings

Instead of the non-uniform Pareto covering, one may also construct uniform coverings using norm balls, such as ℓ_∞ or ℓ_2 -balls (Zhang et al., 2024). Alternatively, one could also cover the simplex in ℓ_1 -norm and then minimize a weighted sum of the risks (Mansour et al., 2021). We compare the coverings in more detail in Appendix C.2. A discussion of other Pareto front approximations can be found in (Zitzler et al., 2008; Vassilvitskii and Yannakakis, 2005; Papadimitriou and Yannakakis, 2000; Compton et al., 2026).

We now explicitly calculate the density in Eq. (1) for a family of example Pareto fronts, and then numerically validate Theorem 1 on this example (cf. Fig. 4). We compare this distribution with the other coverings in Appendix C.2.

Example 1. Let $\mathcal{F} = [\varepsilon, 1 - \varepsilon]$ and $K = 2$. Then, take the risks $\mathcal{R}_1(x) = |x|^p$ and $\mathcal{R}_2(x) = |x - 1|^p$ with $p \in \mathbb{N}$. The density of π with respect to Lebesgue measure λ is proportional to the harmonic mean of the derivatives

$$\frac{d\pi}{d\lambda}(x) \propto \frac{px^{p-1}(1-x)^{p-1}}{x^{p-1} + (1-x)^{p-1}}.$$

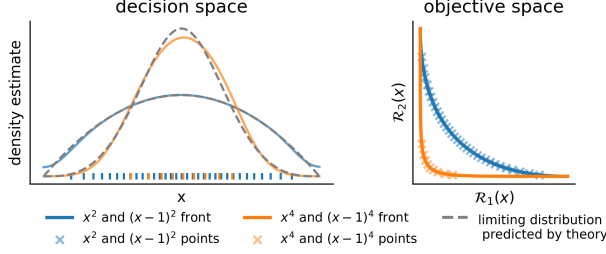


Figure 4. Limiting distribution from Example 1, cf. Fig. 10.

4. Transfer Learning under Monotonicity

In this section, we bound the statistical complexity of learning under (approximate) monotonicity.

4.1. The Statistical Complexity of Transfer Learning

In this section, we discuss two estimators based on the *empirical risk*, defined as the average loss on the i.i.d. sample $\{z_i\}_{i=1}^n \sim Q^{\otimes n}$, that is, $\widehat{\mathcal{R}}_Q(f) := \frac{1}{n} \sum_{i=1}^n \ell(f, z_i)$.

Pareto ERM. The first algorithm is a simple two-step procedure: for some fixed $t > 0$,

- (i) build a minimal t -Pareto set $\{f_1, \dots, f_N\}$, and
- (ii) compute $\widehat{f}_{\text{Par}(t)} \in \arg \min_{f \in \{f_1, \dots, f_N\}} \widehat{\mathcal{R}}_Q(f)$.

We call $\widehat{f}_{\text{Par}(t)}$ the *Pareto Empirical Risk Minimizer*.

Pareto EW. For the second algorithm, we consider again the quotient space $\text{Par}(\mathcal{F}, \mathbf{R})/\mathbf{R}$, and let Assumption 1 hold. Then, we can let π be the prior distribution supported on $\text{Par}(\mathcal{F}, \mathbf{R})/\mathbf{R}$ from Eq. (1), induced by the limiting distribution in Theorem 1. The Exponential Weights (EW) posterior $\widehat{\rho} \ll \pi$ with temperature $\lambda > 0$ is defined by its Radon-Nikodym derivative

$$\frac{d\widehat{\rho}}{d\pi}(f) \propto \exp(-\lambda \widehat{\mathcal{R}}_Q(f)), \quad f \in \text{Par}(\mathcal{F}, \mathbf{R})/\mathbf{R}. \quad (2)$$

We call the aggregated predictor $\widehat{f}_{\widehat{\rho}}(\cdot) := \mathbb{E}_{f \sim \widehat{\rho}}[f(\cdot)]$ the *Pareto Exponential Weights* (Pareto EW).

Note that the t -Pareto set concentrates in different regions of the Pareto set (Fig. 2), which “biases” the ERM according to the geometry. For the Pareto EW, this is even more explicit, as it directly endows the Pareto set with a prior. In both cases, the better trade-offs there are, the more both estimators *hedge* by preferring these trade-offs over other Pareto optimal models.⁷ We formalize the benefits of hedging for transfer in the following upper bounds on the excess risk $\mathcal{E}_Q(\widehat{f}, \mathcal{F}) = \mathcal{R}_Q(\widehat{f}) - \inf_{f \in \mathcal{F}} \mathcal{R}_Q(f)$. Recall that we as-

⁷The bounds do not differ significantly between the two estimators; we include the Pareto EW to highlight the mechanism of hedging depending on the Pareto front through Theorem 1.

sume that the loss is bounded $\ell \in [0, 1]$, $\overline{\omega}_Q(t)$ is the upper modulus defined in Definition 4.

Theorem 2. *With probability at least $1 - \delta$, the Pareto ERM $\widehat{f}_{\text{Par}(t)}$ achieves*

$$\mathcal{E}_Q(\widehat{f}_{\text{Par}(t)}, \mathcal{F}) \leq \overline{\omega}_Q(t) + \sqrt{\frac{2 \log(2N_{\text{Par}}(t, \mathcal{F}, \mathbf{R})/\delta)}{n}}.$$

Moreover, in the setting of Theorem 1, assume that for some $f^* \in \arg \min_{f \in \mathcal{F}} \mathcal{R}_Q(f)$ its benchmark values $\mathbf{R}(f^*)$ are in the relative interior point of $\mathfrak{R}$ and that the loss is convex. Then there exists a $t_0 > 0$ such that for all $t \in (0, t_0]$ and for $\lambda = \sqrt{8n \log((4N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}))/\delta)}$, the EW posterior $\widehat{\rho}$ achieves with probability at least $1 - \delta$

$$\mathcal{E}_Q(\widehat{f}_{\widehat{\rho}}, \mathcal{F}) \leq \overline{\omega}_Q(t) + \sqrt{\frac{2 \log(4N_{\text{Par}}(t, \mathcal{F}, \mathbf{R})/\delta)}{n}}.$$

The proof is in Appendix G.2. First, observe that both estimators are consistent whenever there exists a sequence $t_n \rightarrow 0$ so that $\overline{\omega}_Q(t_n) \rightarrow 0$ as $n \rightarrow \infty$ (i.e., weak monotonicity holds). At the same time, note that the bounds hold even when monotonicity is violated: if $\overline{\omega}_Q(0) > 0$, we simply incur a potentially irreducible approximation error of the Pareto set, as discussed in Section 2.2.

For example, when a decreasing upper bound on $\overline{\omega}_Q$ is known, we can choose t to balance the two terms appearing in Theorem 2, exemplified in the following corollary for a linearly decaying modulus as $t \rightarrow 0$.

Corollary 1. *Suppose that $\overline{\omega}_Q(t) \leq Lt$ with L known to the learner. Define*

$$t_n = \inf \left\{ t > 0 : t^2 \geq \frac{2 \log(2N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}))}{L^2 n} \right\}.$$

Then for any $t > t_n$, with probability at least $1 - \delta$,

$$\mathcal{E}_Q(\widehat{f}_{\text{Par}(t)}, \mathcal{F}) \leq 2Lt + \sqrt{\frac{2 \log(1/\delta)}{n}}.$$

If $n > \frac{2 \log 2}{L^2}$ and $K > 1$, then it holds that $t_n^2 \leq 1 \wedge \frac{K-1}{L^2 n} \log(1 + \frac{16L^2 n}{K-1})$. An analogous bound holds under the additional assumptions of Theorem 2 for the Pareto EW.

The proof is a direct consequence of Theorem 2 and can be found in Appendix G.3. The second part of the corollary uses a worst-case upper bound on the Pareto covering number (Lemma 9) to obtain a uniform bound on t_n . It implies that the complexity of the transfer problem is at most of order $\sqrt{K/n}$ (up to log factors). In Proposition 2 of Appendix E.1, we demonstrate that for strongly convex losses, by using a modified version of Pareto ERM, we can achieve fast rates in the worst case of order K/n .

Mansour et al. (2021) also prove a $\sqrt{K/n}$ rate for target tasks that are mixtures of the sources, i.e., which

are expectations of the same loss over distributions $Q \in \text{conv}(P_1, \dots, P_K)$. Under their mixture assumption, we have $\bar{\omega}(t) \leq t$ (Lemmas 4 and 5) and the corollary applies with the choice of $t_n \asymp \sqrt{K/n}$. Thus our result recovers their worst-case rate, while yielding potentially smaller bounds (by a constant factor characterized by $\mu(\mathfrak{R})$ with μ from Theorem 1) in benign regimes where $\log N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}) \ll K \log(1/t)$ (e.g., when the loss function exhibits strong curvature). Indeed, as a result of using an ℓ_1 -norm discretization of the simplex, their bound is agnostic to the geometry of the Pareto front, whereas ours is adaptive (cf. Fig. 10). In Appendix D, we prove that the separation between the algorithm proposed by Mansour et al. (2021) and Pareto ERM is not only in the upper bound; it is indeed suboptimal even when the target is a mixture.

A matching lower bound. In Mansour et al. (2021, Theorem 7), a matching lower bound of order $\sqrt{K/n}$ is also established. However, it is only under the very worst-case geometry that this lower bound applies. We now show a refined lower bound for any *fixed* Pareto covering number, and that Corollary 1 is tight in the following sense: If we fix t , the Pareto covering number N_{Par} at scale t , and the modulus $\bar{\omega}_Q(t) = t$, we can find a problem instance where the rate obtained by Corollary 1 is tight up to constants.

Theorem 3. *Let $K \geq 4$. Then, for every $5 \leq N \leq \exp(K-1)$, $n \geq 64 \log(N/4)$ and $t \leq \sqrt{\log(N/4)/64n}$, there exists a $(\mathcal{F}, \mathbf{R}, \{\mathcal{R}_Q : Q \in \mathcal{Q}\})$ so that (i) $\bar{\omega}_Q(t) = t$ for all $Q \in \mathcal{Q}$, (ii) $N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}) = N$ and (iii), the minimax excess risk is lower bounded by*

$$\inf_{\hat{f}} \sup_{Q \in \mathcal{Q}} \mathbb{E} \left[\mathcal{E}_Q(\hat{f}, \mathcal{F}) \right] \geq t,$$

where the infimum is taken over all estimators.

The proof is in Appendix G.4. For $\log N = K - 1$, we again obtain in the worst case a rate of $\sqrt{K/n}$, matching Mansour et al. (2021, Theorem 7).

4.2. A Parameter-free Estimator for Finite Classes

In the previous section, we derived upper bounds that are consistent whenever $\bar{\omega}_Q(t) \rightarrow 0$ as $t \rightarrow 0$, i.e., weak monotonicity holds. However, we saw in Fig. 3 that for instance both HELM and VHELM only approximately satisfy monotonicity. In this section, we discuss estimators that can always achieve consistency, but converge faster the closer to exact monotonicity the setting is.

We begin by introducing the following threshold:

$$\underline{\gamma}_Q := \inf \{ t \in [0, 1] : \bar{\omega}_Q(t) \geq 0 \}.$$

It captures the minimum amount that all risks need to be improved so that the target risk is never worse. This quantity

is closely related to the pivot considered by Hanneke and Kpotufe (2024, Definition 15) in the single-source setting. Fig. 3 illustrates this threshold for HELM and VHELM.

Margin ERM. Define the γ -margin pruned class for $\gamma = 0$ as $\mathcal{F}_0 := \text{Par}(\mathcal{F}, \mathbf{R})$ and for $\gamma > 0$ as

$$\mathcal{F}_\gamma = \{ f \in \mathcal{F} : \Gamma(f) \leq \gamma \}, \quad (3)$$

where $\Gamma(f) := -\inf_{f' \in \mathcal{F}} d_{\mathbf{R}}(f', f)$. Intuitively, $\Gamma(f) \leq \gamma$ means that no other $f' \in \mathcal{F}$ can strictly beat f by more than γ in all coordinates at the same time, and f is within a “margin” of γ to the Pareto set. Define the γ -margin ERM as $\hat{f}_\gamma \in \arg \min_{f \in \mathcal{F}_\gamma} \hat{\mathcal{R}}_Q(f)$. When $\gamma > \underline{\gamma}_Q$, the definition of $\underline{\gamma}_Q$ guarantees that at least one optimal model must lie within the margin γ , and hence the margin ERM can be consistent (Appendix E.2). However, this requires oracle knowledge of a valid $\gamma > \underline{\gamma}_Q$. Instead, for *finite* hypothesis classes, we now propose an adaptive choice of the parameter γ based on a version of structural risk minimization.

Adaptive Margin ERM. The adaptive method is summarized by the following simple algorithm.

- (i) For all $f \in \mathcal{F}$ compute the margin $\Gamma(f)$, and sort increasingly to obtain $\gamma_{(1)} \leq \dots \leq \gamma_{(|\mathcal{F}|)}$.⁸
- (ii) Let $m_n := \min\{|\mathcal{F}|, \max\{n, |\text{Par}(\mathcal{F}, \mathbf{R})|\}\}$ and choose $\gamma_n := \gamma_{(m_n)}$.
- (iii) Return $\hat{f}_{\text{adapt}} \in \arg \min_{f \in \mathcal{F}_{\gamma_n}} \hat{\mathcal{R}}_Q(f)$ where $\mathcal{F}_{\gamma_n}$ is defined in Eq. (3).

The adaptive estimator is an instance of the margin ERM with $\gamma = \gamma_n$. Roughly speaking, as n increases, this method runs ERM on an expanding subset of models ordered by their margin to the Pareto set and of size approximately m_n . For large sample sizes n , this strategy recovers regular ERM on all of $\mathcal{F}$. For this strategy, we can prove an adaptive excess risk bound (proof in Appendix G.5).

Proposition 1. *With probability at least $1 - \delta$ it holds that $\mathcal{E}_Q(\hat{f}_{\text{adapt}}, \mathcal{F}) \leq \varepsilon_n + 40\sqrt{\log(8|\mathcal{F}_{\gamma_n}|/\delta)/n}$, for a sample size-dependent error ε_n specified in the proof. Moreover, for all $n \geq n_0 := \min_{\gamma > \underline{\gamma}_Q} |\mathcal{F}_\gamma|$, it holds $\varepsilon_n = 0$.*

If monotonicity holds approximately with a small threshold $\underline{\gamma}_Q$, the error term ε_n vanishes already for small n and the estimator benefits from its inductive bias. On the other hand, it is also guaranteed to eventually match the performance of the ERM for large sample sizes and achieve consistency. This approach can be extended to infinite hypothesis spaces via standard structural risk minimization arguments.

⁸Here $\gamma_{(j)}$ denotes the j -th element of the increasingly sorted list of margins $\{\Gamma(f) : f \in \mathcal{F}\}$.

4.3. Experimental Results on HELM and VHELM

We empirically evaluate the adaptive estimator $\hat{f}_{\text{adapt}}$ (together with natural baselines) on HELM and VHELM. Let $\mathcal{F}$ denote the finite *zoo* of models evaluated on the benchmark scenarios⁹ $\mathcal{S}$. Our experimental setup is as follows: We form *transfer combinations* by selecting K source tasks $S = \{S_1, \dots, S_K\} \subset \mathcal{S}$ and a distinct target task $Q \in \mathcal{S} \setminus S$. For each (S, Q) , we sample (without replacement) n test instances from Q , compute the empirical target risks $\hat{\mathcal{R}}_Q(f)$, and apply each selection rule to obtain estimators $\hat{f}$. For each (S, Q) and each n on a predefined grid, we repeat the target subsampling 500 times (for all transfer combinations) and report the average excess risk where the target risks $\mathcal{R}_Q(\hat{f})$ are evaluated on the full target pool. More details (with also an ablation on K) are in Appendix A.

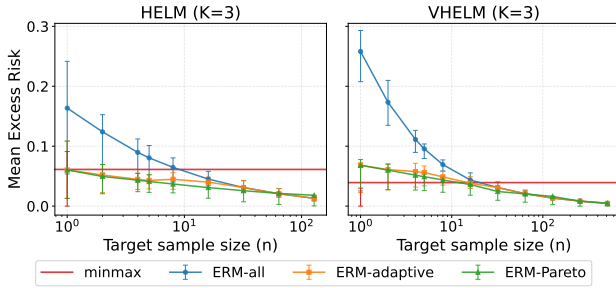


Figure 5. Mean (over all possible (S, Q)) excess risks.

Methods. We compare the following selection rules:

- ERM-all: $\hat{f}_{\text{all}} \in \arg \min_{f \in \mathcal{F}} \hat{\mathcal{R}}_Q(f)$;
- ERM-Pareto: $\hat{f}_{\text{Par}(0)} \in \arg \min_{f \in \text{Par}(\mathcal{F}, \mathbf{R})} \hat{\mathcal{R}}_Q(f)$;
- ERM-adaptive: $\hat{f}_{\text{adapt}} \in \arg \min_{f \in \mathcal{F}_{\gamma_n}} \hat{\mathcal{R}}_Q(f)$;
- Min-Max: $\hat{f}_{\text{rob}} = \arg \min_{f \in \mathcal{F}} \max_{k \in [K]} \mathcal{R}_k(f)$.

The Min-Max does not use target samples and is a popular procedure for robust (worst-case) generalization over mixtures (Sagawa et al., 2019; Mohri et al., 2019).

Results and Discussion. The results are reported in Fig. 5. Across both HELM and VHELM, restricting ERM to the frontier yields large gains at small sample size n : $\hat{f}_{\text{all}}$ suffers from high variance, whereas restricting to the frontier reduces this variance without introducing too large of a bias. By design, $\hat{f}_{\text{adapt}}$ is no worse than $\hat{f}_{\text{all}}$ eventually, but introduces large gains for small sample sizes. These results support the general message of our paper: when source risks induce a meaningful preorder, biasing towards the Pareto frontier can substantially reduce the statistical complexity of transfer, especially at small target sample sizes.

⁹Although HELM and VHELM use the term *scenarios*, we use it interchangeably with *tasks* throughout this section.

5. Model Selection Aggregation

In Section 4 we saw that, under monotonicity, we can achieve a worst-case error of order $\sqrt{K/n}$ for bounded losses, and K/n for strongly convex losses. This can be prohibitive for a large number of benchmarks K , and we may be forced to aim for a weaker guarantee: maybe a model that performs well on *one* of the benchmarks already performs well on the new task. More formally, if we denote the minimizer of benchmark $\mathcal{R}_k$ on $\mathcal{F}$ as f_k , how can we find a model $\hat{f}$ that performs as well as $\arg \min_{k \in [K]} \mathcal{R}_Q(f_k)$?

This is a particular instantiation of classical model selection aggregation (Tsybakov, 2003), in which the comparator class is usually an arbitrary finite dictionary $\{f_1, \dots, f_K\}$. Without any curvature assumptions on the loss, the minimax rate of this problem is known to be $\sqrt{\log(K)/n}$ (Lecué and Rigollet, 2014), and for strongly convex losses it is $\log(K)/n$ (Tsybakov, 2003). Let $\rho \in \mathcal{P}([K]) \cong \Delta^{K-1}$ be a distribution on the dictionary indices, and denote $f_\rho := \sum_{k \in [K]} \rho_k f_k$. Classical aggregators achieve fast rates under strongly convex losses by *hedging*: they return a $\hat{f}_\rho$ in the convex hull of the dictionary. Hedging is necessary because any proper method that selects one of the dictionary elements is minimax suboptimal (Juditsky et al., 2008). However, these methods are completely oblivious to the origin of the dictionary. This raises the question: Can we leverage the benchmarks together with monotonicity for model selection aggregation?

Let us assume throughout this section that $\mathcal{F}$ contains $\text{conv}(f_1, \dots, f_K)$. Then, roughly speaking, both the convex hull and the Pareto set in $\mathcal{F}$ can be identified with the simplex Δ^{K-1} through the following maps: $\Delta^{K-1} \ni \rho \mapsto f_\rho$ and $\Delta^{K-1} \ni \rho \mapsto \psi_\rho$ defined as

$$\begin{aligned} \psi_\rho &= \arg \min_{f \in \mathcal{F}} \max_{k \in [K]} \{\mathcal{R}_k(f) - \mathcal{R}_k(f_\rho)\}, \\ \Psi(\rho) &= \min_{k \in [K]} \{\mathcal{R}_k(f_\rho) - \mathcal{R}_k(\psi_\rho)\} \geq 0. \end{aligned} \quad (4)$$

Notice that $\mathbf{R}(\psi_\rho) \preceq \mathbf{R}(f_\rho) - \Psi(\rho)\mathbf{1}$, and so each point on the convex hull can be dominated by a point on the Pareto set (see Fig. 6). Under the monotonicity assumption, it is therefore natural to return a function on the Pareto set instead of the convex hull, and to hedge there instead. We show that, similar to Section 4, the geometry of the Pareto front determines how much such an alternative strategy improves the excess risk. In Appendix E.3.1, we discuss a simple method that only requires the loss $\ell(f, z)$ to be bounded in $[0, 1]$ and convex in f to benefit from monotonicity. Here, we focus on a potentially more intriguing benefit.

5.1. Fast Rates from Strongly Concave Gap

We now show that an assumption on the separation of the Pareto set from the convex hull is enough to get *fast rates*

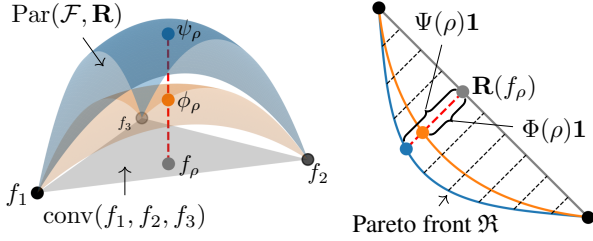


Figure 6. *Left*: the Pareto set is curved away from the convex hull. The functions ψ_ρ, ϕ_ρ are guaranteed to dominate f_ρ . *Right*: visualization of Assumption 2: if we can fit a strongly concave gap Φ between the Pareto front and the image of the convex hull, we obtain fast rates by Theorem 4.

even when the loss is *not strongly convex*. As mentioned, this is in contrast to the classical aggregation setting.

Assumption 2 (Strongly concave Pareto gap). There exists a function $\Phi : \mathcal{P}([K]) \rightarrow [0, \infty)$ such that (i) $\Phi(\delta_k) = 0$ for all k , where δ_k denotes the Dirac delta, (ii) Φ is η -strongly concave: for all $\alpha \in [0, 1]$ and $\rho, \gamma \in \mathcal{P}([K])$, it holds that

$$\begin{aligned} & \Phi(\alpha\rho + (1-\alpha)\gamma) \\ & \geq \alpha\Phi(\rho) + (1-\alpha)\Phi(\gamma) + \frac{\eta}{2}\alpha(1-\alpha)\|f_\rho - f_\gamma\|_{L^2(Q_X)}^2, \end{aligned} \quad (5)$$

and (iii) $\phi_\rho \in \mathcal{F}$ satisfies $\mathbf{R}(\phi_\rho) = \mathbf{R}(f_\rho) - \Phi(\rho)\mathbf{1}$.

By definition, we have $\Phi \leq \Psi$ and the risks $\mathbf{R}(\phi_\rho)$ lie between the image of the convex hull and the Pareto set with strongly concave gap Φ , as illustrated in Fig. 6. In fact, Assumption 2 is weaker than directly assuming strong concavity of Ψ defined in (4); such a function Φ can exist even when Ψ is not strongly concave, e.g., in Example 8. In Appendix E.3.2, we show how the two Examples 7 and 8 satisfy Assumption 2: in the former, Ψ itself is strongly concave, whereas the latter requires the “intermediate” Φ .

We now consider the estimator $\phi_{\hat{\rho}_{\text{Par}}}$ that is inspired by the Q -aggregation estimator (Dai et al., 2012; Lecué and Rigollet, 2014; Mourtada et al., 2023), defined via

$$\hat{\rho}_{\text{Par}} \in \arg \min_{\rho \in \mathcal{P}([K])} \left\{ \frac{1}{2} \widehat{\mathcal{R}}_Q(\phi_\rho) + \frac{1}{2} \mathbb{E}_{k \sim \rho} \widehat{\mathcal{R}}_Q(f_k) \right\}.$$

Theorem 4 (Fast rates from strongly concave gap). *Let Assumption 2 hold with Φ, ϕ and η known to the algorithm. Further assume that $\underline{\omega}_Q(t) \geq t$ and $\overline{\omega}_Q(t) \leq t$ for all $t \in [0, 1]$ (implying MON), that all functions f_ρ, ϕ_ρ and Y are in $[0, 1]$, and that the loss $\ell(f, (x, y)) = \ell(f(x), y)$ is κ -strongly convex in $f(x)$ (with $\kappa = 0$ allowed), as well as*

$$\begin{aligned} & \forall \hat{y}, \hat{y}' \in [0, 1] : |\ell(\hat{y}, y) - \ell(\hat{y}', y)| \leq L_\ell |\hat{y} - \hat{y}'|, \\ & \forall x \in \mathcal{X} : |\phi_\rho(x) - \phi_\gamma(x)| \leq L_\phi |f_\rho(x) - f_\gamma(x)|. \end{aligned}$$

Choose $\lambda = \frac{1}{11L_\ell} \min \left\{ \frac{1}{L_\phi}, \frac{\eta + \kappa}{L_\ell + \eta + \kappa} \right\}$. Then, for all $\delta \in$

$(0, 1)$, with probability at least $1 - \delta$,

$$\mathcal{E}_Q(\phi_{\hat{\rho}_{\text{Par}}}, \{f_k\}_{k=1}^K) \leq \frac{2 \log(K/\delta)}{\lambda n}.$$

The proof is in Appendix G.6 and is very similar to that of Lecué and Rigollet (2014) for the regular Q -aggregation estimator, despite the fact that our estimator maps to a completely different space. In order to compute $\phi_{\hat{\rho}_{\text{Par}}}$, the learner requires access to Φ from Assumption 2 with a certificate of the η -strong concavity with respect to $L^2(Q_X)$. Luckily, this can be a weak requirement, depending on the problem instance. Importantly, *without* assuming any Bernstein-type assumption, nor strong convexity of the loss, the estimator $\phi_{\hat{\rho}_{\text{Par}}}$ achieves fast rates whenever $\eta > 0$.

5.2. Some Intuition

Classical aggregation procedures output a model in the convex hull of the dictionary: to achieve the fast minimax rate $\log(K)/n$, this is required because by hedging (that is, assigning non-zero weight to more than one model), the estimators can exploit the Jensen gap between $\mathcal{R}_Q(f_\rho)$ and $\mathbb{E}_{k \sim \rho} \mathcal{R}_Q(f_k)$ (Alquier, 2024; Mourtada et al., 2023; Audibert, 2004; 2007; 2009; Lecué and Mendelson, 2009), similar to the offset term discussed in the theory of offset Rademacher complexity (Liang et al., 2015; Vijaykumar, 2021; Kanade et al., 2024). Now, by definition of the modulus (Definition 4) and Assumption 2, we have

$$\mathcal{R}_Q(\phi_\rho) \leq \mathcal{R}_Q(f_\rho) - \underline{\omega}_Q(\Phi(\rho)). \quad (6)$$

The intuitive reason we can achieve fast rates is then that combining Eqs. (5) and (6) with $\underline{\omega}_Q(t) \geq t$ yields an “artificial Jensen’s gap” or “modulus of convexity” for all ρ

$$\mathcal{R}_Q(\phi_\rho) - \mathbb{E}_{k \sim \rho} \mathcal{R}_Q(f_k) \leq -\frac{\eta + \kappa}{2} \mathbb{E}_{X \sim Q_X} \text{Var}_{k \sim \rho} f_k(X), \quad (7)$$

which is negative as long as $\eta > 0$ and ρ is in the interior of the simplex (neglecting degenerate cases). The estimator can exploit this gap to achieve fast rates. In Appendix E.3.3 we verify that Theorem 4 can genuinely yield fast rates where standard aggregation methods (in the convex hull) would fail to achieve fast rates.

6. Conclusion

Motivated by the surge in benchmarking, we study transfer learning and model selection aggregation under *monotonicity*. We show that the sample complexity is governed by the Pareto front geometry, and empirically on language benchmarks that biasing toward the frontier helps in the low-data regime. Our bounds assume finite target samples but full benchmark access; extending the results to finite source samples (Hanneke and Kpotufe, 2024) and connecting to multi-objective learning (Súkeník and Lampert, 2024; Wegel et al., 2025) are natural directions for future work.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Additional Experimental Results

In what follows, we describe the experimental protocol in detail and report additional ablation studies.

A.1. Datasets and Methods

We run our experiments on two real-world benchmark suites: HELM (Holistic Evaluation of Language Models) (Liang et al., 2023) and VHELM (Holistic Evaluation of Vision Language Models) (Lee et al., 2024).

HELM. In its complete release, HELM measures (whenever applicable) seven top-level desiderata—*accuracy*, *calibration*, *robustness*, *fairness*, *bias*, *toxicity*, and *efficiency*—over a suite of 16 *core* scenarios, often reporting multiple sub-metrics per desideratum. Whenever a metric admits multiple variants, we use the *primary* metric reported on the HELM leaderboard. Since HELM is a *living* benchmark that evolves over time, we fix a single public snapshot and run all experiments on release v1.0.0 to avoid version drift and minimize missing entries.

Because our goal is not an extensive empirical validation but rather to illustrate that our *monotonicity* assumption is reasonably consistent with practice, we work with the HELM Lite setup. In v1.0.0, HELM Lite contains evaluations of 31 LLMs on 7 scenarios: OPENBOOKQA, GSM8K, LEGALBENCH, MATH, MEDQA, MMLU, and NARRATIVEQA. For each scenario, we extract a *single* scalar performance measure given by its primary leaderboard metric (e.g., exact match for multiple-choice QA, F1 for NARRATIVEQA, and equivalence-based accuracy for MATH). We treat each scenario as providing one “accuracy-style” objective, and convert metrics to risks via a fixed monotone transformation (e.g., $1 - \text{score}$ when larger is better). Finally, to increase variability in the Pareto set, we remove the GPT-4 models (which are substantially

Table 1. Notation.

Symbol	Meaning
$\mathbf{v} \preceq \mathbf{w}$	partial ordering of vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^K$: $v_k \leq w_k$ for all k
$\mathbf{v} \prec \mathbf{w}$	strict partial ordering of vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^K$: $v_k \leq w_k$ for all k and $\mathbf{v} \neq \mathbf{w}$
$\mathbf{v} \ll \mathbf{w}$	strong partial ordering of vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^K$: $v_k < w_k$ for all k
$\mathbf{1}$	all-ones vector $\mathbf{1} = (1, \dots, 1)^\top$
$\mathbf{1}\{\cdot\}$	the indicator function
$\mathcal{F}$	underlying hypothesis space
$\mathbf{R} = (\mathcal{R}_k)_{k=1}^K$	vector of benchmarks / source risks
$\preceq_{\mathbf{R}}$	Partial order induced by $\mathbf{R}$
$\mathcal{R}_Q$	target risk on distribution Q
$\mathcal{E}_Q(\hat{f}, \mathcal{F})$	excess risk $\mathcal{E}_Q(\hat{f}, \mathcal{F}) = \mathcal{R}_Q(\hat{f}) - \inf_{f \in \mathcal{F}} \mathcal{R}_Q(f)$
$\text{Par}(\mathcal{F}, \mathbf{R})$	Pareto set in $\mathcal{F}$ with respect to $\mathbf{R}$ (Definition 2)
$d_{\mathbf{R}}$	Pareto distance (Definition 3)
$\bar{\omega}_Q(t), \underline{\omega}_Q(t)$	upper and lower modulus of monotonicity at scale t (Definition 4)
$\mathfrak{R}$	Pareto front, that is, $\mathbf{R}(\text{Par}(\mathcal{F}, \mathbf{R}))$
$N_{\text{Par}}(t, \mathcal{F}, \mathbf{R})$	Pareto covering number at scale t (Definition 5)
$\mathcal{P}([K])$	the set of distributions on $[K] = \{1, \dots, K\}$
$\triangle^{K-1}$	the $(K-1)$ -dimensional simplex $\{\mathbf{v} \in \mathbb{R}^K : v_k \geq 0, \sum_{k=1}^K v_k = 1\}$
$\hat{f}_{\text{Par}(t)}$	Pareto ERM at scale t (Section 4.1)
$\hat{f}_{\hat{\rho}}$	Pareto EW (Section 4.1)
$\hat{f}_{\text{star}(t)}$	Pareto Star Estimator at scale t (Appendix E.1)
$\hat{f}_{\gamma}$	Margin ERM at margin γ (Section 4.2)
$\hat{f}_{\text{adapt}}$	Adaptive Margin ERM (Section 4.2)
$\phi_{\hat{\rho}_{\text{Par}}}$	Fast rate aggregator (Section 5.1)

stronger than the remaining models), yielding a final model set of size 29. With 7 scenarios, the total number of source/target combinations for a fixed $K \in \{1, \dots, 6\}$ equals $7 \cdot \binom{6}{K}$.

VHELM. We use the v2.0.0 release of VHELM, which evaluates 26 vision-language models across multiple metrics; we focus on the accuracy-style metric and restrict attention to 9 scenarios for which sufficient data are available for all models: A_OKVQA, BLINK, FAIR_FACE, HATEFUL_MEMES, MATH_VISTA, MM_STAR, MMMU, SEED_BENCH, and UNICORN. We discard scenarios with insufficient coverage across models. With 9 scenarios, the total number of source/target combinations for a fixed $K \in \{1, \dots, 8\}$ equals $9 \cdot \binom{8}{K}$.

Baselines. For completeness, we restate here the model-selection methods reported in the experiments.

- ERM-all: $\hat{f}_{\text{all}} \in \arg \min_{f \in \mathcal{F}} \hat{\mathcal{R}}_Q(f)$;
- ERM-Pareto: $\hat{f}_{\text{Par}(0)} \in \arg \min_{f \in \text{Par}(\mathcal{F}, \mathbf{R})} \hat{\mathcal{R}}_Q^{(n)}(f)$ as defined in Section 4.1 (note here that by setting the Parameter $t = 0$ we effectively use the entire Pareto frontier);
- ERM-adaptive: we compute the parameter γ_n and the corresponding pruned hypothesis space $\mathcal{F}_{\gamma_n}$; yielding the adaptive margin ERM $\hat{f}_{\text{adapt}} \in \arg \min_{f \in \mathcal{F}_{\gamma_n}} \hat{\mathcal{R}}_Q(f)$;
- Min-Max: $\hat{f}_{\text{rob}} = \min_{f \in \mathcal{F}} \max_{k \in [K]} \mathcal{R}_k(f)$, the robust estimator that does not use any target data.

Since we assume black-box access to models through benchmark evaluations, we do not compare with representation-based selection methods (Dong et al., 2022; Zhang et al., 2023).

Experimental protocol. Let $\mathcal{F}$ denote the finite set of models evaluated on the benchmark scenarios $\mathcal{S}$ (HELM or VHELM). We form *transfer combinations* by selecting K source scenarios $S = \{S_1, \dots, S_K\} \subset \mathcal{S}$ and a distinct target scenario $Q \in \mathcal{S} \setminus S$. For each (S, Q) , we compute the Pareto set $\text{Par}(\mathcal{F}, \mathbf{R})$ using all available source data. To emulate limited labeled target data, we then sample (without replacement) n target instances from Q , compute the empirical target risks $\hat{\mathcal{R}}_Q(f)$, and apply each selection rule to obtain $\hat{f}$. We evaluate its target risk $\mathcal{R}_Q(\hat{f})$ on the full target pool. For each (S, Q) and each n , we repeat the target subsampling $B = 500$ times and report the average excess risk

$$\hat{\mathcal{E}}_B(\hat{f}) = \frac{1}{B} \sum_{i=1}^B \left(\mathcal{R}_Q(\hat{f}^{(i)}) - \min_{f \in \mathcal{F}} \mathcal{R}_Q(f) \right),$$

where $\hat{f}^{(i)}$ is the model selected on trial i of the subsampling. Finally, for each n in a predefined grid, we aggregate results by averaging $\hat{\mathcal{E}}_B(\hat{f})$ over all transfer combinations.

A.2. Ablation on the Number of Benchmarks

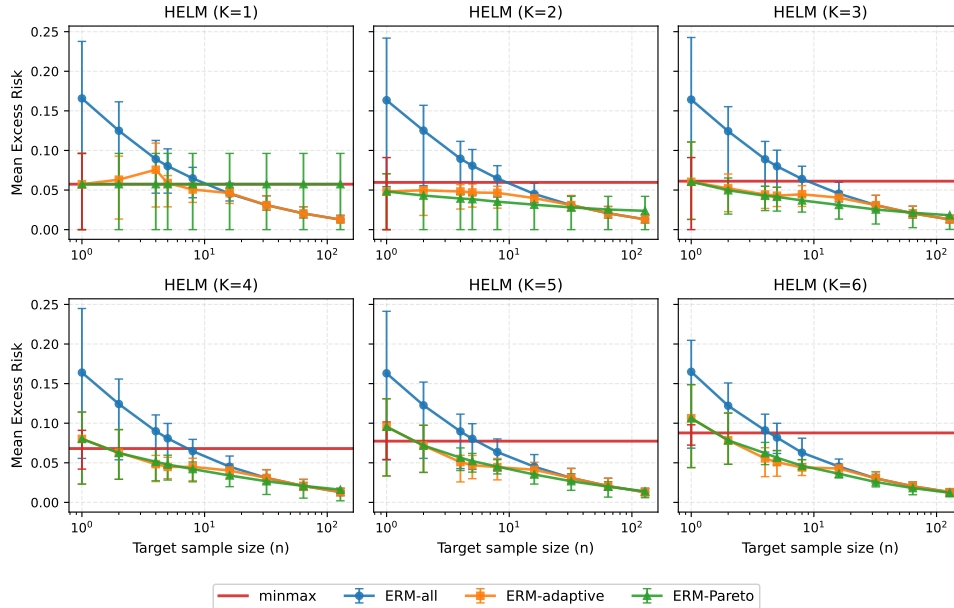


Figure 7. Mean (over all possible combinations of K sources and a target Q) excess risk of different model selection methods. Vertical bars are (25-75 quartiles).

The results of Fig. 5 are presented for a fixed value K of source tasks. Following, we presented the same experiment by varying K . In particular, HELM has 7 scenarios and thus $K \in \{1, \dots, 6\}$; similarly, VHELM has 9 scenarios and $K \in \{1, \dots, 8\}$. We highlight some interesting observations from the plots in Figs. 7 and 8. First of all, the main qualitative pattern is consistent across both benchmark suites: *frontier-biased* selection methods (Pareto ERM and adaptive ERM) provide a smaller excess risk compared to ERM on the full hypothesis class in the low-data regime, where ERM over the full class suffers from substantial variance. More precisely, for small target sample sizes n , both ERM-Pareto and ERM-adaptive significantly improve upon ERM-all for every K . As n grows, the gap narrows and all methods converge, reflecting the fact that estimation noise on the target decreases with n . On the other hand, for bigger values of n , ERM on the Pareto set can be worse; this can be appreciated for example in the case of $K = 1$ for both HELM and VHELM and this is due to the fact that the oracle model $f^* \in \arg \min_{f \in \mathcal{F}} \mathcal{R}_Q(f)$ is not always in the Pareto frontier induced by the source tasks. In the HELM experiments with $K = 1$, we observe an extreme version of this phenomenon: for essentially all combinations the Pareto set is a singleton, and the corresponding model is not the target oracle. Consequently, ERM-Pareto becomes identical to a purely source-based selection rule (it does not depend on n) and its excess risk matches the constant min max baseline.

Finally, the adaptive procedure $\hat{f}_{\text{adapt}}$ of Section 4.2 mitigates this potential bias by enlarging the candidate set with n : by construction $\mathcal{F}_{\gamma_n}$ contains the Pareto set and grows with n , interpolating between strict frontier restriction at very small n

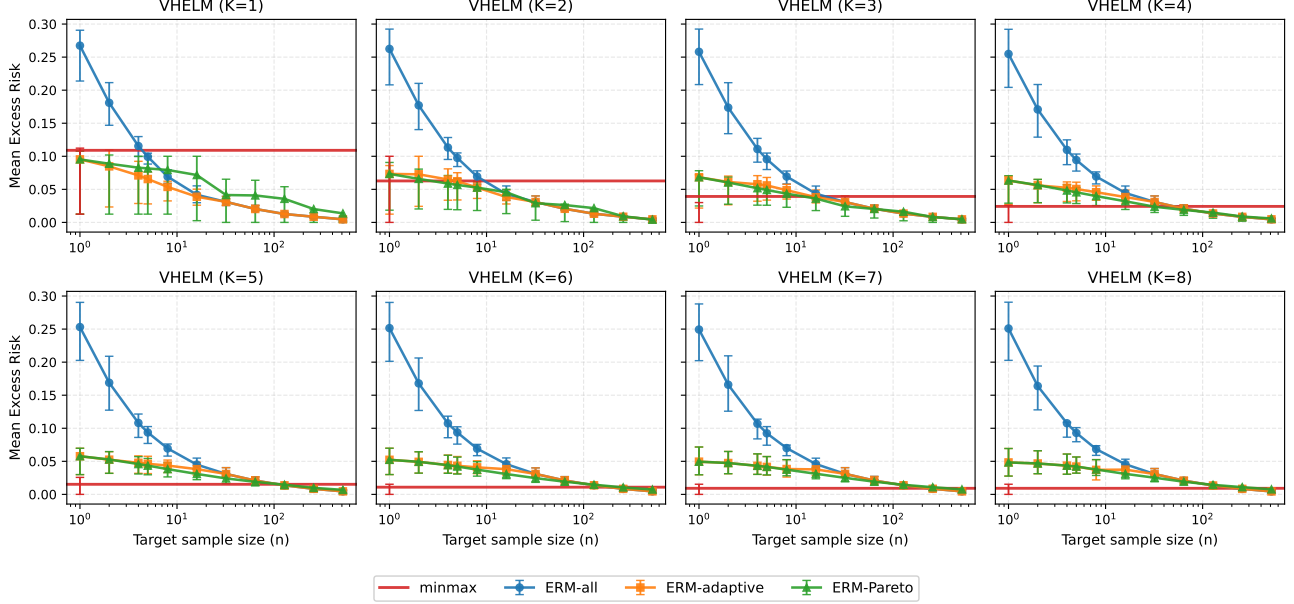


Figure 8. Mean (over all possible combinations of K sources and a target Q) excess risk of different model selection methods. Vertical bars are (25-75 quartiles).

and ERM over $\mathcal{F}$ for larger n . Empirically, this yields performance close to ERM-Pareto in the low-data regime (where variance dominates) while reducing the risk of a plateau when n is large (where approximation bias becomes the limiting factor).

B. Additional Discussions on the Monotonicity Assumption

In this section, we provide some more discussion and results about our monotonicity assumption (MON). We restate here for completeness a couple of previously introduced definitions that will be needed for the discussion. Recall the definitions of monotonicity and Pareto optimality from Section 2.1.

We begin by mentioning that the sufficiency from Lemma 2 can be exploited more explicitly. Recall that by Lemma 2, the representation $\mathbf{R}(f)$ gives sufficient information to compute $\mathcal{R}_Q$, as long as we can learn the mapping s . Prior work has studied algorithms that explicitly learn this map from a fixed class of scalarizations (Mansour et al., 2021). However, this may prove suboptimal, either when the chosen class of scalarizations is unnecessarily complex (see Section 4), or on the contrary, not expressive enough. This can happen, for instance, when the Pareto front is *non-convex* (see Example 2 below). In contrast, our approach completely avoids learning the scalarization s .

B.1. Results for the Pareto Distance and Moduli

We now provide two results about the Pareto distance and the moduli (Definitions 3 and 4). Recall that the Pareto distance measures the smallest amount t needed to improve all objectives $\mathcal{R}_k(f)$ in order to achieve a performance that is no worse than $\mathcal{R}_k(f')$. Note that, for arbitrary functions f and g , $d_{\mathbf{R}}(f, g)$ can be negative and asymmetric. In particular, this is not a metric even when restricted to the Pareto set: symmetry can fail, and $d_{\mathbf{R}}(f, g) = 0$ does not imply $f = g$. On the Pareto front, we can prove the following.

Lemma 3. *Let $f, g \in \mathcal{F}$. The Pareto distance (Definition 3) satisfies $d_{\mathbf{R}}(f, g) = \min\{t \in \mathbb{R} : \mathbf{R}(f) - t\mathbf{1} \preceq \mathbf{R}(g)\}$. Moreover, if $f, g, h \in \text{Par}(\mathcal{F}, \mathbf{R})$ are restricted to the Pareto set, it satisfies the following properties:*

- (i) *Non-negativity:* $d_{\mathbf{R}}(f, g) \geq 0$ and $d_{\mathbf{R}}(f, f) = 0$.
- (ii) *Triangle inequality:* $d_{\mathbf{R}}(f, h) \leq d_{\mathbf{R}}(f, g) + d_{\mathbf{R}}(g, h)$.

Proof. If t is the smallest value satisfying $\mathbf{R}(f) - \mathbf{R}(g) \preceq t\mathbf{1}$, then it must be larger than $\max_{k \in [K]} \mathcal{R}_k(f) - \mathcal{R}_k(g)$, thus $d_{\mathbf{R}}(f, g) \leq \min \{t : \mathbf{R}(f) - t\mathbf{1} \preceq \mathbf{R}(g)\}$. On the other hand, $\mathbf{R}(f) - d_{\mathbf{R}}(f, g)\mathbf{1} \preceq \mathbf{R}(g)$, so that $d_{\mathbf{R}}(f, g) \geq \min \{t : \mathbf{R}(f) - t\mathbf{1} \preceq \mathbf{R}(g)\}$. And so, equality holds. When $f, g, h \in \text{Par}(\mathcal{F}, \mathbf{R})$, we also have:

1. Since $f, g \in \text{Par}(\mathcal{F}, \mathbf{R})$, neither dominates the other. That is, there exists some $k \in [K]$ such that $\mathcal{R}_k(f) \geq \mathcal{R}_k(g)$. The maximum over $k \in [K]$ is there non-negative. That $d_{\mathbf{R}}(f, f) = 0$ is immediate by the definition.
2. By definition, we have $\mathbf{R}(f) - d_{\mathbf{R}}(f, g)\mathbf{1} \preceq \mathbf{R}(g)$ and $\mathbf{R}(g) - d_{\mathbf{R}}(g, h)\mathbf{1} \preceq \mathbf{R}(h)$. Combining these, we obtain that

$$\mathbf{R}(f) - (d_{\mathbf{R}}(f, g) + d_{\mathbf{R}}(g, h))\mathbf{1} \preceq \mathbf{R}(h),$$

which implies that $d_{\mathbf{R}}(f, h) \leq d_{\mathbf{R}}(f, g) + d_{\mathbf{R}}(g, h)$.

That concludes the proof. $\square$

Now, recall the definition of the moduli (Definition 4), which are the quantities that directly follow from the distance $d_{\mathbf{R}}(\cdot, \cdot)$. We provide the following bounds for scalarizations from Lemma 2 that are in a coordinate-separable form.

Lemma 4. Assume (MON) and denote s the scalarization from Lemma 2. If s is of the form $s(\mathbf{r}) = \sum_{k=1}^K q_k g(r_k)$ for some non-negative weights $\mathbf{q} = (q_1, \dots, q_K) \in [0, \infty)^K$ and non-decreasing $g : [0, \infty) \rightarrow \mathbb{R}$, then, for

$$\overline{\Delta}_g(t) = \sup_{u \in [0, 1]} g(u + t) - g(u), \quad \text{and} \quad \underline{\Delta}_g(t) = \inf_{u \in [t, 1]} g(u) - g(u - t),$$

it holds $\overline{\omega}_Q(t) \leq \|\mathbf{q}\|_1 \overline{\Delta}_g(t)$ and $\underline{\omega}_Q(t) \geq \|\mathbf{q}\|_1 \underline{\Delta}_g(t)$. In particular, if s is a weighted average (a.k.a. linear scalarization), then $\overline{\omega}_Q(t) \leq t$ and $\underline{\omega}_Q(t) \geq t$.

While these bounds can be loose, they are essentially tight for weighted averages (except for some edge cases). This lemma also demonstrates what happens under *monotone transformations* of the source risks. For example, if $g(t) = \sqrt{t}$, then the upper modulus is bounded on order of $\sqrt{t}$.

Proof. For any $t \geq 0$, let $f, f' \in \mathcal{F}$ be such that $\mathbf{R}(f) \leq \mathbf{R}(f') + t\mathbf{1}$. Then, using $\mathcal{R}_Q = s \circ \mathbf{R}$ from Lemma 2, the assumption that s is of the form $s(\mathbf{r}) = \sum_{k=1}^K q_k g(r_k)$, and the definition of $\overline{\Delta}_g(t)$, we get that

$$\begin{aligned} \mathcal{R}_Q(f) - \mathcal{R}_Q(f') &= s(\mathbf{R}(f)) - s(\mathbf{R}(f')) && \text{(Lemma 2)} \\ &= \sum_{k=1}^K q_k (g(\mathcal{R}_k(f)) - g(\mathcal{R}_k(f'))) && (s(\mathbf{r}) = \sum_{k=1}^K q_k g(r_k)) \\ &\leq \sum_{k=1}^K q_k (g(\mathcal{R}_k(f') + t) - g(\mathcal{R}_k(f'))) && (g \text{ is non-decreasing and } \mathcal{R}_k(f) \leq \mathcal{R}_k(f') + t) \\ &\leq \sum_{k=1}^K q_k \left(\sup_{u \in [0, 1]} g(u + t) - g(u) \right) && (\mathcal{R}_k(f') \in [0, 1]) \\ &= \|\mathbf{q}\|_1 \overline{\Delta}_g(t). && \text{(Definition of } \overline{\Delta}_g(t)) \end{aligned}$$

By taking the supremum over all such $f, f' \in \mathcal{F}$ and Definition 4 of the upper modulus, it holds that $\overline{\omega}_Q(t) \leq \|\mathbf{q}\|_1 \overline{\Delta}_g(t)$.

Similarly, for any $t \geq 0$, let $f, f' \in \mathcal{F}$ be such that $\mathbf{R}(f) \leq \mathbf{R}(f') - t\mathbf{1}$. Then, using $\mathcal{R}_Q = s \circ \mathbf{R}$ from Lemma 2 and the

assumption that s is of the form $s(\mathbf{r}) = \sum_{k=1}^K q_k g(r_k)$, and the definition of $\underline{\Delta}_g(t)$, we get that

$$\begin{aligned}
 \mathcal{R}_Q(f') - \mathcal{R}_Q(f) &= \sum_{k=1}^K q_k (g(\mathcal{R}_k(f')) - g(\mathcal{R}_k(f))) \\
 &\geq \sum_{k=1}^K q_k (g(\mathcal{R}_k(f')) - g(\mathcal{R}_k(f') - t)) \quad (g \text{ is non-decreasing and } \mathcal{R}_k(f) \leq \mathcal{R}_k(f') - t) \\
 &\geq \sum_{k=1}^K q_k \left(\inf_{u \in [t, 1]} g(u) - g(u - t) \right) \quad (\mathcal{R}_k(f) \leq \mathcal{R}_k(f') - t \text{ implies } \mathcal{R}_k(f') \geq t) \\
 &\geq \|\mathbf{q}\|_1 \underline{\Delta}_g(t).
 \end{aligned}$$

By taking the infimum over all such $f, f' \in \mathcal{F}$ and Definition 4 of the lower modulus, it holds that $\underline{\omega}_Q(t) \geq \|\mathbf{q}\|_1 \underline{\Delta}_g(t)$.

Finally, note that for a weighted average we can take $g(x) = x$ (which clearly satisfies $\overline{\Delta}_g(t) = \underline{\Delta}_g(t) = t$) and $\|\mathbf{q}\|_1 = 1$. $\square$

B.2. Monotonicity in the Statistical Learning Setting

Let $\ell_k(f, z) \in [0, 1]$ be a loss function; in the statistical learning setting, we assume that the source risks are also risks over distributions P_k on $\mathcal{Z}$, that is, $\mathcal{R}_k(f) = \mathbb{E}_{z \sim P_k} \ell_k(f, z)$ for all $k = 1, \dots, K$.

B.2.1. MIXTURE TARGET DISTRIBUTIONS AND MONOTONICITY.

We first state a basic sufficient condition for monotonicity in the statistical learning setting when the target distribution is a convex mixture.

Lemma 5. *If $Q \in \text{conv}(P_1, \dots, P_K)$ and $\ell_k = \ell$ for all k , then $\mathcal{R}_Q$ is monotonic with respect to $\mathbf{R}$, and s from Lemma 2 is a weighted average.*

Proof. Note that for some weights $\mathbf{q} = (q_1, \dots, q_K) \in \Delta^{K-1}$ we can write

$$\mathcal{R}_Q(f) = \int \ell(f, z) dQ(z) = \int \ell(f, z) d\left(\sum_{k=1}^K q_k P_k\right) = \sum_{k=1}^K q_k \int \ell(f, z) dP_k = \sum_{k=1}^K q_k \mathcal{R}_k(f).$$

Clearly, this implies that $\mathcal{R}_Q$ is monotonic in $\mathbf{R}$. $\square$

We now prove a partial converse of Lemma 5 which demonstrates that under fairly strong assumptions (roughly speaking that the span of $\{\ell(f, \cdot) : f \in \mathcal{F}\}$ is expressive enough) the target risk must be a convex combination of the source risks.

Lemma 6 (Mixture from monotonicity). *Let $\ell_k = \ell$ for all k . Define the loss class $\mathcal{L} = \{\ell_f : f \in \mathcal{F}\}$, where $\ell_f \equiv \ell(f, \cdot)$. Let $\mathcal{H} = \text{span}(\mathcal{L})$ form a Hilbert space such that:*

1. *Every function $\phi \in \mathcal{H}$ can be expressed in the form $\phi = \lambda(\ell_f - \ell_{f'})$, where $\lambda \in \mathbb{R}$ and $\ell_f, \ell_{f'} \in \mathcal{L}$,*
2. *The constant function $\mathbf{1}$ is contained in $\mathcal{H}$. Moreover, let $\mathcal{R}_1, \dots, \mathcal{R}_K$ be linearly independent in $\mathcal{H}^*$.*

If $\mathcal{R}_Q$ is weakly monotonic with respect to $(\mathcal{R}_1, \dots, \mathcal{R}_K)$, then

$$\mathcal{R}_Q \in \text{conv}_{\mathcal{H}^*}(\mathcal{R}_1, \dots, \mathcal{R}_K).$$

To explain these assumptions in words, we may view the functionals $\mathcal{R}_1, \dots, \mathcal{R}_K$ and $\mathcal{R}_Q$ as elements in the dual space $\mathcal{H}^*$. The first condition of Lemma 6 states that the model class $\mathcal{F}$ is rich enough so that we can probe the risk functional $\mathcal{R}_Q$ using only pairwise comparisons of models in $\mathcal{F}$, where for $\phi = \lambda(\ell_f - \ell_{f'})$,

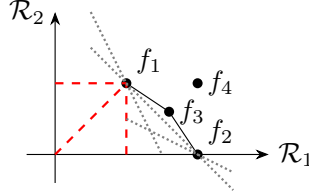
$$\langle \mathcal{R}_Q, \phi \rangle_{\mathcal{H}} = \lambda[\mathcal{R}_Q(f) - \mathcal{R}_Q(f')].$$

The second condition enables a more interpretable and aesthetically pleasing result (though it is not ‘morally’ necessary), although it rules out cases such as $\mathcal{R}_1 = -\mathcal{R}_2$.

Lemma 6 shows that if we require monotonicity to hold for a sufficiently expressive set of models $\mathcal{F}$, then this imposes a lot of constraints on how the new task $\mathcal{R}_Q$ relates to the old ones $\mathbf{R}$. However, there exist numerous meaningful examples where the conclusion of Lemma 6 can fail, especially when the loss function varies or the number K of source risks is large. We exemplify this in the following simple example.

Example 2 (Monotonicity is strictly more general than convex hull). Consider the instances $\mathcal{Z} = \{z_1, z_2, z_3\}$ and models $\mathcal{F} = \{f_1, f_2, f_3, f_4\}$, incurring the following losses:

ℓ	f_1	f_2	f_3	f_4
z_1	0.5	1.0	0.8	1.0
z_2	0.5	0	0.3	0.5
z_3	0.5	0.5	0.3	0.0



Let $P_1 = \delta_{z_1}, P_2 = \delta_{z_2}$. Then for any $\tilde{Q} \in \text{conv}(P_1, P_2)$ we have that $\mathbb{E}_{z \sim \tilde{Q}} \ell(f, z)$ is minimized by either f_1 or f_2 , as visualized by the gray dotted lines. But if $Q = \text{Uniform}(\mathcal{Z})$, then f_3 is the target minimizer. Importantly, $\mathcal{R}_Q$ remains monotonic in the sources.

This and many other examples (e.g., with varying losses) show that, even in the statistical learning setting where source risks can be written as expected losses, we generally cannot conclude that $\mathcal{R}_Q$ or Q must be in the convex hull of the source risks $\mathcal{R}_k$ or source distributions P_k , so our setting is strictly more general.

Finally, before proving Lemma 6, the following lemma is needed:

Lemma 7. Let $\mathcal{H}$ be a Hilbert space containing vectors $u, v_1, \dots, v_N$ and $\ker(v) := \ker(w \mapsto \langle w, v \rangle_{\mathcal{H}}) = v^{\perp}$. Then:

$$u \in \text{span}(v_1, \dots, v_N) \iff \bigcap_{n \in [N]} \ker(v_n) \subset \ker(u).$$

Proof. Let $V = \text{span}(v_1, \dots, v_N)$. Suppose that $u \in V$ where $u = \sum_n \lambda_n v_n$. Let $w \in \bigcap \ker(v_n)$. Then $w \in \ker(u)$ since $\langle w, u \rangle_{\mathcal{H}} = \sum_{n \in [N]} \lambda_n \langle w, v_n \rangle_{\mathcal{H}} = 0$. Suppose on the other hand that $u \notin V$. Since $\mathcal{H}$ is a Hilbert space, u decomposes into components in V and $V^{\perp}$, $u = u_V + u_{V^{\perp}}$, where the orthogonal component is nonzero, $u_{V^{\perp}} \neq 0$. Then, $u_{V^{\perp}} \in \bigcap \ker(v_n)$, but $u_{V^{\perp}} \notin \ker(u)$. $\square$

Proof of Lemma 6. Throughout, we view $\mathcal{R}_1, \dots, \mathcal{R}_K$ and $\mathcal{R}_Q$ as elements of the dual space $\mathcal{H}^*$. Lemma 7 shows that:

$$\bigcap_{k \in [K]} \ker(\mathcal{R}_k) \subset \ker(\mathcal{R}_Q) \implies \mathcal{R}_Q \in \text{span}_{\mathcal{H}^*}(\mathcal{R}_1, \dots, \mathcal{R}_K). \quad (8)$$

Assume for now that the left-hand side holds. The right-hand side almost gives the result, except that we need to ensure that $\mathcal{R}_Q$ is not just a linear span, but a convex combination. Let $\mathcal{R}_Q = \sum \lambda_k \mathcal{R}_k$. Then, in fact, it is a convex combination:

By the second condition of Lemma 6, $\mathbf{1} \in \mathcal{H}$. Thus, we obtain

$$1 = \mathcal{R}_Q(\mathbf{1}) = \sum_{k \in [K]} \lambda_k \mathcal{R}_k(\mathbf{1}) = \sum_{k \in [K]} \lambda_k.$$

Fix any $j \in [K]$. We show that $\lambda_j \geq 0$. By linear independence, the risk functionals $\mathcal{R}_k$ ranging over $k \in [K]$ are linearly independent. Lemma 7 implies that there exists $\phi \in \mathcal{H}$ such that:

$$\phi \in \bigcap_{k \in [K] \setminus \{j\}} \ker(\mathcal{R}_k) \quad \text{and} \quad \phi \notin \ker(\mathcal{R}_j).$$

By the first condition of Lemma 6, there exist $f, f' \in \mathcal{F}$ such that $\phi \in \text{span}(\ell_f - \ell_{f'})$. For all $k \neq j$, we deduce from $\phi \in \ker(\mathcal{R}_k)$ and from $\phi \notin \ker \mathcal{R}_j$ that: $\mathcal{R}_k(f) = \mathcal{R}_k(f')$ and $\mathcal{R}_j(f) \neq \mathcal{R}_j(f')$. Swapping f and f' if needed, this shows that f is strictly $\mathbf{R}$ -dominated by f' . We also obtain

$$\mathcal{R}_Q(f) - \mathcal{R}_Q(f') = \lambda_j(\mathcal{R}_j(f) - \mathcal{R}_j(f')).$$

Since $\mathcal{R}_j(f) - \mathcal{R}_j(f') \neq 0$, weak monotonicity forces $\lambda_j \geq 0$.

To finish the proof, we just need to show that the left-hand side of the implication in Equation (8) holds, namely, $\bigcap_{k \in [K]} \ker(\mathcal{R}_k) \subset \ker(\mathcal{R}_Q)$. Let $\phi \in \bigcap \ker(\mathcal{R}_k)$. Again, by the first condition of Lemma 6, there are $f, f' \in \mathcal{F}$ such that $\phi \in \text{span}(\ell_f - \ell_{f'})$. This implies that $\mathcal{R}_k(f) = \mathcal{R}_k(f')$ for all $k \in [K]$. By Lemma 2, weak monotonicity implies that $\mathcal{R}_Q(f) = \mathcal{R}_Q(f')$. Thus, $\phi \in \ker(\mathcal{R}_Q)$. $\square$

B.2.2. COVARIATE AND LABEL SHIFT.

Assume that the source and target risks share the same feature space $\mathcal{X}$, label space $\mathcal{Y}$, and loss function ℓ , but may differ in their distributions. Let $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, let $z = (x, y)$, and write $\ell(f, z) := \ell(f(x), y)$. Let $(X, Y) \sim Q$ and $(X^k, Y^k) \sim P_k$ for $k \in [K]$.

We say that Q satisfies *covariate shift* relative to $P_1, \dots, P_K$ if, for every $k \in [K]$, the regular conditional law of Y^k given X^k under P_k coincides with the regular conditional law of Y given X under Q , $P_{k,X}$ -almost surely. Similarly, we say that Q satisfies *label shift* relative to $P_1, \dots, P_K$ if, for every $k \in [K]$, the regular conditional law of X^k given Y^k under P_k coincides with the regular conditional law of X given Y under Q , $P_{k,Y}$ -almost surely.

Lemma 8. *Covariate shift and label shift are each neither necessary nor sufficient for monotonicity.*

The examples that we use below to prove this lemma highlight the following difference: monotonicity is more robust to benign structural changes (like noise, Example 3) but more sensitive to marginal shifts (Examples 4 and 5).

Proof. We prove the claim via three examples. Throughout, we consider a single source domain when constructing counterexamples; this is without loss of generality since failure (or validity) of weak monotonicity for $K = 1$ implies the same for general K by ignoring the other sources.

Example 3 (Covariate and label shift are not necessary for weak monotonicity). Weak monotonicity may hold even if the regular conditional laws of Y^1 given X^1 under P_1 and of Y given X under Q differ. Consider binary classification with 0–1 loss. Let $X^1 \sim \text{Ber}(\frac{1}{2})$ and $Y^1 = X^1$ almost surely. Define the target by $X = X^1$ and $Y = Y^1 \oplus N$, where $N \sim \text{Ber}(p)$ with $0 < p < \frac{1}{2}$, independent of (X^1, Y^1) . Then covariate shift fails since, for instance, the conditional law of Y^1 given $X^1 = 1$ is δ_1 , whereas the conditional law of Y given $X = 1$ is $(1 - p)\delta_1 + p\delta_0$. Similarly, label shift fails since the conditional law of X^1 given $Y^1 = 1$ under P_1 is δ_1 , whereas the conditional law of X given $Y = 1$ under Q is $(1 - p)\delta_1 + p\delta_0$. Moreover, for every classifier f , $\mathcal{R}_Q(f) = \mathbb{P}_{(X,Y) \sim Q}(f(X) \neq Y) = (1 - p)\mathbb{P}_{(X^1, Y^1) \sim P_1}(f(X^1) \neq Y^1) + p\mathbb{P}_{(X^1, Y^1) \sim P_1}(f(X^1) = Y^1) = p + (1 - 2p)\mathcal{R}_1(f)$. Since $1 - 2p > 0$, $\mathcal{R}_1(f) \leq \mathcal{R}_1(f')$ implies $\mathcal{R}_Q(f) \leq \mathcal{R}_Q(f')$. Thus weak monotonicity holds although covariate and label shift fail.

Example 4 (Covariate shift alone is not sufficient for weak monotonicity). Even if the regular conditional laws of Y^1 given X^1 under P_1 and of Y given X under Q coincide, weak monotonicity may fail. Consider binary classification with 0–1 loss, let $\mathcal{X} = \{a, b\}$, and let $Y^1 \equiv 0$ and $Y \equiv 0$ almost surely. Thus covariate shift holds. Let $\mathbb{P}(X^1 = a) = 0.9$, $\mathbb{P}(X^1 = b) = 0.1$, $\mathbb{P}(X = a) = 0.1$, and $\mathbb{P}(X = b) = 0.9$. Define classifiers $f(a) = 0, f(b) = 1$ and $f'(a) = 1, f'(b) = 0$. Then $\mathcal{R}_1(f) = \mathbb{P}_{(X^1, Y^1) \sim P_1}(f(X^1) \neq Y^1) = 0.1 < 0.9 = \mathbb{P}_{(X^1, Y^1) \sim P_1}(f'(X^1) \neq Y^1) = \mathcal{R}_1(f')$, whereas $\mathcal{R}_Q(f) = \mathbb{P}_{(X,Y) \sim Q}(f(X) \neq Y) = 0.9 > 0.1 = \mathbb{P}_{(X,Y) \sim Q}(f'(X) \neq Y) = \mathcal{R}_Q(f')$. Hence weak monotonicity fails.

The next example is essentially identical to Example 4, but for label shift.

Example 5 (Label shift alone is not sufficient for weak monotonicity). Even if the regular conditional laws of X^1 given Y^1 under P_1 and of X given Y under Q coincide, weak monotonicity may fail. Let $\mathcal{X} = \{x_0\}$, $\mathcal{Y} = \{0, 1\}$, and use 0–1 loss. Then label shift holds trivially. Let $\mathbb{P}(Y^1 = 0) = 0.9$ and $\mathbb{P}(Y = 0) = 0.1$. For the constant classifiers $f \equiv 0$ and $f' \equiv 1$, we have $\mathcal{R}_1(f) = \mathbb{P}_{(X^1, Y^1) \sim P_1}(f(X^1) \neq Y^1) = 0.1 < 0.9 = \mathbb{P}_{(X^1, Y^1) \sim P_1}(f'(X^1) \neq Y^1) = \mathcal{R}_1(f')$, whereas $\mathcal{R}_Q(f) = \mathbb{P}_{(X,Y) \sim Q}(f(X) \neq Y) = 0.9 > 0.1 = \mathbb{P}_{(X,Y) \sim Q}(f'(X) \neq Y) = \mathcal{R}_Q(f')$. Hence weak monotonicity fails.

Together, these three examples show that covariate shift and label shift are each neither necessary nor sufficient for weak monotonicity. $\square$

C. Additional Discussion of the Pareto Covering

C.1. Computation of a Pareto covering

In general, *exactly* computing a minimal t -Pareto cover is computationally hard (Zitzler et al., 2008; Papadimitriou and Yannakakis, 2000; Chvatal, 1979). For finite hypothesis classes, the problem becomes much simpler to analyze: Let M be the number of models, K the number of source benchmarks, and $P \leq M$ the size of the Pareto set. For $t = 0$ (used in our experiments), we only need to compute the exact Pareto set. This can be done by checking pairwise dominance across all models, which requires at most $O(KM^2)$ comparisons. In practice, this may already yield a good candidate set. For $t > 0$, one can first restrict attention to the Pareto set and then build a t -Pareto cover on top using standard greedy set-cover heuristics to obtain a small approximate cover with a logarithmic size overhead and computational cost $O(KP^2)$ (Chvatal, 1979). Notice that a logarithmic size overhead is essentially irrelevant in the upcoming bounds.

C.2. Comparison of Different Coverings

We begin by providing a more in-depth comparison with the two other ways of covering a Pareto front (beyond the Pareto covering from Definition 5) that we discussed in Section 3.

Norm covering. The first option is to construct a covering in some norm, such as the ℓ_2 -norm. An example of this is visualized in Fig. 9 on the right. It is clear that this method of covering adapts to the geometry, and if the front has low intrinsic volume, this covering number will be small. And indeed, perhaps unsurprisingly, the ℓ_2 -norm covering converges to a uniform distribution with respect to the Hausdorff measure under some regularity conditions (Borodachov et al., 2007). It also has other appealing properties, some of them discussed in Zhang et al. (2024). From a statistical perspective, however, when a notion of monotonicity holds, it is not necessary to cover the Pareto front point-wise in this norm. After all, if a point lies far away in ℓ_2 -norm, but is almost dominated by another, there is no need to spend any statistical budget on it. In particular, this is reflected in the fact that when better trade-offs become available, this covering number can actually *increase*, as visualized in Fig. 9, and numerically validated in Fig. 10d. Moreover, the distribution induced by this kind of covering does not avoid improper regions, see also Fig. 10. As demonstrated in Section 4, this is contrary to the actual statistical hardness under monotonicity, so these covering numbers do not offer a good perspective on the problem.

Simplex ℓ_1 -covering. Another option is to cover the simplex $\triangle^{K-1}$ in ℓ_1 -norm, as done in Mansour et al. (2021) (cf. Appendix D), and use it as weights for a weighted average. In particular, standard covering results show that creating a t -covering $\Lambda \subset \triangle^{K-1}$ in ℓ_1 -norm requires a set of size $|\Lambda| = \Theta(t^{-(K-1)})$. Because, under the assumption that the Pareto front is *convex*, it can be fully recovered by solving optimization problems of the form

$$f_\lambda \in \arg \min_{f \in \mathcal{F}} \sum_{k=1}^K \lambda_k \mathcal{R}_k(f),$$

it is natural to cover the Pareto front with points $\{\mathbf{R}(f_\lambda) : \lambda \in \Lambda\}$, see center panel of Fig. 9. However, this approach has three major limitations. Firstly, the *number* of covering points is completely oblivious to the Pareto front, even when it only contains one point. Hence bounds using this covering directly cannot show adaptivity to the geometry. Moreover, as demonstrated in Lemma 9 below, the Pareto covering number is never larger than $\Theta(t^{-(K-1)})$. Secondly, while the induced distribution on the Pareto set changes with the front, it can produce unnecessary redundancies and emphasis on “almost dominated” parts of the front, as visualized in Fig. 10. Informally, it is not hard to see that under suitable regularity conditions, the limiting distribution of such a covering as $t \rightarrow 0$ is the push-forward of the uniform distribution on $\triangle^{K-1}$ with the map $\lambda \mapsto \mathbf{R}(f_\lambda)$. In the setting of Example 1 for $p > 1$, this corresponds to a density on $[\varepsilon, 1 - \varepsilon]$ given by

$$\frac{d \pi_{\text{simplex-}\ell_1}}{d \lambda}(x) \propto \frac{x^{p-2}(1-x)^{p-2}}{(x^{p-1} + (1-x)^{p-1})^2},$$

where by abuse of notation here λ denotes the Lebesgue measure. Note that this is different from the limiting distribution of the Pareto covering number. And thirdly, when the front is non-convex, and only monotonicity holds, it may not reach the optimal solution (although this is ruled out under the stronger mixture assumption from Mansour et al. (2021)).

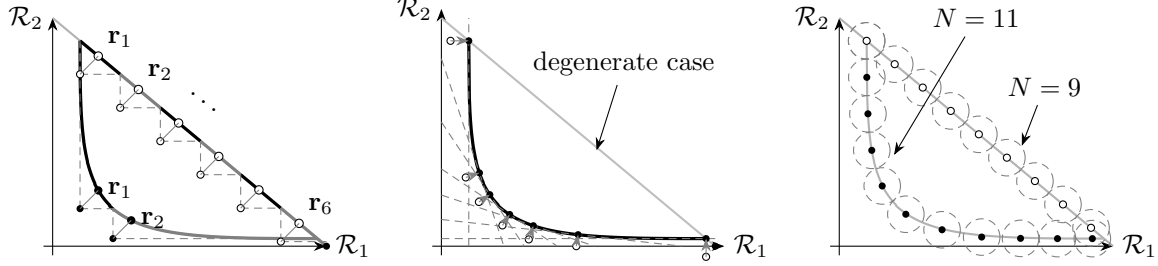


Figure 9. Comparison of different coverings. *Left*: Pareto covering. *Center*: ℓ_1 -simplex covering and weighted sum. *Right*: ℓ_2 -covering.

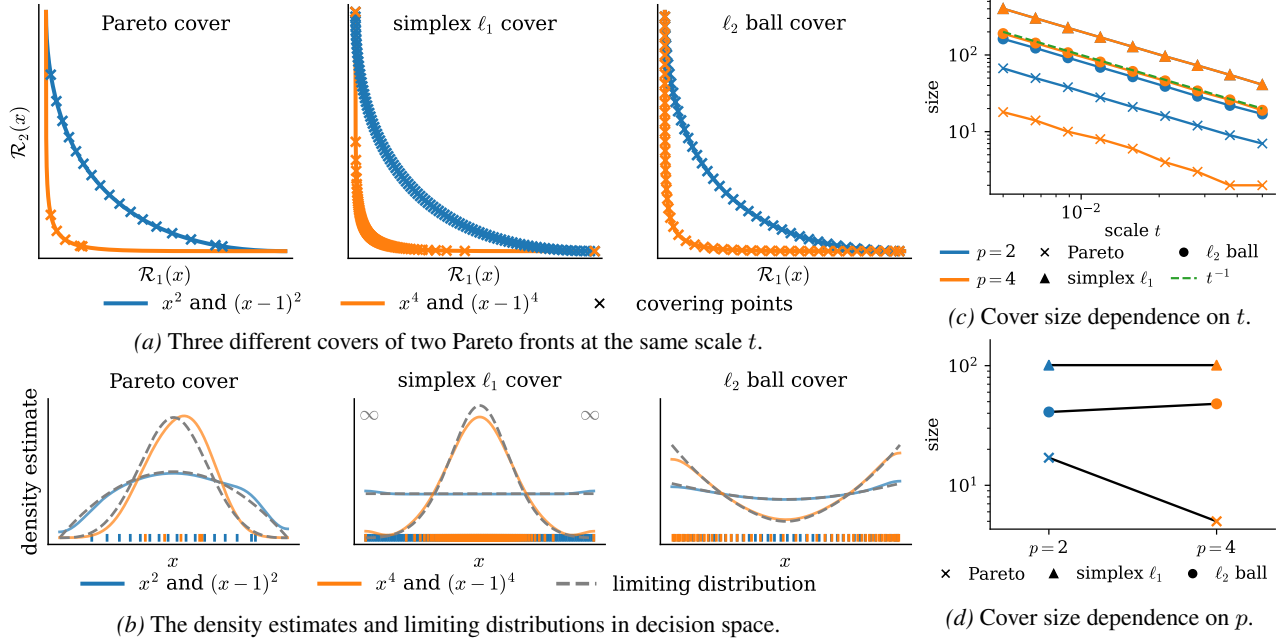


Figure 10. (a) For a fixed scale t and two different Pareto fronts from Example 1, we plot a Pareto covering and two different covers from Section 3.2 and Appendix C.2. (b) We plot a kernel density estimate and the limiting distribution from Theorem 1 (for the other two limiting distributions, see Appendix C.2). (c) As a function of the scale t , the covering numbers are all of order $t^{-(K-1)} = t^{-1}$, but the constant factor depends on the geometry and the covering method. (d) The Pareto covering number decreases with more favorable trade-offs, whereas the simplex ℓ_1 cover is indifferent, and the ℓ_2 -norm cover even increases.

In Fig. 10, we visualize the setting of Example 1 for the values $p = 2$ and $p = 4$. Fig. 10a shows the Pareto fronts with numerically computed minimal Pareto covering as well as the two different coverings from above. Fig. 10b shows that each covering method induces a different distribution over the decision space, and depends differently on the Pareto front geometry. Clearly, the Pareto covering number puts more mass on parts corresponding to the “elbow” of the Pareto fronts. Fig. 10c shows that, unsurprisingly, all coverings scale as $t^{-(K-1)} = t^{-1}$, but there is a constant factor gap between them, with the Pareto covering being much smaller at the same scale. Therefore, the other two coverings can be “unnecessarily large”. Finally, Figs. 10b and 10d together show that only the Pareto covering adapts to the Pareto front geometry as one may expect: the ℓ_2 -norm covering number *increases* as more favorable trade-offs become available ($p = 4$), and even puts more mass on improper regions. At the same time, naturally the simplex ℓ_1 -covering number is completely agnostic to the geometry of the front, and while the induced distributions are more similar to those of the Pareto covering, they still do not avoid improper regions in the case $p = 4$.

C.3. A Worst-case Bound on the Pareto Covering Number

Recall the definition of Pareto covering number from Definition 5. Here we prove a bound that is independent of $\mathbf{R}$ and hence worst-case over all possible Pareto front geometries.

Lemma 9. Suppose that $\mathbf{R} : \mathcal{F} \rightarrow [0, 1]^K$. Then $N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}) \leq \left(1 + \frac{1}{t}\right)^{K-1}$ for all $t \in (0, 1]$.

Proof. It is a well-known fact that we can cover the cube $[0, 1]^K$ in ℓ_∞ -norm by $(1 + 1/t)^K$ points, which would immediately imply the weaker bound $N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}) \leq \left(1 + \frac{1}{t}\right)^K$. The difference here is that we want to show an exponent of $K - 1$ rather than K .

Fix $t \in (0, 1]$ and define the grid

$$G = \left\{t, 2t, \dots, \left\lceil \frac{1}{t} \right\rceil t\right\} \subset [0, 1 + t].$$

Then the grid has cardinality at most $|G| = \left\lceil \frac{1}{t} \right\rceil \leq 1 + \frac{1}{t}$. By replicating this grid on the first $K - 1$ coordinates, we obtain the net G^{K-1} . Now, for each point in the grid $\mathbf{a} = (a_1, \dots, a_{K-1}) \in G^{K-1}$ define the slice

$$S_{\mathbf{a}} = \{\mathbf{r} \in \mathbf{R}(\mathcal{F}) : r_j \leq a_j \forall j \leq K - 1\}.$$

We ignore all $\mathbf{a}$ for which $S_{\mathbf{a}} = \emptyset$. For the rest, we let $\alpha(\mathbf{a}) = \inf\{r_K : \mathbf{r} \in S_{\mathbf{a}}\} \in [0, 1]$. By definition of the infimum, there exists some $\mathbf{r}(\mathbf{a}) \in S_{\mathbf{a}}$ such that in the K th coordinate it holds $r_K(\mathbf{a}) \leq \alpha(\mathbf{a}) + t$. Now pick any $f_{\mathbf{a}} \in \mathcal{F}$ with $\mathbf{R}(f_{\mathbf{a}}) = \mathbf{r}(\mathbf{a})$ (which is possible since $\mathbf{r}(\mathbf{a}) \in \mathbf{R}(\mathcal{F})$) and let $\mathcal{I} \subseteq G^{K-1}$ be the index set of all $\mathbf{a}$ with $S_{\mathbf{a}} \neq \emptyset$. Then

$$|\mathcal{I}| \leq |G|^{K-1} \leq \left(1 + \frac{1}{t}\right)^{K-1}.$$

We claim that $\{f_{\mathbf{a}}\}_{\mathbf{a} \in \mathcal{I}}$ is a t -Pareto cover of $\mathcal{F}$: Take any $f \in \mathcal{F}$ and write $\mathbf{r}' = \mathbf{R}(f) \in [0, 1]^K$. Choose $\mathbf{a} \in G^{K-1}$ by

$$a_j := t \max \left\{1, \left\lceil \frac{r'_j}{t} \right\rceil\right\}, \quad \forall j \in [K - 1].$$

Then it holds that $r'_j \leq a_j \leq r'_j + t$ for all $j \leq K - 1$, and hence $\mathbf{r}' \in S_{\mathbf{a}}$. In particular, we obtain that $S_{\mathbf{a}} \neq \emptyset$, so we know that $\mathbf{a} \in \mathcal{I}$. Since $\mathbf{r}(\mathbf{a}) \in S_{\mathbf{a}}$, we have $r_j(\mathbf{a}) \leq a_j \leq r'_j + t$ for all $j \leq K - 1$. Moreover, because $\mathbf{r}' \in S_{\mathbf{a}}$, we get $\alpha(\mathbf{a}) \leq r'_K$, hence $r_K(\mathbf{a}) \leq \alpha(\mathbf{a}) + t \leq r'_K + t$. Therefore, in all coordinates, $r_j(\mathbf{a}) \leq r'_j + t, j \in [K]$, i.e.,

$$\mathbf{R}(f_{\mathbf{a}}) = \mathbf{r}(\mathbf{a}) \preceq \mathbf{r}' + t\mathbf{1} = \mathbf{R}(f) + t\mathbf{1}.$$

Thus for every f there exists some $f_{\mathbf{a}}, \mathbf{a} \in \mathcal{I}$ with $\mathbf{R}(f_{\mathbf{a}}) \preceq \mathbf{R}(f) + t\mathbf{1}$, proving that $N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}) \leq |\mathcal{I}| \leq (1 + 1/t)^{K-1}$. Note that technically we require the t -Pareto set to only contain Pareto optimal models, which our construction does not guarantee. However, we may simply replace any $f_{\mathbf{a}}$ by a function that is Pareto optimal and dominates $f_{\mathbf{a}}$ using Lemma 1. That concludes the proof. $\square$

D. Separation from LMSA by Mansour et al. (2021) on Mixture Distributions

In this section, we compare the Pareto ERM algorithm (Section 4.1) with the *Limited Target Data Multiple Source Adaptation* (LMSA) algorithm from (Mansour et al., 2021) from a learning perspective. As mentioned in Section 3.2 and Appendix C.2, if monotonicity holds but the target is not a mixture of the sources, it is obvious that the Pareto ERM can outperform LMSA simply because LMSA cannot reach all parts of the Pareto frontier, as Example 2 demonstrates. And even if the target is a mixture (and by Lemma 5 monotonicity holds), the upper bound of order $\sqrt{K \log(n)/n}$ by Mansour et al. (2021) for the LMSA algorithm can be much worse (but not better) than the bound from Corollary 1 for Pareto ERM.

However, this does not yet imply that LMSA actually performs worse than Pareto ERM on mixture distributions; only that our bound is adaptive. At the same time, Fig. 10 shows that the algorithms do actually fundamentally work differently; and in particular, that the covering used by LMSA can be overly redundant in certain parts of the Pareto frontier. We now demonstrate that this indeed can have an effect on the *statistical* rates achieved by LMSA, and there is a true separation.

For simplicity, we show this separation in the case $K = 2$; we leave the interesting but more involved case of $K > 2$ for future work. To that end, let us first introduce the LMSA algorithm formally.

LMSA for $K = 2$. Let $\mathfrak{G}$ be the set of uniform grids on $[0, 1]$ with varying widths $h > 0$:

$$\mathfrak{G} = \{(\theta + h\mathbb{Z}) \cap [0, 1] : 0 < h \leq 1, \theta \in [0, h)\}.$$

For $\Lambda \in \mathfrak{G}$, $\text{LMSA}(\Lambda)$ is defined as

- (i) Compute the set of minimizers for each fixed set of weights

$$\mathcal{F}_\Lambda(\mathbf{R}) := \left\{ f_\lambda : f_\lambda \in \arg \min_{f \in \mathcal{F}} \lambda \mathcal{R}_1(f) + (1 - \lambda) \mathcal{R}_2(f), \lambda \in \Lambda \right\},$$

where we use any fixed tie-breaking rule (the choice does not matter for the following result).

- (ii) Return the empirical target-risk minimizer $\hat{f}_\Lambda \in \arg \min_{f \in \mathcal{F}_\Lambda(\mathbf{R})} \hat{\mathcal{R}}_Q(f)$ over $\mathcal{F}_\Lambda(\mathbf{R})$.

We get the following separation result.

Theorem 5 (Separation from LMSA on mixture distributions). *There are universal constants $c, C > 0$ such that, for all sufficiently large sample sizes $n \in \mathbb{N}$, there exists a class $\mathcal{C}_n$ of instances $(\mathcal{F}, \mathbf{R}, \mathcal{R}_Q)$ with source risks in $[0, 1]^2$ and target losses bounded in $[0, 1]$ satisfying: (i) Each target risk is an exact mixture of the two source risks: for every instance there is a $\lambda^* \in [0, 1]$ such that $\mathcal{R}_Q(f) = \lambda^* \mathcal{R}_1(f) + (1 - \lambda^*) \mathcal{R}_2(f)$ for all $f \in \mathcal{F}$. (ii) Every point of the Pareto front is supported by a weighted sum of source objectives. (iii) Even after optimizing the grid, the $\text{LMSA}(\Lambda)$ estimator is lower bounded by*

$$\inf_{\Lambda \in \mathfrak{G}} \sup_{(\mathcal{F}, \mathbf{R}, \mathcal{R}_Q) \in \mathcal{C}_n} \mathbb{E} \left[\mathcal{E}_Q(\hat{f}_\Lambda, \mathcal{F}) \right] \geq c \sqrt{\frac{\log n}{n}}.$$

- (iv) Uniformly over $\mathcal{C}_n$, Pareto ERM with scale $1/n$ satisfies

$$\sup_{(\mathcal{F}, \mathbf{R}, \mathcal{R}_Q) \in \mathcal{C}_n} \mathbb{E} \left[\mathcal{E}_Q(\hat{f}_{\text{Par}(1/n)}, \mathcal{F}) \right] \leq \frac{C}{\sqrt{n}}.$$

The proof is in the following Appendix D.1. Note that the gap is only of order $\sqrt{\log n}$, which is essentially the largest gap one may hope for in the case $K = 2$. We suspect that a similar construction for $K > 2$ would yield a bigger gap (depending on K), but we leave this to future work.

D.1. Proof of Theorem 5

Set $\gamma := a_0 \sqrt{\log(n)/n}$, with $a_0 > 0$ a sufficiently small universal constant, and set $M := \lfloor 1/\gamma \rfloor \geq 1$. Write $s_\lambda(\mathbf{r}) = \lambda r_1 + (1 - \lambda) r_2$ for linear scalarizations with weight λ on the first objective. The only geometric fact we use is that if two vectors $\mathbf{r}, \mathbf{r}'$ differ as $\mathbf{r}' - \mathbf{r} = \Delta(1, -\theta/(1 - \theta))^\top$ with some $\Delta > 0, \theta \in (0, 1)$ then the scalarizations of $\mathbf{r}, \mathbf{r}'$ are tied at exactly θ , that is, $s_\theta(\mathbf{r}) = s_\theta(\mathbf{r}')$, because for any $\lambda \in [0, 1]$ we have

$$s_\lambda(\mathbf{r}') - s_\lambda(\mathbf{r}) = \lambda \Delta - (1 - \lambda) \frac{\Delta \theta}{1 - \theta} = \Delta \frac{\lambda(1 - \theta) - (1 - \lambda)\theta}{1 - \theta} = \Delta \frac{\lambda - \theta}{1 - \theta}. \quad (9)$$

We make a case distinction depending on the grid width h . The main effort of the lower bound is in the fine case (small h), as we need to carefully construct a Pareto set on which LMSA will use an overly redundant covering (in a statistically meaningful way), while Pareto ERM remains statistically efficient. For coarse grids, we can invoke a simpler approximation error argument.

Case 1: Fine grids. Suppose $\Lambda \in \mathfrak{G}$ has width $h \leq 1/(40M)$. Then Λ contains M points $\lambda_1 > \dots > \lambda_M$ in $(1/5, 1/3)$ because $(1/3 - 1/5)/h = 2/(15h) > \frac{1}{40h}(1 + 1/M) \geq M + 1$. Also, Λ contains at least one point $\lambda^* \in (11/20, 3/5)$, since $(3/5 - 11/20)/h = 1/(20h) = 2/(40h) \geq 2M \geq 2$.

Construction of Pareto front. Let $\mathbf{g} = \mathbf{r}_1 = (1/2, 1/2)$, choose $\theta_L = 2/3, \theta_R = 1/2$, and define

$$\mathbf{r}_0 = \mathbf{g} - \gamma \frac{1 - \theta_L}{\theta_L - \lambda^*} \begin{pmatrix} 1 \\ -\theta_L/(1 - \theta_L) \end{pmatrix}, \quad \mathbf{r}_2 = \mathbf{g} + \gamma \frac{1 - \theta_R}{\lambda^* - \theta_R} \begin{pmatrix} 1 \\ -\theta_R/(1 - \theta_R) \end{pmatrix}. \quad (10)$$

Then $\mathbf{g}$ is selected over $\mathbf{r}_0, \mathbf{r}_2$ by scalarizing with λ^* , because both $\mathbf{r}_0$ and $\mathbf{r}_2$ have scalarized risk $s_{\lambda^*}(\mathbf{g}) + \gamma$:

$$\begin{aligned} s_{\lambda^*}(\mathbf{g}) &= \lambda^* \frac{1}{2} + (1 - \lambda^*) \frac{1}{2} = \frac{1}{2}, \\ s_{\lambda^*}(\mathbf{r}_0) &= \lambda^* \left(\frac{1}{2} - \gamma \frac{1 - \theta_L}{\theta_L - \lambda^*} \right) + (1 - \lambda^*) \left(\frac{1}{2} + \gamma \frac{\theta_L}{\theta_L - \lambda^*} \right) = \frac{1}{2} + \gamma, \\ s_{\lambda^*}(\mathbf{r}_2) &= \lambda^* \left(\frac{1}{2} + \gamma \frac{1 - \theta_R}{\lambda^* - \theta_R} \right) + (1 - \lambda^*) \left(\frac{1}{2} - \gamma \frac{\theta_R}{\lambda^* - \theta_R} \right) = \frac{1}{2} + \gamma. \end{aligned} \quad (11)$$

Now we “attach a bad cluster” of points below $\mathbf{r}_2$. Set $\vartheta_0 = 2/5$, choose $\vartheta_j = (\lambda_{j+1} + \lambda_j)/2$ for $j = 1, \dots, M-1$, and let $\mathbf{b}_0 := \mathbf{r}_2$ and recursively for $j = 1, \dots, M$

$$\mathbf{b}_j = \mathbf{b}_{j-1} + \frac{1}{10nM} \left(-\vartheta_{j-1}/(1 - \vartheta_{j-1}) \right)$$

We visualize the construction in Fig. 11. Importantly, the cluster $\{\mathbf{r}_2, \mathbf{b}_1, \dots, \mathbf{b}_M\}$ lies within a ℓ_∞ -ball of radius much smaller than $1/n$:

$$\begin{aligned} \sum_{j=1}^M |b_{j,1} - b_{j-1,1}| &= \sum_{j=1}^M \frac{1}{10nM} = \frac{1}{10n}, \\ \sum_{j=1}^M |b_{j,2} - b_{j-1,2}| &= \sum_{j=1}^M \frac{1}{10nM} \frac{\vartheta_{j-1}}{1 - \vartheta_{j-1}} \leq \frac{1}{15n} \end{aligned} \quad (12)$$

where we used that $\vartheta_0 = 2/5$ and $\vartheta_j \leq 2/5$ for all j . Since, by definition of $\mathbf{b}_j$ and Eq. (9), $\mathbf{b}_{j-1}$ and $\mathbf{b}_j$ tie for scalarization weight ϑ_{j-1} (that is, $s_{\vartheta_{j-1}}(\mathbf{b}_j) = s_{\vartheta_{j-1}}(\mathbf{b}_{j-1})$) and, for $j < M$, $\mathbf{b}_j$ and $\mathbf{b}_{j+1}$ tie for weight ϑ_j , the point $\mathbf{b}_j$ is selected by minimizing $s_{\lambda_j}(\mathbf{r})$ over $\{\mathbf{r}_0, \mathbf{g}, \mathbf{r}_2, \mathbf{b}_1, \dots, \mathbf{b}_M\}$. Also, by combining Eqs. (11) and (12) and some more calculations we can see that

$$\gamma \leq s_{\lambda^*}(\mathbf{b}_j) - s_{\lambda^*}(\mathbf{g}) \leq \gamma + \frac{1}{n}, \quad j = 1, \dots, M. \quad (13)$$

It is also clear that all points are supported by a weighted sum of source objective: $\mathbf{r}_0$ by sufficiently large weight λ , $\mathbf{g}$ at λ^* , and $\mathbf{b}_1, \dots, \mathbf{b}_M$ each with weights between each tie weights.

Construction of distributions and risks. Let $\mathcal{F} = \{\mathbf{r}_0, \mathbf{g}, \mathbf{r}_2, \mathbf{b}_1, \dots, \mathbf{b}_M\}$ and identify each hypothesis with its vector of source risks, so that $\mathbf{R}(\mathbf{r}) = \mathbf{r}$. We can now realize these risks by actual source distributions. For an observation Z , we define the whole loss vector $(L_{\mathbf{r}})_{\mathbf{r} \in \mathcal{F}} \in [0, 1]^{\mathcal{F}}$ where $L_{\mathbf{r}} = \ell(\mathbf{r}, Z)$. We may choose Z and ℓ so that the distribution of L is as follows: Under source $k \in \{1, 2\}$, $L_{\mathbf{g}}$ is the constant $1/2$, $L_{\mathbf{r}_0}$ and $L_{\mathbf{r}_2}$ are the constants $r_{0,k}$ and $r_{2,k}$, and the coordinates $L_{\mathbf{b}_1}, \dots, L_{\mathbf{b}_M}$ are mutually independent Bernoulli random variables with

$$\mathbb{P}_k(L_{\mathbf{b}_j} = 1) = b_{j,k}, \quad \mathbb{P}_k(L_{\mathbf{b}_j} = 0) = 1 - b_{j,k}.$$

We can then see that then exactly $\mathbf{R}(\mathbf{r}) = \mathbf{r}$ for all $\mathbf{r} \in \mathcal{F}$:

$$\begin{aligned} \mathbf{R}(\mathbf{r}_0) &= \mathbb{E}[L_{\mathbf{r}_0}] = \mathbf{r}_0, \quad \mathbf{R}(\mathbf{g}) = \mathbb{E}[L_{\mathbf{g}}] = \mathbf{g}, \quad \mathbf{R}(\mathbf{r}_2) = \mathbb{E}[L_{\mathbf{r}_2}] = \mathbf{r}_2, \\ \mathbf{R}(\mathbf{b}_j) &= \mathbb{E}[L_{\mathbf{b}_j}] = \left(\frac{\mathbb{P}_1(L_{\mathbf{b}_j} = 1)}{\mathbb{P}_2(L_{\mathbf{b}_j} = 1)} \right) = \mathbf{b}_j, \quad j = 1, \dots, M. \end{aligned}$$

We choose the target distribution is the mixture $Q = \lambda^* P_1 + (1 - \lambda^*) P_2$, equivalently it first draws $T \in \{1, 2\}$ with $\mathbb{P}(T = 1) = \lambda^*$ and then draws the loss vector L from P_T . Hence, for every $\mathbf{r} \in \mathcal{F}$,

$$\mathcal{R}_Q(\mathbf{r}) = \lambda^* r_1 + (1 - \lambda^*) r_2 = s_{\lambda^*}(\mathbf{r}).$$

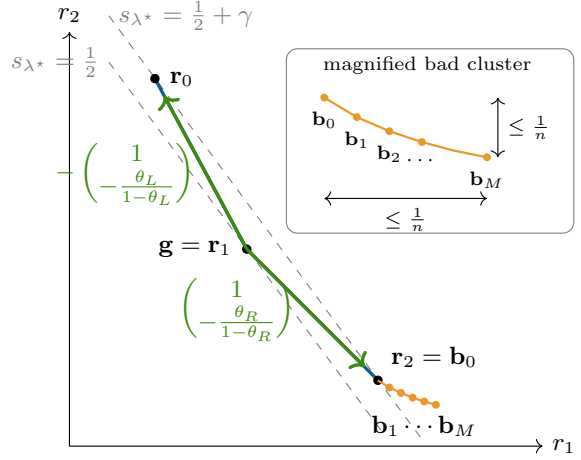


Figure 11. The Pareto front for fine grids.

Lower bound for LMSA. Let $T_1, \dots, T_n$ be the source indices in the target sample and $N_1 := \sum_{i=1}^n \mathbf{1}\{T_i = 1\}$. On the event $E := \{|N_1/n - \lambda^*| \leq \gamma/4\}$, which has probability at least $1/2$ for all large n : By Hoeffding's inequality and $\gamma = a_0 \sqrt{\log(n)/n}$ we have that

$$\mathbb{P}(E^c) \leq 2 \exp\left(-2n \frac{\gamma^2}{16}\right) = 2n^{-a_0^2/8}, \quad (14)$$

so for n large enough, $\mathbb{P}(E) \geq 1/2$. On E write $q := N_1/n$. Conditional on the source indices, $\widehat{\mathcal{R}}_Q(\mathbf{b}_j)$ is the average of independent (but not necessarily identically distributed) Bernoulli variables with parameters in $[1/3, 2/3]$, because all coordinates of the front lie in this interval:

$$\widehat{\mathcal{R}}_Q(\mathbf{b}_j) = \frac{1}{n} \sum_{i=1}^n \underbrace{\mathbf{1}\{T_i = 1\} (L_{\mathbf{b}_j})_1}_{\sim \text{Ber}(b_{j,1}) \text{ conditional on } T_i} + \underbrace{\mathbf{1}\{T_i = 2\} (L_{\mathbf{b}_j})_2}_{\sim \text{Ber}(b_{j,2}) \text{ conditional on } T_i} =: \frac{1}{n} \sum_{i=1}^n X_{j,i}.$$

Hence, conditional on the source indices $T_1, \dots, T_n$, the mean of $\widehat{\mathcal{R}}_Q(\mathbf{b}_j)$ is

$$\mathbb{E} \left[\widehat{\mathcal{R}}_Q(\mathbf{b}_j) \mid T_1, \dots, T_n \right] = qb_{j,1} + (1-q)b_{j,2} = s_{\lambda^*}(\mathbf{b}_j) + (q - \lambda^*)(b_{j,1} - b_{j,2}).$$

Since $\mathbf{b}_j \in [0, 1]^2$ and $|q - \lambda^*| \leq \gamma/4$ on E , $|(q - \lambda^*)(b_{j,1} - b_{j,2})| \leq \frac{\gamma}{4}$. Combining this with $1/2 + \gamma \leq s_{\lambda^*}(\mathbf{b}_j) \leq 1/2 + \gamma + 1/n$ from Eq. (13) gives, for all large n ,

$$\frac{1}{2} + \frac{3\gamma}{4} \leq \mathbb{E} \left[\widehat{\mathcal{R}}_Q(\mathbf{b}_j) \mid T_1, \dots, T_n \right] \leq \frac{1}{2} + \frac{5\gamma}{4} + \frac{1}{n} \leq \frac{1}{2} + \frac{3}{2}\gamma.$$

We now use the celebrated Berry-Esseen theorem (see [Petrov \(1975\)](#) for different versions).

Lemma 10 (Theorem 3 in Section 5.2 of [Petrov \(1975\)](#)). *There exists a positive universal constant $C > 0$ such that the following is true. Suppose $X_1, \dots, X_n$ are independent (but not necessarily identically distributed) real-valued random variables with means μ_i , variance σ_i^2 , and third moment $\rho_i = \mathbb{E}[|X_i - \mu_i|^3] < \infty$. Then for the standard normal c.d.f. Φ it holds*

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\sum_{i=1}^n [X_i - \mu_i]}{\sqrt{\sum_{i=1}^n \sigma_i^2}} \leq x \right) - \Phi(x) \right| \leq C \frac{\sum_{i=1}^n \rho_i}{(\sum_{i=1}^n \sigma_i^2)^{3/2}}.$$

We can apply this to our setting. Specifically, let $S_j := n\widehat{\mathcal{R}}_Q(\mathbf{b}_j) = \sum_{i=1}^n X_{j,i}$, $\mu_{j,i} = \mathbb{E}[X_{j,i} \mid T_1, \dots, T_n]$, $m_j := \mathbb{E}[S_j \mid T_1, \dots, T_n] = \sum_{i=1}^n \mu_{j,i}$, $v_j^2 := \text{Var}(S_j \mid T_1, \dots, T_n) = \sum_{i=1}^n \text{Var}(X_{j,i} \mid T_1, \dots, T_n)$, $\rho_{j,i} = \mathbb{E}[|X_{j,i} - \mu_{j,i}|^3 \mid T_1, \dots, T_n]$ and $a_j := m_j - n/2$. Then, on E ,

$$\frac{3n\gamma}{4} \leq a_j \leq \frac{3n\gamma}{2} \quad \text{and} \quad \frac{2n}{9} \leq v_j^2 \leq \frac{n}{4} \quad \text{and} \quad \sum_{i=1}^n \rho_{j,i} \leq n.$$

The variance bounds use $p(1-p) \in [2/9, 1/4]$ for $p \in [1/3, 2/3]$. We obtain that

$$\frac{\sum_{i=1}^n \rho_{j,i}}{v_j^3} \leq \frac{n}{(2n/9)^{3/2}} \leq \frac{10}{\sqrt{n}}. \quad (15)$$

Let $x_j := a_j/v_j$ and notice $0 < x_j \leq \frac{9}{2\sqrt{2}}\sqrt{n}\gamma$. and Lemma 10 yields

$$\begin{aligned} \mathbb{P} \left(S_j \leq \frac{n}{2} - 1 \mid T_1, \dots, T_n \right) &= \mathbb{P} \left(\sum_{i=1}^n [X_{j,i} - \mu_{j,i}] \leq \frac{n}{2} - m_j - 1 \mid T_1, \dots, T_n \right) \\ &= \mathbb{P} \left(\frac{\sum_{i=1}^n [X_{j,i} - \mu_{j,i}]}{v_j} \leq \frac{\frac{n}{2} - m_j}{v_j} - \frac{1}{v_j} \mid T_1, \dots, T_n \right) \\ &= \mathbb{P} \left(\frac{\sum_{i=1}^n [X_{j,i} - \mu_{j,i}]}{v_j} \leq -x_j - \frac{1}{v_j} \mid T_1, \dots, T_n \right) \\ &\geq \Phi \left(-x_j - \frac{1}{v_j} \right) - \frac{10}{\sqrt{n}}. \end{aligned} \quad (\text{Lemma 10 and (15)})$$

As shown in Gasull and Utzet (2014, bottom of page 15), Mills' ratio can be approximated to yield the lower bound $\Phi(-y) \geq \kappa(1+y)^{-1} \exp(-y^2/2)$ (for $\kappa = 1/\sqrt{2\pi}$) for every $y > 0$. Since $x_j \leq \frac{9}{2\sqrt{2}}\sqrt{n}\gamma$ for all large n and $1/v_j \leq \sqrt{9/(2n)}$, we know that for large enough n we have $x_j + 1/v_j \leq 5\sqrt{n}\gamma$. The last display then yields the bound

$$\begin{aligned} \mathbb{P}\left(\widehat{\mathcal{R}}_Q(\mathbf{b}_j) < \frac{1}{2} \mid T_1, \dots, T_n\right) &\geq \frac{c}{1 + \sqrt{n}\gamma} \exp(-13n\gamma^2) - \frac{10}{\sqrt{n}} && (\text{for a small enough } c > 0) \\ &= \frac{c}{1 + a_0\sqrt{\log n}} \exp(-13a_0^2 \log n) - \frac{10}{\sqrt{n}} && (\sqrt{n}\gamma = a_0\sqrt{\log n}) \\ &\geq \frac{c}{1 + a_0\sqrt{\log n}} n^{-13a_0^2} - \frac{10}{\sqrt{n}} \\ &\geq c' \sqrt{\frac{\log n}{n}} && (\text{for } 13a_0^2 < 1/2 \text{ and large enough } n) \\ &= \frac{c'}{a_0} \gamma \geq c_0 \frac{1}{M}. && (\text{by definition } M^{-1} \leq 2\gamma) \end{aligned}$$

Thus, after adjusting constants appropriately, uniformly over all source indices in E ,

$$\mathbb{P}\left(\widehat{\mathcal{R}}_Q(\mathbf{b}_j) < \frac{1}{2} \mid T_1, \dots, T_n\right) \geq \frac{c_0}{M}.$$

Conditional on the source indices, the events $\{\widehat{\mathcal{R}}_Q(\mathbf{b}_j) < 1/2\}$ are independent across j . Hence

$$\begin{aligned} \mathbb{P}\left(\min_{1 \leq j \leq M} \widehat{\mathcal{R}}_Q(\mathbf{b}_j) < \frac{1}{2}\right) &\geq \mathbb{E}\left[\mathbf{1}_E \mathbb{P}\left(\bigcup_{j=1}^M \left\{\widehat{\mathcal{R}}_Q(\mathbf{b}_j) < \frac{1}{2}\right\} \mid T_1, \dots, T_n\right)\right] \\ &= \mathbb{P}(E) \mathbb{E}\left[\left(1 - \prod_{j=1}^M \left(1 - \mathbb{P}\left(\widehat{\mathcal{R}}_Q(\mathbf{b}_j) < \frac{1}{2} \mid T_1, \dots, T_n\right)\right)\right) \mid E\right] \\ &\geq \frac{1}{2} \left(1 - \left(1 - \frac{c_0}{M}\right)^M\right) \geq c_1. && (\text{by (14) and the lower bound above}) \end{aligned}$$

On this event, empirical minimization over $\mathcal{F}_\Lambda(\mathbf{R})$ selects a bad vertex because $\mathbf{g} \in \mathcal{F}_\Lambda(\mathbf{R})$ has empirical risk exactly $1/2$. Hence, by Eq. (13) and because all other vertices have excess risk at least γ , we have

$$\mathbb{E}_Q\left[\mathcal{E}_Q(\widehat{f}_\Lambda, \mathcal{F})\right] \geq \mathbb{P}\left(\min_{1 \leq j \leq M} \widehat{\mathcal{R}}_Q(\mathbf{b}_j) < \frac{1}{2}\right) \gamma = c_1 a_0 \sqrt{\frac{\log n}{n}}.$$

Upper Bound for Pareto ERM. It is easy to see that any minimal $1/n$ -Pareto cover must contain $\mathbf{g}$ and must be of size 3: Since the bad cluster $\{\mathbf{r}_2 = \mathbf{b}_0, \mathbf{b}_1, \dots, \mathbf{b}_M\}$ has coordinate diameter at most $1/n$ (Eq. (12)), choosing any element will cover the cluster in Pareto distance. Moreover, no other point can cover $\mathbf{r}_0$ or $\mathbf{g}$ at scale $1/n$ since in each coordinate the distance is at least $\gamma \gg 1/n$, so $\mathbf{r}_0$ and $\mathbf{g}$ must be contained in any minimal Pareto covering. For instance, the set $\{\mathbf{r}_0, \mathbf{g}, \mathbf{r}_2\}$ is a minimal $1/n$ -Pareto cover.

Empirical target-risk minimization over $\{\mathbf{r}_0, \mathbf{g}, \mathbf{r}_2\}$ has finite-class estimation cost at most $\sqrt{2 \log(6)/n}$ for losses in $[0, 1]$. In particular, by Theorem 2 (since $\bar{\omega}_Q(1/n) \leq 1/n$ by Lemmas 4 and 5), we get for all $u > 0$

$$\mathbb{P}\left(\mathcal{E}_Q(\widehat{f}_{\text{Par}(1/n)}, \mathcal{F}) > \frac{1}{n} + u\right) \leq 6 \exp\left(-\frac{nu^2}{2}\right).$$

By simple tail integration we obtain that

$$\mathbb{E}_Q\left[\mathcal{E}_Q(\widehat{f}_{\text{Par}(1/n)}, \mathcal{F})\right] \leq \frac{1}{n} + \int_0^\infty 6 \exp(-nu^2/2) \, du \leq \frac{4}{\sqrt{n}}.$$

Case 2: Coarse grids. Now suppose the grid width is $h > 1/(40M)$. There is an interval $[\theta_R, \theta_L] \subset [11/20, 3/5]$ with $[\theta_R, \theta_L] \cap \Lambda = \emptyset$ of length at least $\min\{h/2, 1/40\}$, since $3/5 - 11/20 = 1/20$. Set the parameter $\delta = (\theta_L - \theta_R)/10$. Let λ^* be its midpoint, set $\mathbf{g} = (1/2, 1/2)$, and define $\mathbf{r}_0, \mathbf{r}_2$ by (10) but with these values of θ_L, θ_R and γ replaced by δ . Then $\mathbf{g}$ is the target minimizer and both neighbors have target excess δ , by the same calculation as Eq. (11):

$$s_{\lambda^*}(\mathbf{r}_0) = s_{\lambda^*}(\mathbf{r}_2) = s_{\lambda^*}(\mathbf{g}) + \delta.$$

In this case we do not add the bad cluster to the Pareto frontier.

Lower bound for LMSA. Since Λ does not intersect $[\theta_R, \theta_L]$, the good vector $\mathbf{g}$ is not in $\mathcal{F}_\Lambda(\mathbf{R})$. Every selected candidate is $\mathbf{r}_0$ or $\mathbf{r}_2$. We can, again, realize the sources by deterministic loss coordinates $L_{\mathbf{r}} = r_k$ under source k , and let the target again draw source 1 with probability λ^* and source 2 otherwise. Then $\mathcal{R}_Q(\mathbf{r}) = s_{\lambda^*}(\mathbf{r})$, and deterministically,

$$\mathcal{E}_Q(\hat{f}_\Lambda, \mathcal{F}) \geq \delta \geq \frac{1}{10} \min\left\{\frac{h}{2}, \frac{1}{40}\right\} \geq \frac{1}{800M} \geq \frac{1}{800}\gamma = \frac{a_0}{800} \sqrt{\frac{\log n}{n}}.$$

Upper bound for Pareto ERM. The coarse instances already have only three vectors so the bound follows immediately from the same calculation as above.

Take $\mathcal{C}_n^{\text{mix}}$ to be the union over fine and coarse instances constructed above. Then, combining the fine and coarse cases show that

$$\inf_{\Lambda \in \mathfrak{G}} \sup_{(\mathcal{F}, \mathbf{R}, \mathcal{R}_Q) \in \mathcal{C}_n^{\text{mix}}} \mathbb{E}_Q \left[\mathcal{E}_Q(\hat{f}_\Lambda, \mathcal{F}) \right] \geq c \sqrt{\frac{\log n}{n}} \quad \text{while} \quad \sup_{(\mathcal{F}, \mathbf{R}, \mathcal{R}_Q) \in \mathcal{C}_n^{\text{mix}}} \mathbb{E}_Q \left[\mathcal{E}_Q(\hat{f}_{\text{Par}(1/n)}, \mathcal{F}) \right] \leq \frac{C}{\sqrt{n}}.$$

That concludes the proof.

E. Additional Learning Results

Here we provide an additional result for Section 4 on transfer learning when the loss is strongly convex in Appendix E.1, examples and a general bound for monotonicity above a threshold in Appendix E.2, and a learning bound for model selection aggregation with general loss functions and examples for Theorem 4 in Appendix E.3.

E.1. Fast Rates in Transfer Learning

While for general loss functions running ERM on the Pareto covering is sufficient, for strongly convex losses we would like to achieve fast rates instead. In this section, we assume that the loss $\ell(\hat{y}, y)$ is κ -strongly convex in $\hat{y}$ and L -Lipschitz on $[-1, 1]$. Luckily, we can achieve those fast rates by running an aggregation procedure on a t -Pareto set. In particular, we consider the algorithm that for some fixed $t > 0$

- (i) builds a minimal t -Pareto set $\{f_1, \dots, f_N\}$ and
- (ii) computes the *star estimator* (Audibert, 2007; 2009) on the covering, that is,

$$\begin{aligned} \hat{f}_{\text{star}(t)} &\in \arg \min_{f \in \text{star}(\hat{f}_{\text{Par}(t)}, \{f_k\}_{k=1}^N)} \hat{\mathcal{R}}_Q(f) \\ \text{where } \text{star}(\hat{f}_{\text{Par}(t)}, \{f_k\}_{k=1}^N) &= \left\{ \alpha \hat{f}_{\text{Par}(t)} + (1 - \alpha) f_i : i \in [N], \alpha \in [0, 1] \right\}, \end{aligned}$$

and recall that $\hat{f}_{\text{Par}(t)}$ is the Pareto ERM on the same t -Pareto set.

Proposition 2. *There exists a constant $C > 0$, so that for every $t > 0$, with probability at least $1 - \delta$, the Pareto star-aggregator $\hat{f}_{\text{star}(t)}$ achieves*

$$\mathcal{R}_Q(\hat{f}_{\text{star}(t)}) - \inf_{f \in \mathcal{F}} \mathcal{R}_Q(f) \leq \bar{\omega}_Q(t) + \frac{CL^2 \log(N_{\text{Par}}(t, \mathcal{F}, \mathbf{R})/\delta)}{\kappa n}.$$

Proof. Let $N = N_{\text{Par}}(t, \mathcal{F}, \mathbf{R})$ and $\{f_1, \dots, f_N\}$ be the minimal t -Pareto set in $\mathcal{F}$ on which we compute the Pareto star aggregator $\hat{f}_{\text{star}(t)}$. By definition, for every $f \in \mathcal{F}$ there is a f_j with $\mathcal{R}_k(f_j) \leq \mathcal{R}_k(f) + t$ for all k . By Definition 4, we know that hence $\mathcal{R}_Q(f_j) \leq \mathcal{R}_Q(f) + \bar{\omega}_Q(t)$.

Let $f^* \in \arg \min_{f \in \mathcal{F}} \mathcal{R}_Q(f)$. By applying the main result of Kanade et al. (2024) to the star estimator (see their Appendix A.1), there exists a universal constant $C > 0$ so that with probability at least $1 - \delta$

$$\mathcal{R}_Q(\hat{f}_{\text{star}(t)}) - \min_{j \in [N]} \mathcal{R}_Q(f_j) \leq \frac{CL^2 \log(N/\delta)}{\kappa n} =: r.$$

Hence, the same error decomposition as in the proof of Theorem 2 yields that

$$\mathcal{R}_Q(\hat{f}_{\text{star}(t)}) - \mathcal{R}_Q(f^*) = \underbrace{\mathcal{R}_Q(\hat{f}_{\text{star}(t)}) - \min_{j \in [N]} \mathcal{R}_Q(f_j)}_{\leq r} + \underbrace{\min_{j \in [N]} \mathcal{R}_Q(f_j) - \mathcal{R}_Q(f^*)}_{\leq \bar{\omega}_Q(t)},$$

and the bound of Proposition 2 follows. $\square$

E.2. More on the Margin ERM

Recall that in Section 4.1, we derived upper bounds that are consistent whenever $\bar{\omega}_Q(t) \rightarrow 0$ as $t \rightarrow 0$ and hence monotonicity holds. However, the latter need not hold; instead the upper and lower moduli may stay bounded away from zero near the origin. Recall from Section 4.2, that we defined $\underline{\gamma}_Q := \inf \{t \in [0, 1] : \underline{\omega}_Q(t) \geq 0\}$. We now demonstrate this in an example where the lower modulus and the threshold $\underline{\gamma}_Q$ have a closed form solution. We then prove a general bound on the excess risk of the margin ERM when $\gamma > \underline{\gamma}_Q$.

Example 6 (Monotonicity above a threshold). Consider the hypothesis space $\mathcal{F} = \{f_\theta \equiv \theta \in [0, 1]^2 : \theta_1 + \theta_2 \geq 1\}$, the source risks $\mathcal{R}_1(f_\theta) = \theta_1, \mathcal{R}_2(f_\theta) = \theta_2$ (e.g., expected absolute loss) and, for $\gamma_0 \in (0, 1/2)$, the target risk $\mathcal{R}_Q(f_\theta) = |\theta_1 + \theta_2 - (1 + \gamma_0)|$. Then a calculation shows that for all $t \geq 0$ (cf. Fig. 12)

$$\underline{\omega}_Q(t) = \begin{cases} -\gamma_0 & t \leq \gamma_0/2, \\ 2(t - \gamma_0) & \gamma_0/2 < t \leq 1/2, \\ 1 & t > 1/2. \end{cases}$$

so that $\underline{\gamma}_Q = \gamma_0$ and the pruned sets are $\mathcal{F}_\gamma = \{\theta : 1 \leq \theta_1 + \theta_2 \leq 1 + 2\gamma\}$.

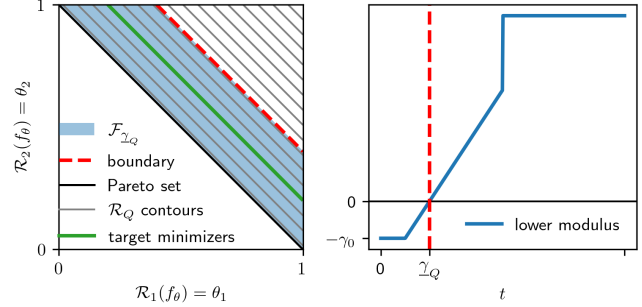


Figure 12. Left: the objective space of Example 6 with the modulus-pruned set $\mathcal{F}_{\gamma_0}$, the Pareto front and contour lines of the target risk. Right: the lower modulus as a function of t in this example.

In Example 6, monotonicity is not satisfied because $\underline{\omega}_Q(0) < 0$ and hence also $\bar{\omega}_Q(t)$ stays bounded away from zero. See Fig. 12 for a visualization, and Fig. 3 for the moduli of HELM and VHELM.

Notice that in Example 6 the threshold satisfies $\underline{\gamma}_Q = \gamma_0 < \infty$. Hence, any functions f, f' for which $\mathbf{R}(f) \ll \mathbf{R}(f') - \gamma_0 \mathbf{1}$ satisfy $\mathcal{R}_Q(f) \leq \mathcal{R}_Q(f')$, and hence we may prune f' from the candidate set, resulting in $\mathcal{F}_{\underline{\gamma}_Q}$, visualized in Fig. 12. Of course, in Example 6, if we knew $\underline{\gamma}_Q$, we could improve upon this approach; here it just serves as an easy example.

Also recall from Section 4.2 that for $\gamma > 0$ the pruned hypothesis space is defined as

$$\mathcal{F}_\gamma = \{f \in \mathcal{F} : \text{there exists no } f' \in \mathcal{F} \text{ such that } \mathbf{R}(f') \ll \mathbf{R}(f) - \gamma \mathbf{1}\},$$

and the margin ERM $\hat{f}_\gamma$ is defined by running ERM on $\mathcal{F}_\gamma$; that is, $\hat{f}_\gamma \in \arg \min_{f \in \mathcal{F}_\gamma} \hat{\mathcal{R}}_Q(f)$. We obtain the following result, which is also used in the proof of the parameter-free strategy Proposition 1.

Proposition 3. For any $\gamma > \underline{\gamma}_Q$, with probability at least $1 - \delta$, the margin ERM $\hat{f}_\gamma$ achieves

$$\mathcal{R}_Q(\hat{f}_\gamma) - \inf_{f \in \mathcal{F}} \mathcal{R}_Q(f) \leq 4\mathcal{R}_n(\ell \circ \mathcal{F}_\gamma) + 2\sqrt{\frac{2\log(4/\delta)}{n}},$$

where $\mathcal{R}_n(\ell \circ \mathcal{F}_\gamma)$ denotes the standard Rademacher complexity of the pruned hypothesis class composed with the loss

$$\mathcal{R}_n(\ell \circ \mathcal{F}) = \mathbb{E}_{z_1^n \sim Q^{\otimes n}} \mathbb{E}_\sigma \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i \ell(f, z_i) \right|$$

with i.i.d. Rademacher random variables $\sigma = (\sigma_1, \dots, \sigma_n)$.

Proof. Let $f^* \in \arg \min_{f \in \mathcal{F}} \mathcal{R}_Q(f)$. Then

$$\mathcal{R}_Q(\hat{f}_\gamma) - \mathcal{R}_Q(f^*) = \mathcal{R}_Q(\hat{f}_\gamma) - \inf_{f \in \mathcal{F}_\gamma} \mathcal{R}_Q(f) + \inf_{f \in \mathcal{F}_\gamma} \mathcal{R}_Q(f) - \mathcal{R}_Q(f^*).$$

Recalling that $\hat{f}_\gamma$ is the empirical risk minimizer on $\mathcal{F}_\gamma$, standard ERM analysis for bounded losses (Shalev-Shwartz and Ben-David, 2014, Theorem 26.5) shows that with probability at least $1 - \delta$

$$\mathcal{R}_Q(\hat{f}_\gamma) - \inf_{f \in \mathcal{F}_\gamma} \mathcal{R}_Q(f) \leq 4\mathcal{B}_n(\ell \circ \mathcal{F}_\gamma) + 2\sqrt{\frac{2 \log(4/\delta)}{n}}.$$

We argue now that $\inf_{f \in \mathcal{F}_\gamma} \mathcal{R}_Q(f) - \mathcal{R}_Q(f^*) = 0$, in other words, that a minimizer of the target risk lies in $\mathcal{F}_\gamma$. First, recall that

$$\gamma_Q := \inf \{t \geq 0 : \omega_Q(t) \geq 0\},$$

and $t \mapsto \omega_Q(t)$ is a non-decreasing function. Hence, for $\gamma > \gamma_Q$, we know that any functions f, f' with $\mathbf{R}(f) \preceq \mathbf{R}(f') - \gamma \mathbf{1}$ satisfy $\mathcal{R}_Q(f) \leq \mathcal{R}_Q(f')$ by Definition 4 of the lower modulus. Recalling that $\mathcal{F}_\gamma$ is defined as

$$\mathcal{F}_\gamma = \{f \in \mathcal{F} : \text{there exists no } f' \in \mathcal{F} \text{ such that } \mathbf{R}(f') \ll \mathbf{R}(f) - \gamma \mathbf{1}\}$$

we can consider two cases: if $f^* \in \mathcal{F}_\gamma$, then we are done. If f^* is not in $\mathcal{F}_\gamma$, then there is another function $f \in \mathcal{F}$ so that $\mathbf{R}(f) \ll \mathbf{R}(f^*) - \gamma \mathbf{1}$, and hence $\mathcal{R}_Q(f) \leq \mathcal{R}_Q(f^*)$, so f must also be a minimizer of $\mathcal{R}_Q$, call that minimizer $f_2^* = f$. By induction, we obtain a chain f_i^* of minimizers, each satisfying $\mathbf{R}(f_i^*) \ll \mathbf{R}(f_{i-1}^*) - \gamma \mathbf{1}$. Since $\gamma > \gamma_Q \geq 0$, and $\mathbf{R}$ is bounded from below, this chain cannot be infinite, which means there exists an i so that $f_i^* \in \mathcal{F}_\gamma$, yielding the claim. $\square$

E.3. Model Selection Aggregation

In this section, we further discuss the model selection aggregation problem from Section 5 and provide some additional results and examples.

First, note that under monotonicity and a convex loss, it holds by definition for any estimator $\hat{\rho}$ that $\mathcal{R}_k(\psi_{\hat{\rho}}) \leq \mathcal{R}_k(f_{\hat{\rho}}) - \Psi(\hat{\rho})$, which then implies the following gap between the risks of $\psi_{\hat{\rho}}$ and $f_{\hat{\rho}}$:

$$\mathcal{R}_Q(\psi_{\hat{\rho}}) \leq \mathcal{R}_Q(f_{\hat{\rho}}) - \omega_Q(\Psi(\hat{\rho})) \leq \mathbb{E}_{k \sim \hat{\rho}} \mathcal{R}_Q(f_k) - \omega_Q(\Psi(\hat{\rho})). \quad (16)$$

The first inequality holds by definition of ω_Q (Definition 4), and the second is Jensen's inequality. If $\mathcal{F}$ contains the convex hull $\text{conv}(f_1, \dots, f_K)$, it holds that $\omega_Q(\Psi(\rho)) \geq 0$ for all ρ under monotonicity.

E.3.1. TILTED EXPONENTIAL WEIGHTS

In this section, we discuss a *tilted exponential weights* estimator, defined for $\lambda, J \geq 0, \pi \in \mathcal{P}([K])$ as

$$\hat{\rho}_{\text{tilt}} := \arg \min_{\rho \in \mathcal{P}([K])} \left\{ \mathbb{E}_{k \sim \rho} \hat{\mathcal{R}}_Q(f_k) + \frac{\text{KL}(\rho, \pi)}{\lambda} - J\Psi(\rho) \right\},$$

where we recall the definition of Ψ from Eq. (4). Note here that for $J = 0$ we recover the regular exponential weights estimator (Alquier, 2024). We call it a *tilted* exponential weights posterior, because the function Ψ tilts the objective towards the center of the simplex, depending on where the gap between the Pareto front and the convex hull, as measured by Ψ , is largest.

To state the bound for this estimator, we use the following *maximal additive gain* $G_J(Q)$, defined as

$$G_J(Q) := \sup_{\rho \in \mathcal{P}([K])} \left\{ J\Psi(\rho) - \left(\mathbb{E}_{k \sim \rho} \mathcal{R}_Q(f_k) - \min_{k \in [K]} \mathcal{R}_Q(f_k) \right) \right\}.$$

Under monotonicity (MON), $G_J(Q) \in [0, J]$: $G_J(Q) \geq 0$ follows from choosing ρ as a point mass on the best function in the dictionary and noting that then $\Psi(\rho) = 0$, and $G_J(Q) = 0$ can be attained, e.g., when one of the dictionary elements is optimal in $\mathcal{F}$, that is, $f_k \in \arg \min_{f \in \mathcal{F}} \mathcal{R}_Q(f)$. On the other hand $G_J(Q) \leq J$ follows from boundedness of the loss.

Proposition 4. Let the loss $\ell(f, z)$ be bounded in $[0, 1]$ and convex in f , and assume that $\underline{\omega}_Q(t) \geq Jt$ for all $t \geq 0$, where J is known to the algorithm, in particular implying (MON). Choose the prior π as uniform and $\lambda = \sqrt{8n \log(2K/\delta)}$. Then it holds with probability at least $1 - \delta$ that

$$\mathcal{R}_Q(\psi_{\hat{\rho}_{\text{tilt}}}) - \min_{k \in [K]} \mathcal{R}_Q(f_k) \leq \sqrt{\frac{2 \log(2K/\delta)}{n}} - G_J(Q).$$

The first term is the minimax rate $\sqrt{\log(K)/n}$ from the standard aggregation setting with convex loss. However, the bound can be negative for large $G_J(Q)$ (respectively, large n), because we are in an improper setting. But it's clear that when one of the functions in the dictionary is optimal in all of $\mathcal{F}$, then $G_J(Q)$ will be zero and there is no gain.

Proof. From the assumption that $\underline{\omega}_Q(t) \geq Jt$ and Eq. (16), we know that $\mathcal{R}_Q(\psi_\rho) \leq \mathbb{E}_{k \sim \rho} \mathcal{R}_Q(f_k) - J\Psi(\rho)$. Combining this with Theorem 2.1 in Alquier (2024) yields that

$$\mathbb{P}\left(\forall \rho \in \mathcal{P}([K]) : \mathcal{R}_Q(\psi_\rho) \leq \mathbb{E}_{k \sim \rho} \widehat{\mathcal{R}}_Q(f_k) + \frac{\lambda}{8n} + \frac{\text{KL}(\rho, \pi) + \log(1/\delta)}{\lambda} - J\Psi(\rho)\right) \geq 1 - \delta.$$

Minimizing the bound over ρ , we get the tilted posterior; that is,

$$\mathbb{P}\left(\mathcal{R}_Q(\psi_{\hat{\rho}_{\text{tilt}}}) \leq \inf_{\rho \in \mathcal{P}([K])} \left\{ \mathbb{E}_{k \sim \rho} \widehat{\mathcal{R}}_Q(f_k) + \frac{\lambda}{8n} + \frac{\text{KL}(\rho, \pi) + \log(1/\delta)}{\lambda} - J\Psi(\rho) \right\}\right) \geq 1 - \delta.$$

To turn this into an oracle inequality, we can use the same trick as Alquier (2024, Theorem 4.2), that is, plugging in their Equation (4.2) and using a union bound yields that

$$\mathbb{P}\left(\mathcal{R}_Q(\psi_{\hat{\rho}_{\text{tilt}}}) \leq \inf_{\rho \in \mathcal{P}([K])} \left\{ \mathbb{E}_{k \sim \rho} \mathcal{R}_Q(f_k) + \frac{\lambda}{4n} + 2 \frac{\text{KL}(\rho, \pi) + \log(2/\delta)}{\lambda} - J\Psi(\rho) \right\}\right) \geq 1 - \delta.$$

Now note that for a uniform prior π on $[K]$ we have that $\text{KL}(\rho, \pi) = \sum_{k=1}^K \rho_k \log(K\rho_k) \leq \log K$ for all $\rho \in \Delta^{K-1}$. Plugging this and the choice of λ into the previous bound, we get that with probability $1 - \delta$,

$$\mathcal{R}_Q(\psi_{\hat{\rho}_{\text{tilt}}}) \leq \inf_{\rho \in \mathcal{P}([K])} \left\{ \mathbb{E}_{k \sim \rho} \mathcal{R}_Q(f_k) + \sqrt{\frac{2 \log(2K/\delta)}{n}} - J\Psi(\rho) \right\}.$$

Subtracting $\min_{k \in [K]} \mathcal{R}_Q(f_k)$ on both sides and plugging in the definition of $G_J(Q)$ yields the result. $\square$

E.3.2. EXAMPLES FOR FAST RATES

We describe two examples of where Assumption 2 holds: in the first one we may use $(\Phi, \phi) = (\Psi, \psi)$ from Eq. (4), whereas in the latter we have to construct it separately.

In the first example, any estimator whose range lies in the convex hull will have excess risk lower bounded by $1/\sqrt{n}$ in the worst case, while our estimator achieves the fast rate of order $1/n$. The same construction also serves as the basis for the construction that appears in the proof of the lower bound in Theorem 6, essentially by stacking multiple copies of this problem when $K > 2$.

Example 7. We begin by constructing a tuple $(\mathbf{R}, \mathcal{F}, \{\mathcal{R}_Q : Q \in \mathcal{Q}\})$ describing the problem; we then verify the assumptions of Theorem 4. Let $K = 2$ and $\mathcal{X} = \{x_1, x_2\}$. We define two distributions P_1, P_2 : Under P_1 let $X = x_1$ and $Y = 0$ almost surely, whereas under P_2 we let $X = x_2$ and $Y = 1$ almost surely. Using absolute loss, we then find that $\mathcal{R}_1(f) = \mathbb{E}_{(X,Y) \sim P_1} |f(X) - Y| = f(x_1)$ and $\mathcal{R}_2(f) = \mathbb{E}_{(X,Y) \sim P_2} |f(X) - Y| = 1 - f(x_2)$ for any function $f : \mathcal{X} \rightarrow [0, 1]$. Take the dictionary $\{f_1 \equiv 0, f_2 \equiv 1\}$ so that the convex combination always outputs $f_\rho = \rho_1 f_1 + \rho_2 f_2 = \rho_2$ and the benchmark values of the convex combination are $\mathbf{R}(f_\rho) = (\rho_2, \rho_1)$. Now let the hypothesis space be

$$\mathcal{F} = \text{conv}(\{f_1, f_2\}) \cup \{g_u : u \in [0, 1]\}$$

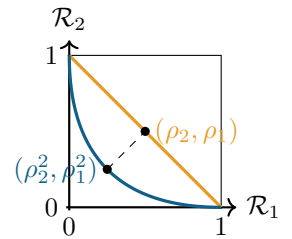


Figure 13. Construction of Example 7.

where we define the functions g_u as $g_u(x_1) = u^2$ and $g_u(x_2) = 2u - u^2$, so that $\mathbf{R}(g_u) = (u^2, (1 - u)^2)$; see Fig. 13. Let the target risk be $\mathcal{R}_Q = q_1 \mathcal{R}_1 + q_2 \mathcal{R}_2$ for $\mathbf{q} \in \Delta^1$ which we may obtain by also using absolute loss and target distribution $Q \in \mathcal{Q} = \{q_1 P_1 + q_2 P_2 : \mathbf{q} \in \Delta^1\}$. Then, on the convex hull of the dictionary, we have that $\mathcal{R}_Q(f_\rho) = q_1 \rho_2 + q_2 \rho_1$.

Due to using absolute loss, standard aggregation methods are not guaranteed to achieve a fast rate in this setting (and in fact, Theorem 6 uses the same idea of this setting to prove this formally). Our estimator, on the other hand, achieves a fast rate.

To that end, we now verify the assumptions of Theorem 4. We begin with Assumption 2 by showing that Ψ from Eq. (4) itself is strongly concave and hence we may choose $(\Phi, \phi) = (\Psi, \psi)$. Let us compute the map $\rho \mapsto \psi_\rho$ parameterizing the Pareto set. It is not hard to see that the minimizer ψ_ρ is given by $\psi_\rho = g_{\rho_2}$ and $\Psi(\rho) = \rho_1 \rho_2$. Indeed, we have that $\mathcal{R}_1(g_u) - \mathcal{R}_1(f_\rho) = u^2 - \rho_2$ and $\mathcal{R}_2(g_u) - \mathcal{R}_2(f_\rho) = (1 - u)^2 - \rho_1$. And for $u = \rho_2$, by the fact that $\rho_1 + \rho_2 = 1$, these two terms are balanced and equal to $\rho_2^2 - \rho_2 = -\rho_1 \rho_2$. Hence, we also obtain that $\mathbf{R}(\psi_\rho) = (\rho_2^2, \rho_1^2)$. A calculation now verifies that $\Psi(\rho) = \rho_1 \rho_2$ is 2-strongly concave in $\|f_\rho - f_\gamma\|_2^2 = (\rho_2 - \gamma_2)^2$ and $\mathbf{R}(\psi_\rho) = \mathbf{R}(f_\rho) - \Psi(\rho)\mathbf{1}$, and so the pair (ψ, Ψ) satisfies Assumption 2. Further, we can verify the other assumptions of Theorem 4: first, notice that in this setting we have $\bar{w}_Q(t) \leq t$ by Lemma 4, and the absolute loss is $L_\ell = 1$ -Lipschitz on $[0, 1]$. Second, since $|\psi_\rho(x_1) - \psi_\gamma(x_1)| = |g_{\rho_2}(x_1) - g_{\gamma_2}(x_1)| = |\rho_2^2 - \gamma_2^2| \leq 2|\rho_2 - \gamma_2| = 2|f_\rho(x_1) - f_\gamma(x_1)|$, and analogously for x_2 we have $|\psi_\rho(x_2) - \psi_\gamma(x_2)| \leq 2|f_\rho(x_2) - f_\gamma(x_2)|$, the last requirement holds with $L_\phi = 2$.

Consequently, this is a setting in which we may apply Theorem 4. That is, our estimator $\phi_{\hat{\rho}_{\text{Par}}} = \psi_{\hat{\rho}_{\text{Par}}}$ achieves excess risk of order at most $1/n$ for all distributions $Q \in \mathcal{Q} = \{q_1 P_1 + q_2 P_2 : \mathbf{q} \in \Delta^1\}$: For some constant $c > 0$ and with probability at least $1 - \delta$, we have

$$\mathcal{E}_Q(\phi_{\hat{\rho}_{\text{Par}}}, \{f_1, f_2\}) \leq \frac{c \log(1/\delta)}{n}.$$

The next example is in the setting of linear regression with absolute loss. We consider the case of $K = d = 2$ for the example to be simpler to follow; it can be extended to any dimension d and number of source tasks K . As opposed to Example 7, now the map Ψ itself is not strongly concave. We visualize Example 8 in Fig. 14.

Example 8. Again, we begin by constructing the problem setting. Let $K = d = 2$ and consider the hypothesis space $\mathcal{F} = \{\mathbf{x} \mapsto \langle \mathbf{x}, \mathbf{w} \rangle : \mathbf{w} \in B_2^2\}$. Let P_1, P_2 be two joint distributions of the random vectors $X \in B_2^d$ and $Y \in [-1, 1]$, defined via $X = \Sigma_k^{1/2} U$, where $U \sim \text{Uniform}(\mathbb{S}^{d-1})$ and $Y = \langle X, \mathbf{w}_k \rangle$ with ground truths $\mathbf{w}_1 = \mathbf{e}_1, \mathbf{w}_2 = -\mathbf{e}_1$ and shape matrices

$$\Sigma_1 = \frac{\pi^2}{16(1+\sigma)} \begin{pmatrix} 1 & \sigma \\ \sigma & 1 \end{pmatrix} \quad \text{and} \quad \Sigma_2 = \frac{\pi^2}{16(1+\sigma)} \begin{pmatrix} 1 & -\sigma \\ -\sigma & 1 \end{pmatrix}$$

where $\sigma \in [0, 1)$. Note that the largest eigenvalues of both shape matrices Σ_k are bounded by 1 so that $|Y| = |\langle X, \mathbf{w}_k \rangle| \leq 1$ almost surely. Further, we consider the risk $\mathcal{R}_k(\mathbf{w}) = \mathbb{E}_{(X,Y) \sim P_k} |\langle X, \mathbf{w} \rangle - Y|$ and notice that $\mathcal{R}_k(\mathbf{w}) = \frac{2}{\pi} \sqrt{(\mathbf{w} - \mathbf{w}_k)^\top \Sigma_k (\mathbf{w} - \mathbf{w}_k)}$. Therefore, the two benchmark risks are given by

$$\mathcal{R}_1(\mathbf{w}) = c_\sigma \sqrt{(w_1 - 1)^2 + 2\sigma(w_1 - 1)w_2 + w_2^2}, \quad \mathcal{R}_2(\mathbf{w}) = c_\sigma \sqrt{(w_1 + 1)^2 - 2\sigma(w_1 + 1)w_2 + w_2^2},$$

where $c_\sigma = \frac{1}{2\sqrt{1+\sigma}}$. The scaling of Σ_k ensures that $\mathcal{R}_k$ maps into $[0, 1]$. We can now write $\mathbf{R}(\mathbf{w}) = (\mathcal{R}_1(\mathbf{w}), \mathcal{R}_2(\mathbf{w}))$. Define for $\rho \in \Delta^1$ the convex combination $\mathbf{w}_\rho = \rho_1 \mathbf{w}_1 + \rho_2 \mathbf{w}_2$ and notice that $\mathbf{R}(\mathbf{w}_\rho) = c_\sigma (1 - (\rho_1 - \rho_2), 1 + (\rho_1 - \rho_2))$. Finally, define the family of distributions $Q_{\mathbf{q}} = q_1 P_1 + q_2 P_2$ for $\mathbf{q} \in \Delta^1$. A calculation shows that then

$$\mathcal{R}_{\mathbf{q}}(\mathbf{w}) \equiv \mathcal{R}_{Q_{\mathbf{q}}}(\mathbf{w}) := \mathbb{E}_{(X,Y) \sim Q_{\mathbf{q}}} |\langle X, \mathbf{w} \rangle - Y| = q_1 \mathcal{R}_1(\mathbf{w}) + q_2 \mathcal{R}_2(\mathbf{w}).$$

Now, given this setting, in the following Lemma 11 we verify that Assumption 2 is satisfied, and show that weights $\mathbf{v}_\rho \in B_2^2$ exist that satisfy $\mathbf{R}(\mathbf{v}_\rho) = \mathbf{R}(\mathbf{w}_\rho) - \Phi(\rho)\mathbf{1}$.

Lemma 11. *The function $\Phi : \Delta^1 \rightarrow [0, \infty)$ defined as $\Phi(\rho) = \eta \frac{\pi^2}{8(1+\sigma)} \rho_1 \rho_2$ with the constant*

$$\eta = \frac{8\sqrt{1+\sigma}}{\pi^2} \left(1 - \sqrt{\frac{1-\sigma^2}{1+3\sigma^2}}\right) \left(1 + \sqrt{\frac{1-\sigma^2}{1+3\sigma^2}}\right)^{-1}$$

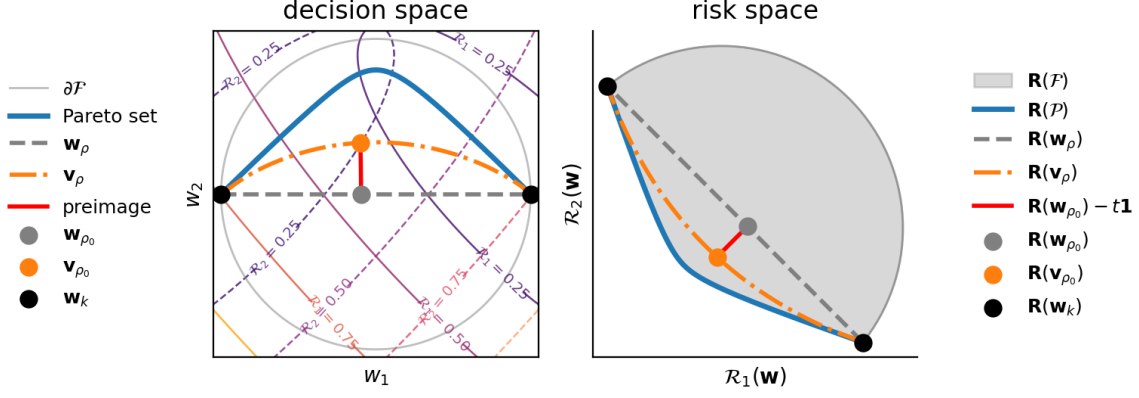


Figure 14. The two-dimensional Example 8 and Lemma 11 visualized. On the left the decision space with contour lines of the two benchmarks, and on the right the risk space.

satisfies Assumption 2: (i) $\Phi(\mathbf{e}_k) = 0$ for $k \in \{1, 2\}$, (ii) $\Phi(\rho)$ is η -strongly concave w.r.t. $\|\langle \cdot, \mathbf{w}_\rho - \mathbf{w}_\gamma \rangle\|_{L^2(Q)}$, that is, for all $\rho, \gamma \in \Delta^1$, $\alpha \in [0, 1]$,

$$\Phi(\alpha\rho + (1-\alpha)\gamma) \geq \alpha\Phi(\rho) + (1-\alpha)\Phi(\gamma) + \frac{\eta}{2}\alpha(1-\alpha)\frac{\pi^2}{4(1+\sigma)}(\rho_1 - \gamma_1)^2,$$

and (iii) the weights $\mathbf{v}_\rho \in B_2^2$ that satisfy the equation $\mathbf{R}(\mathbf{v}_\rho) = \mathbf{R}(\mathbf{w}_\rho) - \Phi(\rho)\mathbf{1}$ exist and are given by

$$\mathbf{v}_\rho = \left(\frac{(\rho_1 - \rho_2)(1 - \Phi(\rho)/c_\sigma)}{1 - \sigma v_2}, v_2 \right)$$

where v_2 is given through the polynomial equation $P(v_2) = 0$ and P is given in Eq. (17).

Note that $\eta > 0$ iff $\sigma > 0$, which is exactly the case when the Pareto set and the convex hull do not coincide and the set $\{\mathbf{v}_\rho : \rho \in \Delta^1\}$ “bends away” from the convex hull. This is visualized in Fig. 14, where we also show that in this case, we do not output models in the Pareto set, because the Pareto front does not exhibit sufficient curvature.¹⁰

Proof of Lemma 11. First, denote the risk minimizer by $\mathbf{w}_\mathbf{q} = \arg \min_{\mathbf{w} \in B_2^2} q_1 \mathcal{R}_1(\mathbf{w}) + q_2 \mathcal{R}_2(\mathbf{w})$ (assume for now that $\mathbf{w}_\mathbf{q}$ is in the interior of B_2^2). Moreover, we can write the $L^2(Q)$ -norm of $\langle \cdot, \mathbf{w}_\rho - \mathbf{w}_\gamma \rangle$, because $\mathbf{w}_\rho - \mathbf{w}_\gamma = (2(\rho_1 - \gamma_1), 0)$, as $\|\langle \cdot, \mathbf{w}_\rho - \mathbf{w}_\gamma \rangle\|_{L^2(Q)}^2 = \sum_{k=1}^K q_k \|\mathbf{w}_\rho - \mathbf{w}_\gamma\|_{\Sigma_k}^2 = \frac{\pi^2}{4(1+\sigma)}(\rho_1 - \gamma_1)^2$ independently of $\mathbf{q}$.

The proof of the first point then is obvious. The strong concavity of Φ holds because the function $g(p) := \Phi((p, 1-p)) = \eta \frac{\pi^2}{8(1+\sigma)} p(1-p)$ has second derivative $g''(p) = -\frac{\pi^2}{4(1+\sigma)}\eta$ and is therefore $\frac{\pi^2}{4(1+\sigma)}\eta$ -strongly concave in squared distance on $[0, 1]$. Hence Φ is η -strongly concave in $\frac{\pi^2}{4(1+\sigma)}(\rho_1 - \gamma_1)^2$.

Finally, to show that $\mathbf{v}_\rho$ exists, fix some ρ and denote $a = \rho_1 - \rho_2$, $t = \Phi(\rho)$. The condition that $\mathbf{R}(\mathbf{v}) = \mathbf{R}(\mathbf{w}_\rho) - \Phi(\rho)\mathbf{1}$ is equivalent to the system of two equations

$$\begin{aligned} c_\sigma \sqrt{(v_1 - 1)^2 + 2\sigma(v_1 - 1)v_2 + v_2^2} &= c_\sigma(1 - a) - t \\ c_\sigma \sqrt{(v_1 + 1)^2 - 2\sigma(v_1 + 1)v_2 + v_2^2} &= c_\sigma(1 + a) - t \end{aligned}$$

Subtracting one from the other yields $v_1(1 - \sigma v_2) = a(1 - t/c_\sigma)$. For $v_2 \in [-1, 1]$ and $\sigma \in [0, 1]$ we know $1 - \sigma v_2 > 0$ and so

$$v_1 = \frac{a(1 - t/c_\sigma)}{1 - \sigma v_2}.$$

¹⁰While Assumption 2 is satisfied, the Lipschitz condition of Theorem 4 is not, so we cannot apply it black box. However, it is not hard to see that with some additional work, fast rates can also be achieved here.

Substituting that back into the first equality from our system of equations yields

$$P(v_2) = \sigma^3 v_2^4 - (\sigma^4 + 3\sigma^2)v_2^3 + (3\sigma^3 + 3\sigma)v_2^2 - (3\sigma^2 + 1)v_2 + \sigma(1 - a^2(1 - t/c_\sigma)^2) = 0. \quad (17)$$

A sign test with $v_2 = -1$ and $v_2 = 1$ shows that this quartic has a solution on $(-1, 1)$, and with some additional but elementary work, it can be shown that there exists a solution that satisfies $\|\mathbf{v}\|_2 \leq 1$. $\square$

E.3.3. A SEPARATION FROM AGGREGATION IN THE CONVEX HULL

Finally, we verify that Theorem 4 can genuinely yield fast rates where standard aggregation methods (in the convex hull) would fail to achieve fast rates. We prove Theorem 6 in Appendix G.7.

Theorem 6. *Assume $K \geq 4$ and $\log(K)/n \leq 1$. There exists a problem setting $(\mathcal{F}, \mathbf{R}, \{\mathcal{R}_Q : Q \in \mathcal{Q}\})$ where $\mathcal{Q}$ is a family of distributions such that (i) any estimator $f_{\hat{\rho}}$ in the convex hull has worst-case risk over $Q \in \mathcal{Q}$ lower bounded by*

$$\sup_{Q \in \mathcal{Q}} \mathbb{E} \left[\mathcal{E}_Q(f_{\hat{\rho}}, \{f_k\}_{k=1}^K) \right] \geq \frac{1}{38} \sqrt{\frac{\log K}{n}},$$

and (ii) the assumptions of Theorem 4 are satisfied with (ϕ, Φ) chosen as (ψ, Ψ) from Eq. (4) and the constants $L_\ell = 1, L_\phi = 2, \eta = 2, \kappa = 0$, and $\lambda = 1/22$ for all $Q \in \mathcal{Q}$. Hence, our estimator achieves

$$\sup_{Q \in \mathcal{Q}} \mathbb{E} \left[\mathcal{E}_Q(\phi_{\hat{\rho}_{\text{Par}}}, \{f_k\}_{k=1}^K) \right] \leq 22 \frac{\log K}{n}.$$

The proof idea is based on Example 7, and constructs approximately $\log_2 K$ copies of the problem.

F. Proof of Theorem 1

Throughout this section, let $\mathfrak{R}$ be the Pareto front associated with $\text{Par}(\mathcal{F}, \mathbf{R})$ that is *nice* (Assumption 1). Theorem 1 computes the mass of certain subsets $A \subset \mathfrak{R}$ with respect to the limiting distribution of the Pareto covering as its granularity goes to zero. In particular, these subsets A are assumed to be *Jordan measurable*, which means that their boundaries have zero measure (see Definition 11 below). We begin with a few remarks on these assumptions.

Niceness ensures that the Pareto front is a smooth manifold; the smoothness structure on the boundary of $\mathfrak{R}$ is handled in the usual way by smoothly extending the manifold within $\mathbb{R}^K$ (Lee, 2003). It also lower bounds each component of the normal vector $\mathbf{n}(\mathbf{r})$ away from zero, where $n_k(\mathbf{r}) > \eta > 0$ for all $k \in [K]$ and $\mathbf{r} \in \mathfrak{R}$. To interpret this assumption, suppose that there were an interior point $\mathbf{r}$ in $\mathfrak{R}$ whose normal vector $\mathbf{n}(\mathbf{r})$ achieves zero on some component $n_k(\mathbf{r}) = 0$, as exemplified in Fig. 16a. Such a point is an *improper* trade-off, which intuitively means that it is almost dominated: it is possible to improve on some other objective at the cost of only an infinitesimal amount of utility in the k th objective (Geoffrion, 1968). In this sense, the lower bound condition ensures that points in $\mathfrak{R}$ are not ‘barely’ Pareto optimal.

Jordan measurability is a standard assumption that helps us avoid pathological situations, such as when the subset $A \subset \mathfrak{R}$ is a space-filling curve or a countable dense set in $\mathfrak{R}$. In these cases, the Hausdorff measure of A is zero, while its covering numbers at all scales coincide with that of $\mathfrak{R}$; the connection between the Hausdorff measure and limiting distribution necessarily breaks down for such irregular sets.

As a high-level roadmap, the first section Appendix F.1 provides some technical machinery for working with smooth Pareto fronts through approximations via linear Pareto fronts. In linear fronts, Pareto balls correspond to simplexes, so the next section Appendix F.2 works out the asymptotics of simplex coverings. Finally, Appendix F.3 proves Theorem 1 by passing through a limit of linear approximations of the nice Pareto front. See Fig. 15 for an overview. All technical proofs are provided afterwards in Appendix F.4.



(a) A point excluded by the normal vector condition.

(b) The linear approximation in Lemma 12.

Figure 16. (a): An example of a point that is excluded by the normal vector condition, since the second coordinate $n_2(\mathbf{r}) = 0$ vanishes. (b): A visualization of the local linear approximation of $\mathfrak{R}$ at $\mathbf{r}_0$ constructed in Lemma 12. The thick curved line is a smooth Pareto front $\mathfrak{R}$. It is approximated by a tangent space $T_{\mathbf{r}_0}\mathfrak{R}$, which has normal direction $\mathbf{n}(\mathbf{r}_0)$. The lines connecting $\mathfrak{R}$ with its tangent space represent the local approximation $\phi : U \rightarrow V$, where $U \subset \mathfrak{R}$ and $V \subset T_{\mathbf{r}_0}\mathfrak{R}$.

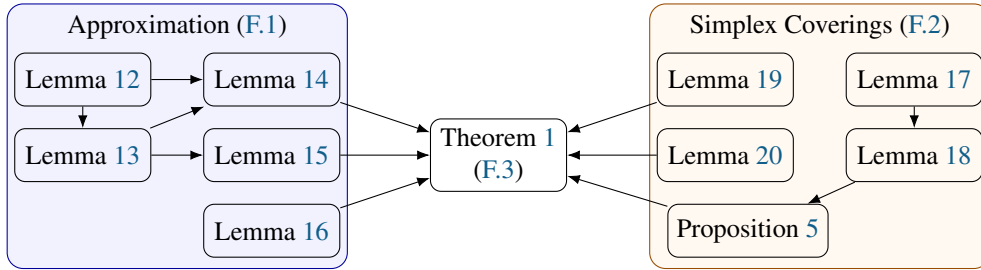


Figure 15. Logical proof structure of Theorem 1.

F.1. Approximations of Nice Pareto Fronts

Before we prove Theorem 1, we first provide five basic results about nice Pareto fronts. The proofs of these results are fairly standard: they use of Taylor’s theorem and compactness. We further defer them to Appendix F.4.

These results are about approximating the Pareto front linearly. First, we show that we can approximate the Pareto front *locally* at one point with its tangent space (Lemma 12). We then use this to establish that the Pareto distance and the Euclidean distance are equivalent for any given Pareto front (Lemma 13). These two results together let us prove the main approximation, Lemma 14, that establishes a *global* approximation consisting of a finite number of maps mapping into local tangent spaces. Importantly, these approximations distort the Pareto quasi-metric between the Pareto front and tangent spaces arbitrarily little. Moreover, Lemmas 12 and 13 let us prove a first *coarse* bound on the Pareto covering number (Lemma 15). Finally, we prove a basic result about covering number under distortions (Lemma 16), which is independent of the previous steps but in the main proof will be applied to the approximation maps (which incur little distortion). This approach generally allows us to prove results for linear fronts and extend them to smooth fronts.

We begin with Lemma 12 which shows that a smooth Pareto front $\mathfrak{R}$ can locally be approximated as a manifold by a linear subspace, its tangent space, to arbitrary accuracy. The proof of Lemma 12 is in Appendix F.4.1.

Lemma 12 (Local linear approximation of smooth Pareto fronts). *Let $\mathfrak{R} \subset \mathbb{R}^K$ be a smooth Pareto front. Fix any $\mathbf{r}_0 \in \mathfrak{R}$ and define the tangent space at $\mathbf{r}_0$ as the hyperplane:*

$$T_{\mathbf{r}_0}\mathfrak{R} := \{\mathbf{r}_0 + \mathbf{v} \in \mathbb{R}^K : \mathbf{v} \cdot \mathbf{n}(\mathbf{r}_0) = 0\}.$$

For any $\varepsilon > 0$, there are subsets U relatively open in $\mathfrak{R}$ and V relatively open in $T_{\mathbf{r}_0}\mathfrak{R}$, and a diffeomorphism $\phi : U \rightarrow V$ such that for any 1-Lipschitz-smooth function $f : \mathbb{R}^K \rightarrow \mathbb{R}$, the following holds for all $\mathbf{r}, \mathbf{r}' \in U$:

$$|(f(\mathbf{r}) - f(\mathbf{r}')) - (f(\phi(\mathbf{r})) - f(\phi(\mathbf{r}')))| \leq \varepsilon \|\mathbf{r} - \mathbf{r}'\|_2. \quad (18)$$

See Fig. 16b for this construction. For asymptotic or non-quantitative results, the following equivalence between Pareto and Euclidean distances is useful and follows from Lemma 12: Lemma 13 shows that the Pareto distance is bi-Lipschitz with the standard Euclidean distance on $\mathfrak{R}$. In particular, these two distances induce the same topology on the Pareto front.

Lemma 13 (Comparability of Pareto and Euclidean distances). *Let d be the Pareto distance on a nice Pareto front $\mathfrak{R}$. There is a constant $c_{\mathfrak{R}} > 0$ depending on $\mathfrak{R}$ such that*

$$c_{\mathfrak{R}} \cdot \|\mathbf{r} - \mathbf{r}'\|_2 \leq d(\mathbf{r}, \mathbf{r}') \leq \|\mathbf{r} - \mathbf{r}'\|_2, \quad \forall \mathbf{r}, \mathbf{r}' \in \mathfrak{R}. \quad (19)$$

The proof of Lemma 13 is in Appendix F.4.2. Combining Lemmas 12 and 13 we can get the main approximation result Lemma 14. It provides a piecewise linear approximation to the Pareto front. This approximation distorts distances, but it turns out that by increasing the number of pieces, we can incur arbitrarily little distortion. By *distortion*, we mean:

Definition 6 (ε -distortion). Let (U, d_U) and (V, d_V) be quasi-metric spaces. A bijection $\phi : U \rightarrow V$ incurs ε -distortion if

$$(1 - \varepsilon) \cdot d_V(\phi(\mathbf{r}), \phi(\mathbf{r}')) \leq d_U(\mathbf{r}, \mathbf{r}') \leq (1 + \varepsilon) \cdot d_V(\phi(\mathbf{r}), \phi(\mathbf{r}')), \quad \forall \mathbf{r}, \mathbf{r}' \in U.$$

When $\varepsilon = 0$, we say that ϕ is an *isometry*.

In particular, Lemma 14 shows that a nice Pareto front $(\mathfrak{R}, d)$ can be partitioned into finitely many pieces, where each piece is arbitrarily well-approximated as a *quasi-metric space* by a corresponding linear front.

Lemma 14 (Piecewise linear approximation of smooth Pareto fronts). *Let $\mathfrak{R} \subset \mathbb{R}^K$ be a nice Pareto front and let d be the Pareto distance. For any $\varepsilon > 0$, there exist a finite collection of disjoint open sets $U_1, \dots, U_M \subset \mathfrak{R}$ where the diameter of each set U_m is at most ε and the following hold:*

- For each $m \in [M]$, there is a diffeomorphism $\phi_m : (U_m, d) \rightarrow (V_m, d)$ that achieves ε -distortion, where $V_m \subset \mathbb{R}^K$ is contained in some hyperplane, i.e.

$$V_m \subset \{\mathbf{v} \in \mathbb{R}^K : \mathbf{n}_m \cdot \mathbf{v} = 0\},$$

and each coordinate of $\mathbf{n}_m$ is positive. Moreover, the determinant of the Jacobian is bounded $1 - \varepsilon < |J\phi_m| < 1 + \varepsilon$ on all of U_m .

- The remainder region $Z = \mathfrak{R} \setminus \bigcup U_m$ is contained in the finite union of the boundaries of a set of balls:

$$Z \subset \bigcup_{m \in [M]} \partial B_m,$$

where each B_m is a ball in $\mathfrak{R}$ with respect to the Euclidean distance, and is $\mathcal{H}^{K-1}$ -measure zero.

The proof of Lemma 14 is in Appendix F.4.3.

As both a useful and illuminating step, we now show that the comparability of the Pareto and Euclidean distances in Lemma 13 also lets us obtain a coarse upper and lower bound on the Pareto covering number with respect to the usual Euclidean covering number; Lemma 15 shows that the Pareto and Euclidean covering numbers in $\mathfrak{R}$ are comparable.

Lemma 15 (Coarse bound on Pareto covering numbers). *Let $(\mathfrak{R}, d)$ be a nice Pareto front and its Pareto distance. Further, denote by $N_2(t, A)$ the t -covering number of A with respect to the Euclidean distance. There is some $C_{\mathfrak{R}} > 0$ depending on $\mathfrak{R}$ so that for all $A \subset \mathfrak{R}$, the Pareto and Euclidean covering numbers equivalent up to multiplicative constants:*

$$C_{\mathfrak{R}} \cdot N_2(t, A) \leq N_{\text{Par}}(t, A) \leq N_2(t, A).$$

Moreover, let $U \subset \mathfrak{R}$ be a ball in $\mathfrak{R}$ with respect to the Euclidean distance and let ∂U be its boundary. Then, the growth rate of the t -Pareto covering number of ∂U is strictly dominated by t^{K-1} :

$$\lim_{t \rightarrow 0} N_{\text{Par}}(t, \partial U) \cdot t^{K-1} = 0.$$

The proof of Lemma 15 is in Appendix F.4.4. This result follows almost immediately from Lemma 13. The second part is a corollary of the first; we can bound the Pareto covering number of low-dimensional sets in $\mathfrak{R}$. This is easy to do since asymptotically, the Pareto and Euclidean covering numbers qualitatively behave the same.

To relate the covering numbers on the Pareto front to those on the linear approximation, we need to quantify how much an ε -distortion affects the covering numbers. The last lemma of this technical preparation allows us to control exactly that; we prove it in Appendix F.4.5. Importantly, recall that the Pareto distance is a quasi-metric on the Pareto manifold by Lemma 3.

Lemma 16 (Covering numbers under distortions). *Let $\phi : (U, d_U) \rightarrow (V, d_V)$ be a map between quasi-metric spaces incurring ε -distortion. For each $A \subset U$, let $N_U(t, A)$ denote the t -covering number of A , and let $N_V(t, \phi(A))$ be the t -covering number of $\phi(A)$ under their respective quasi-metrics. Then:*

$$N_V(t/(1 - \varepsilon), \phi(A)) \leq N_U(t, A) \leq N_V(t/(1 + \varepsilon), \phi(A)).$$

These results let us reduce covering of the smooth Pareto front to covering of its piecewise linear approximation. In the next section we hence focus on how to cover linear fronts.

F.2. Simplex Coverings

In this section, we bound the covering number for linear Pareto fronts. It turns out that Pareto balls in linear fronts are simplices (Fig. 17 and Eq. (20) below), so this problem reduces to one of computing *simplex covering numbers*. We focus on covering subsets $A \subset \mathbb{R}^\kappa$ of a κ -dimensional linear space with simplices. In the context of Theorem 1, $\kappa = K - 1$ is the dimension of the linear front.

Definition 7 (Simplex). A subset $\Omega \subset \mathbb{R}^\kappa$ is a κ -dimensional *simplex* if it is the convex hull of $\kappa + 1$ points in general linear position. That is, there exist $\mathbf{w}_0, \dots, \mathbf{w}_\kappa \in \mathbb{R}^\kappa$ such that any collection of κ vectors is linearly independent, and

$$\Omega = \text{conv}(\mathbf{w}_0, \dots, \mathbf{w}_\kappa).$$

Throughout, let Vol denote the κ -dimensional Lebesgue measure. And whenever $A, B \subset \mathbb{R}^\kappa$ are subsets, $\mathbf{v} \in \mathbb{R}^\kappa$ is a vector, and $t > 0$ is a scalar, we use the notation: $A + B = \{\mathbf{a} + \mathbf{b} : \mathbf{a} \in A \text{ and } \mathbf{b} \in B\}$, $A + \mathbf{v} = \{\mathbf{a} + \mathbf{v} : \mathbf{a} \in A\}$ and $tA = \{t\mathbf{a} : \mathbf{a} \in A\}$.

In particular, it is not hard to see that Pareto balls in nice linear Pareto fronts are always simplices: Take the linear front $V = \{\mathbf{v} \in \mathbb{R}^K : \mathbf{v} \cdot \mathbf{n} = 0\}$ where $n_k > 0$ for all k . Then the Pareto ball centered at zero can be written as

$$\begin{aligned} \{\mathbf{v} \in V : d(\mathbf{0}, \mathbf{v}) \leq t\} &= t \{\mathbf{v} \in V : v_k \geq -1 \text{ for all } k \in [K]\} \\ &= t \text{conv}(\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(K)}) \quad \text{with} \quad w_k^{(j)} = \begin{cases} -1 & j \neq k, \\ \frac{1 \cdot \mathbf{n}}{n_k} - 1 & j = k. \end{cases} \end{aligned} \quad (20)$$

And hence it coincides with a simplex, which we denote $\Omega_{\mathbf{n}}$. We plot two views of such simplices in Fig. 18.

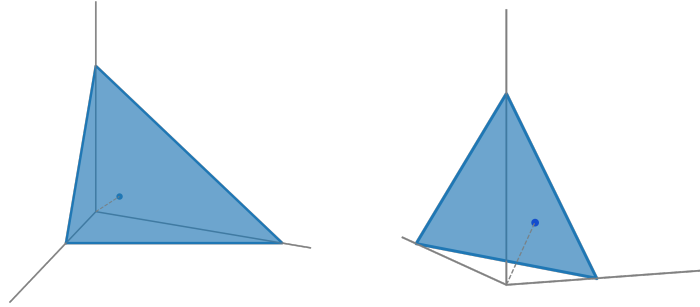


Figure 18. Two different views of a Pareto ball in a linear Pareto front. This is a higher-dimensional analog of Figure 2, see also Fig. 17.

A standard result in geometric measure theory shows that if $A \subset \mathbb{R}^\kappa$ is a closed κ -dimensional rectifiable set, then its Minkowski content, defined as $\mathcal{M}(A) = \limsup_{t \rightarrow 0} \text{Vol}(A + tB)$ where B is the unit ℓ_∞ -ball, coincides with the Hausdorff measure (Federer, 1996, Theorem 3.2.39). This tells us that the asymptotic covering number is governed by the Minkowski content. This section first provides an analogous result, Proposition 5, but for the setting where the covering is done with general compact sets $\mathbb{R}^\kappa$, before specializing to κ -dimensional simplices.

A covering limit via periodic translative covering densities. We now define efficient or ‘economical’ periodic coverings of $\mathbb{R}^\kappa$ by a convex set Ω , which we can think of as a repetitive tiling of $\mathbb{R}^\kappa$ by many translated copies of Ω . We are interested in the *periodic translative covering density* $\theta(\Omega)$, which measures on average how many copies of Ω cover a random point in space. If Ω can tessellate the space, then the optimal density is one. But usually, these translations of Ω will need to overlap in order to cover the space. When Ω is a convex body, the value $\theta(\Omega)$ exists and is upper bounded by $O(\kappa \ln \kappa)$, as established in Rogers (1957).

Before we define these periodic coverings, we recall the standard notion of a lattice (capturing the periodicity of the covering). The following is based off of the reference Micciancio and Goldwasser (2002).

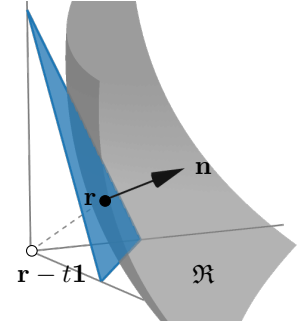


Figure 17. The Pareto ball in the tangent space of a Pareto front is a simplex.

Definition 8 (Lattice, Micciancio and Goldwasser (2002)). A (full-rank) lattice $\Lambda \subset \mathbb{R}^\kappa$ is a set

$$\Lambda = \{z_1 \mathbf{v}_1 + \dots + z_\kappa \mathbf{v}_\kappa : z_1, \dots, z_\kappa \in \mathbb{Z}\},$$

where $V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_\kappa] \in \mathbb{R}^{\kappa \times \kappa}$ form a linearly independent set called a *basis* of Λ . The *fundamental parallelepiped* spanned by the basis V is defined as the set

$$P(V) = \{\lambda_1 \mathbf{v}_1 + \dots + \lambda_\kappa \mathbf{v}_\kappa : \lambda_1, \dots, \lambda_\kappa \in [0, 1)\}.$$

The *determinant* of the lattice is the κ -dimensional volume of $P(V)$, given by $\det \Lambda = |\det V|$.

It is well-established that the determinant of any fixed lattice is independent of the choice of basis (Micciancio and Goldwasser, 2002). We now define periodic coverings and the covering density $\theta(\Omega)$, following the exposition given by Naszódi (2018).

Definition 9 (Periodic translative covering density). Let $\Omega \subset \mathbb{R}^\kappa$ be convex, $\Lambda \subset \mathbb{R}^\kappa$ be a lattice, and let $S \subset \mathbb{R}^\kappa$ be a finite set. The *periodic arrangement of translates*

$$\mathcal{C}(\Omega, \Lambda, S) = \{\Omega + \mathbf{v} + \mathbf{s} : \mathbf{v} \in \Lambda \text{ and } \mathbf{s} \in S\}$$

is an *periodic Ω -covering* of $\mathbb{R}^\kappa$ if it covers $\mathbb{R}^\kappa$. The *periodic translative covering density* of Ω is the following infimum taken over all periodic Ω -coverings:

$$\theta(\Omega) := \inf \{\delta(\mathcal{C}) : \mathcal{C} \text{ is an periodic } \Omega\text{-covering of } \mathbb{R}^\kappa\}, \quad \text{where } \delta(\mathcal{C}) = \frac{|S| \cdot \text{Vol}(\Omega)}{\det \Lambda},$$

where the infimum is taken over all lattices Λ and finite sets S , and $\delta(\mathcal{C})$ is called the *density* of $\mathcal{C}$.

Definition 10 (Ω -covering number). Let $\Omega \subset \mathbb{R}^\kappa$ be a compact set whose interior contains $\mathbf{0} \in \text{int}(\Omega)$. Given any $\mathbf{w} \in \mathbb{R}^\kappa$, we say that $\Omega + \mathbf{w}$ is a translation of Ω *centered* at $\mathbf{w}$. Let $A \subset \mathbb{R}^\kappa$ be any set. The Ω -*covering number* $N(A; \Omega)$ of A is defined as the minimal number of translations of Ω centered at points in A needed to cover A , i.e.,

$$N(A; \Omega) := \inf_{S \subset A} \{|S| : A \subset \bigcup_{\mathbf{s} \in S} \Omega + \mathbf{s}\}.$$

Before we prove the main result of this section (Proposition 5), we provide two lemmata. We leave the proofs to Appendix F.4. The first lemma counts the number of centers of a periodic covering $\mathcal{C}(\Omega, \Lambda, S)$ that lands inside any fundamental parallelepiped of Λ . This number is at most $|S|$, and in fact, equality holds unless there are distinct points $\mathbf{s}, \mathbf{s}' \in S$ that are perfectly spaced apart so that $\mathbf{s} - \mathbf{s}' \in \Lambda$. The proof is in Appendix F.4.7.

Lemma 17 (Centers in the fundamental parallelepiped). Let $\Omega \subset \mathbb{R}^\kappa$ be a compact set whose interior contains $\mathbf{0} \in \text{int}(\Omega)$. Suppose that $\mathcal{C} = \mathcal{C}(\Omega, \Lambda, S)$ is an Ω -periodic covering of $\mathbb{R}^n$. Let V be any basis of Λ and let $P(V)$ be its fundamental parallelepiped. Let $\mathcal{C}(V) = \{\mathbf{v} + \mathbf{s} : \mathbf{v} \in \Lambda, \mathbf{s} \in S\} \cap P(V)$ be the set of centers contained in $P(V)$. Then $|\mathcal{C}(V)| \leq |S|$.

We use Lemma 17 to prove the second lemma which is a simple case of the upcoming Proposition 5, when restricted to boxes; as it presents a significant first step, we mention it here explicitly.

Lemma 18 (Ω -covering number of boxes). Let $\Omega \subset \mathbb{R}^\kappa$ be a compact set with $\mathbf{0} \in \text{int}(\Omega)$. Let $A = [0, 1]^\kappa$. Then:

$$\lim_{t \rightarrow 0} N(A; t\Omega) \cdot t^\kappa = \frac{\theta(\Omega) \cdot \text{Vol}(A)}{\text{Vol}(\Omega)}.$$

The proof of Lemma 18 is in Appendix F.4.8. The following result then generalizes Lemma 18 from boxes to general Jordan measurable sets A . To that end, recall the definition of Jordan measurable sets.

Definition 11 (Jordan measurable set). A set A in a metric measure space is *Jordan measurable* if A is bounded and the boundary of A has measure zero.

Intuitively, we can generalize Lemma 18 to Jordan measurable sets, because the boundary of some set A having measure zero implies that there is a countable disjoint union of boxes B_i contained in A so that $A \setminus \bigcup B_i$ has volume zero. We obtain Proposition 5, which (perhaps surprisingly) to the best of our knowledge seems to not have been established in the literature.

Proposition 5 (Ω -covering number of Jordan measurable sets). *Let $\Omega \subset \mathbb{R}^\kappa$ be compact with $\mathbf{0} \in \text{int}(\Omega)$. Let $A \subset \mathbb{R}^\kappa$ be a Jordan measurable set, so that it is bounded and has measure zero boundary $\text{Vol}(\partial A) = 0$. Then:*

$$\lim_{t \rightarrow 0} N(A; t\Omega) \cdot t^\kappa = \frac{\theta(\Omega) \cdot \text{Vol}(A)}{\text{Vol}(\Omega)}.$$

The proof makes use of Lemma 18 in the way described above; we defer it to Appendix F.4.9.

Application to simplices. We now show that the form of the limit in Proposition 5 is special when we cover with *simplices*. In particular, it turns out that the covering density $\theta(\Omega)$ of all simplices is equal. It follows from the fact that any two simplices Ω and Ω' are affinely isomorphic: the same isomorphism maps periodic Ω -coverings into periodic Ω' -coverings.

Lemma 19 (Covering density of simplices is unique). *Let $\Omega, \Omega' \subset \mathbb{R}^\kappa$ be two κ -dimensional simplices. Then $\theta(\Omega) = \theta(\Omega')$.*

The proof is in Appendix F.4.6. As the final result of this section, we compute the volume of a simplex corresponding to the unit Pareto ball on a linear Pareto front with normal vector $\mathbf{n}$. Effectively, this is where the form of the density in Theorem 1 comes from, as we can plug it into Proposition 5. The proof is in Appendix F.4.10 and is just a calculation.

Lemma 20 (Volume of the simplex). *Let $\mathbf{n} = (n_1, \dots, n_K)$ be a normal vector where $\|\mathbf{n}\|_2 = 1$ and $n_k > 0$ for all $k \in [K]$. Define, as in Eq. (20), the set $\Omega_{\mathbf{n}} := \{\mathbf{v} \in \mathbb{R}^K : \mathbf{v} \cdot \mathbf{n} = 0\} \cap \{\mathbf{v} \in \mathbb{R}^K : v_k \geq -1\}$. Then, $\Omega_{\mathbf{n}}$ is a $(K-1)$ -dimensional simplex and for $t \geq 0$, the $(K-1)$ -dimensional Hausdorff volume of $t\Omega_{\mathbf{n}}$ is*

$$\mathcal{H}^{K-1}(t\Omega_{\mathbf{n}}) = \frac{1}{(K-1)!} \frac{(\mathbf{1} \cdot \mathbf{n})^{K-1}}{\prod_{k \in [K]} n_k} \cdot t^{K-1}.$$

F.3. Main Proof of Theorem 1

We are now ready to prove Theorem 1. Let $A \subset \mathfrak{R}$ be a Jordan measurable set in a nice Pareto front.

Let θ_K be the covering density (Definition 9) of $(K-1)$ -dimensional simplices, which is independent of the choice of simplex by Lemma 19. As shown in (Rogers, 1957), this covering density satisfies $1 \leq \theta_K \leq O(K \log K)$. Choosing $C_K = \theta_K \cdot (K-1)!$ we have

$$\beta(\mathbf{n}) = \theta_K \cdot (K-1)! \cdot \frac{\prod_{k \in [K]} n_k}{(\mathbf{1} \cdot \mathbf{n})^{K-1}}. \quad (21)$$

For any $\varepsilon > 0$, by Lemma 14 we can construct a piecewise linear Pareto manifold achieving ε -distortion. In particular, there is a decomposition $\{\phi_m : U_m \rightarrow V_m : m \in [M]\}$ where U_m has diameter at most ε , V_m is contained in the hyperplane normal to $\mathbf{n}_m = \mathbf{n}(\mathbf{r}_m)$ for some $\mathbf{r}_m \in U_m$, and there is bounded distortion of the volume $1 - \varepsilon < |J\phi_m| < 1 + \varepsilon$ for all m . We use the following notation: we denote the remainder region by $Z = \mathfrak{R} \setminus \bigcup U_m$ and let N_m denote the covering number in the linear Pareto front V_m under the Pareto distance. For any normal vector $\mathbf{n}$, we denote by $\Omega_{\mathbf{n}}$ the simplex defined in Lemma 20, by which we also know that

$$\mathcal{H}^{K-1}(\Omega_{\mathbf{n}}) = \frac{1}{(K-1)!} \frac{(\mathbf{1} \cdot \mathbf{n})^{K-1}}{\prod_{k \in [K]} n_k} = \frac{\theta_K}{\beta(\mathbf{n})}. \quad (22)$$

Upper bound. We can compute an upper bound on the t -Pareto covering number:

$$\begin{aligned} N_{\text{Par}}(t, A) &\leq \sum_{m \in [M]} N_{\text{Par}}(t, A \cap U_m) + N_{\text{Par}}(t, Z) \\ &\leq \sum_{m \in [M]} N_m(t/(1+\varepsilon), \phi_m(A \cap U_m)) + N_{\text{Par}}(t, Z), \end{aligned}$$

where the first step follows because the covering number is subadditive, and the second applies Lemma 16. Let $\pi_m : V_m \rightarrow \mathbb{R}^{K-1}$ be any injective affine map. We can use the simplex covering notation from Appendix F.2 and Proposition 5 to obtain

an upper bound on the (rescaled) first term

$$\begin{aligned}
 \limsup_{t \rightarrow 0} N_m(t/(1+\varepsilon), \phi_m(A \cap U_m)) \cdot t^{K-1} &\stackrel{(i)}{=} \limsup_{t \rightarrow 0} N(\pi_m(\phi_m(A \cap U_m)), (t/(1+\varepsilon))\pi_m(\Omega_{\mathbf{n}_m})) \cdot t^{K-1} \\
 &\stackrel{(ii)}{=} \frac{\theta_K \cdot \text{Vol}(\pi_m(\phi_m(A \cap U_m)))}{\text{Vol}(\pi_m(\Omega_{\mathbf{n}_m}))} \cdot (1+\varepsilon)^{K-1} \\
 &\stackrel{(iii)}{=} \frac{\theta_K \cdot \mathcal{H}^{K-1}(\phi_m(A \cap U_m))}{\mathcal{H}^{K-1}(\Omega_{\mathbf{n}_m})} \cdot (1+\varepsilon)^{K-1} \\
 &\stackrel{(iv)}{\leq} \frac{\theta_K \cdot \mathcal{H}^{K-1}(A \cap U_m)}{\mathcal{H}^{K-1}(\Omega_{\mathbf{n}_m})} \cdot (1+\varepsilon)^K \\
 &\stackrel{(v)}{=} \beta(\mathbf{n}_m) \cdot \mathcal{H}^{K-1}(A \cap U_m) \cdot (1+\varepsilon)^K.
 \end{aligned}$$

Here (i) uses the fact that the Pareto covering number is the same as the simplex covering by the derivations in Appendix F.2, (ii) applies Proposition 5, (iii) uses the fact that π_m is affine and applies the area formula (Theorem 3.2.3 in (Federer, 1996)). Moreover, (iv) again applies the area formula and the fact that the determinant of the Jacobian is bounded $|J\phi_m| \leq 1+\varepsilon$, so

$$\mathcal{H}^{K-1}(\phi_m(A \cap U_m)) = \int_{\phi_m(A \cap U_m)} d\mathcal{H}^{K-1} = \int_{A \cap U_m} |J\phi_m| d\mathcal{H}^{K-1} \leq (1+\varepsilon) \cdot \mathcal{H}^{K-1}(A \cap U_m),$$

and (v) substitutes in Equation (22). Further, we can bound the second term using Lemma 15, so we have the following limit:

$$\limsup_{t \rightarrow 0} N_{\text{Par}}(t, Z) \cdot t^{K-1} = 0.$$

Therefore, the t -Pareto covering number $N_{\text{Par}}(t, A)$ satisfies:

$$\begin{aligned}
 \limsup_{t \rightarrow 0} N_{\text{Par}}(t, A) \cdot t^{K-1} &\leq (1+\varepsilon)^K \cdot \sum_{m \in [M]} \beta(\mathbf{n}_m) \cdot \mathcal{H}^{K-1}(A \cap U_m) \\
 &= (1+\varepsilon)^K \cdot \int_A \sum_{m \in [M]} \beta(\mathbf{n}(\mathbf{r}_m)) \cdot \mathbf{1}_{U_m} d\mathcal{H}^{K-1}.
 \end{aligned} \tag{23}$$

The function $(\beta \circ \mathbf{n})(\mathbf{r})$ is uniformly continuous, since $(\beta \circ \mathbf{n})(\mathbf{r})$ is smooth and $\mathfrak{R}$ is compact. It follows that for every $\delta > 0$, there is a sufficiently small $\varepsilon > 0$ such that whenever $\|\mathbf{r} - \mathbf{r}'\|_2 < \varepsilon$, then $|(\beta \circ \mathbf{n})(\mathbf{r}) - (\beta \circ \mathbf{n})(\mathbf{r}')| < \delta$. In particular, since each U_m has diameter at most ε , we have that the integrand in Equation (23) is bounded above by $\beta(\mathbf{n}(\mathbf{r})) + \delta$. We obtain

$$\limsup_{t \rightarrow 0} N_{\text{Par}}(t, A) \cdot t^{K-1} \leq (1+\varepsilon)^K \cdot \int_{A \setminus Z} (\beta(\mathbf{n}(\mathbf{r})) + \delta) d\mathcal{H}^{K-1}(\mathbf{r}).$$

Finally, by sending $\delta \rightarrow 0$, which also forces $\varepsilon \rightarrow 0$, we deduce the upper bound:

$$\limsup_{t \rightarrow 0} N_{\text{Par}}(t, A) \cdot t^{K-1} \leq \int_{A \setminus Z} \beta(\mathbf{n}(\mathbf{r})) d\mathcal{H}^{K-1}(\mathbf{r}) = \int_A \beta(\mathbf{n}(\mathbf{r})) d\mathcal{H}^{K-1}(\mathbf{r}),$$

where in the last step we use the fact that Z is $\mathcal{H}^{K-1}$ -measure zero.

Lower bound. We now compute a matching lower bound. For each $s > 0$ and U_m , define the set of points in U_m that are s -close to the boundary of U_m in the following sense:

$$\partial U_m^s = \{\mathbf{r} \in U_m : \exists \mathbf{r}' \in \mathfrak{R} \setminus U_m \text{ s.t. } d(\mathbf{r}', \mathbf{r}) < s\}.$$

Notice that ∂U_m^s converges to $\emptyset$ as s goes to zero: for each $\mathbf{r}$, once s becomes sufficiently small, $\mathbf{r} \notin \partial U_m^s$. Fix any $s > 0$ and Jordan measurable $A \subset \mathfrak{R}$. For all $0 < t < s$, we have:

$$N_{\text{Par}}(t, A) \geq \sum_{m \in [M]} N_{\text{Par}}(t, A \cap U_m) - N_{\text{Par}}(t, \partial U_m^s).$$

To see this, let $S \subset A$ be any minimal t -Pareto covering of A and let $S_m \subset \partial U_m^s$ be a minimal t -Pareto covering of ∂U_m^s . We claim that the set $(S \cap U_m) \cup S_m$ forms a t -Pareto covering of $A \cap U_m$. Let's assume the claim for now. It implies the upper bound

$$|S \cap U_m| + |S_m| \geq N_{\text{Par}}(t, A \cap U_m).$$

We deduce the above inequality by taking a summation over $m \in [M]$, and combining with the facts that

$$N_{\text{Par}}(t, A) = \sum_m |S \cap U_m| + |S \cap Z| \geq \sum_m |S \cap U_m| \quad \text{and} \quad |S_m| = N_{\text{Par}}(t, \partial U_m^s).$$

As for the claim, let $\mathbf{r} \in A \cap U_m$. Then, either it is covered by an element of $S \cap U_m$, or it was covered by an element of S centered at a point $\mathbf{r}' \notin U_m$. In this case, $d(\mathbf{r}', \mathbf{r}) < t < s$, which implies that $\mathbf{r} \in \partial U_m^s$, and it is covered by an element of S_m .

Applying Lemma 16 and multiplying through by t^{K-1} , we have:

$$N_{\text{Par}}(t, A) \cdot t^{K-1} \geq \sum_{m \in [M]} N_m(t/(1-\varepsilon), \phi_m(A \cap U_m)) \cdot t^{K-1} - N_m(t/(1+\varepsilon), \phi_m(\partial U_m^s)) \cdot t^{K-1},$$

where recall that N_m gives the covering number for sets in V_m . By Proposition 5, the limit of the right-hand side exists as $t \rightarrow 0$, and so we obtain that:

$$\begin{aligned} & \liminf_{t \rightarrow 0} N_{\text{Par}}(t, A) \cdot t^{K-1} \\ & \geq \sum_{m \in [M]} \frac{\theta_K \cdot \mathcal{H}^{K-1}(\phi_m(A \cap U_m))}{\mathcal{H}^{K-1}(\Omega_{\mathbf{n}_m})} \cdot (1-\varepsilon)^{K-1} - \frac{\theta_K \cdot \mathcal{H}^{K-1}(\phi_m(\partial U_m^s))}{\mathcal{H}^{K-1}(\Omega_{\mathbf{n}_m})} \cdot (1+\varepsilon)^{K-1}. \end{aligned}$$

Let $Z^s = \bigcup \partial U_m^s \cup Z$. Once again, relying on the uniform continuity of $(\beta \circ \mathbf{n})(\mathbf{r})$, we can apply the same argument used for the upper bound to deduce that for all $\delta > 0$,

$$\begin{aligned} & \liminf_{t \rightarrow 0} N_{\text{Par}}(t, A) \cdot t^{K-1} \\ & \geq (1-\varepsilon)^K \cdot \int_{A \setminus Z} (\beta(\mathbf{n}(\mathbf{r})) - \delta) \, d\mathcal{H}^{K-1}(\mathbf{r}) - (1+\varepsilon)^K \cdot \int_{Z^s} (\beta(\mathbf{n}(\mathbf{r})) + \delta) \, d\mathcal{H}^{K-1}(\mathbf{r}). \end{aligned}$$

This holds for all $s > 0$, so we can also let s go to zero, where Z^s converges to Z . Since $\mathcal{H}^{K-1}(Z) = 0$, not only does the second term vanishes, but for the first term, it is equivalent to taking the integral over A instead of $A \setminus Z$. We obtain:

$$\liminf_{t \rightarrow 0} N_{\text{Par}}(t, A) \cdot t^{K-1} \geq (1-\varepsilon)^K \cdot \int_A (\beta(\mathbf{n}(\mathbf{r})) - \delta) \, d\mathcal{H}^{K-1}(\mathbf{r}).$$

Since this holds for all $\delta > 0$, we can let δ go to zero. This also forces ε to zero:

$$\liminf_{t \rightarrow 0} N_{\text{Par}}(t, A) \cdot t^{K-1} \geq \int_A \beta(\mathbf{n}(\mathbf{r})) \, d\mathcal{H}^{K-1}(\mathbf{r}).$$

Since the limit infimum and limit supremum coincide, the limit exists and is given by the theorem statement.

F.4. Technical Proofs

F.4.1. PROOF OF LEMMA 12

Choose any set of smooth coordinates $\psi : \tilde{U} \rightarrow \mathbb{R}^{K-1}$ centered at $\mathbf{r}_0$, so that $\tilde{U} \subset \mathfrak{R}$ is an open set relative to $\mathfrak{R}$ containing $\mathbf{r}_0$, and $\psi(\mathbf{r}_0) = 0$. By shrinking $\tilde{U}$ if needed, we may assume that ψ is L -bi-Lipschitz:

$$\frac{1}{L} \|\mathbf{r} - \mathbf{r}'\|_2 \leq \|\psi(\mathbf{r}) - \psi(\mathbf{r}')\|_2 \leq L \|\mathbf{r} - \mathbf{r}'\|_2, \quad \forall \mathbf{r}, \mathbf{r}' \in \tilde{U}.$$

Define the following map $\theta : \mathbb{T}_{\mathbf{r}_0} \mathfrak{R} \rightarrow \mathbb{R}^{K-1}$, which is also L -bi-Lipschitz:

$$\theta(\mathbf{r}_0 + \mathbf{v}) = \nabla \psi(\mathbf{r}_0) \mathbf{v}.$$

Define $\phi : U \rightarrow V$ by $\phi = \theta^{-1} \circ \psi$ where we set $U \subset \tilde{U}$ later and let $V = \phi(U)$. Also see Figure 16.

We now show that ϕ approximately preserves smooth maps. Let $f : \mathbb{R}^K \rightarrow \mathbb{R}$ be any smooth, 1-Lipschitz map. Let $\mathbf{r}, \mathbf{r}' \in U \subset \tilde{U}$ come from a sufficiently small region around $\mathbf{r}_0$. Taylor's theorem implies that:

$$\begin{aligned} f(\mathbf{r}) - f(\mathbf{r}') &= (f \circ \psi^{-1})(\psi(\mathbf{r})) - (f \circ \psi^{-1})(\psi(\mathbf{r}')) \\ &= \nabla(f \circ \psi^{-1})(\psi(\mathbf{r}'))(\psi(\mathbf{r}) - \psi(\mathbf{r}')) + o(\|\psi(\mathbf{r}) - \psi(\mathbf{r}')\|_2) \end{aligned}$$

For short, let $\hat{\mathbf{r}} = \phi(\mathbf{r})$ and $\hat{\mathbf{r}}' = \phi(\mathbf{r}')$. Taylor's theorem also implies that:

$$\begin{aligned} f(\hat{\mathbf{r}}) - f(\hat{\mathbf{r}}') &= (f \circ \theta^{-1})(\psi(\mathbf{r})) - (f \circ \theta^{-1})(\psi(\mathbf{r}')) \\ &= \nabla(f \circ \theta^{-1})(\psi(\mathbf{r}'))(\psi(\mathbf{r}) - \psi(\mathbf{r}')) + o(\|\psi(\mathbf{r}) - \psi(\mathbf{r}')\|_2) \end{aligned}$$

Let $M_f(\mathbf{r}') = \nabla(f \circ \psi^{-1})(\psi(\mathbf{r}')) - \nabla(f \circ \theta^{-1})(\psi(\mathbf{r}'))$. The term in the absolute value of Equation (18) is:

$$(f(\mathbf{r}) - f(\mathbf{r}')) - (f(\hat{\mathbf{r}}) - f(\hat{\mathbf{r}}')) = M_f(\mathbf{r}')(\psi(\mathbf{r}) - \psi(\mathbf{r}')) + o(\|\psi(\mathbf{r}) - \psi(\mathbf{r}')\|_2).$$

By bi-Lipschitzness, we have that $\|\psi(\mathbf{r}) - \psi(\mathbf{r}')\|_2 = \Theta(\|\mathbf{r} - \mathbf{r}'\|_2)$, and so we have that:

$$|(f(\mathbf{r}) - f(\mathbf{r}')) - (f(\hat{\mathbf{r}}) - f(\hat{\mathbf{r}}'))| \leq \|M_f(\mathbf{r}')\|_2 \cdot O(\|\mathbf{r} - \mathbf{r}'\|_2).$$

It suffices to show that when $U \subset \tilde{U}$ is a sufficiently small region around $\mathbf{r}_0$, then $\|M_f(\mathbf{r}')\|_2$ also becomes arbitrarily small for all 1-Lipschitz-smooth f . Indeed, this holds:

$$\begin{aligned} &\|M_f(\mathbf{r}')\|_2 \\ &\stackrel{(i)}{=} \|\nabla f(\psi^{-1}(\psi(\mathbf{r}')))\nabla\psi^{-1}(\psi(\mathbf{r}')) - \nabla f(\theta^{-1}(\psi(\mathbf{r}')))\nabla\theta^{-1}(\psi(\mathbf{r}'))\| \\ &\stackrel{(ii)}{=} \|\nabla f(\psi^{-1}(\psi(\mathbf{r}')))\nabla\psi^{-1}(\psi(\mathbf{r}')) - \nabla f(\psi^{-1}(\psi(\mathbf{r}')))\nabla\theta^{-1}(\psi(\mathbf{r}')) \\ &\quad + \nabla f(\psi^{-1}(\psi(\mathbf{r}')))\nabla\theta^{-1}(\psi(\mathbf{r}')) - \nabla f(\theta^{-1}(\psi(\mathbf{r}')))\nabla\theta^{-1}(\psi(\mathbf{r}'))\| \\ &\stackrel{(iii)}{\leq} \|\nabla f(\mathbf{r}')\| \cdot \|\nabla\psi^{-1}(\psi(\mathbf{r}')) - \nabla\theta^{-1}(\psi(\mathbf{r}'))\| + \|\nabla f(\mathbf{r}') - \nabla f(\theta^{-1}(\psi(\mathbf{r}')))\| \cdot \|\nabla\theta^{-1}(\psi(\mathbf{r}'))\| \\ &\stackrel{(iv)}{\leq} \|\nabla\psi^{-1}(\psi(\mathbf{r}')) - \nabla\psi^{-1}(\psi(\mathbf{r}_0))\| + L \cdot \|\mathbf{r}' - \phi(\mathbf{r}')\|, \end{aligned}$$

where (i) applies the chain rule, (ii) adds and subtracts the inner terms, (iii) applies triangle inequality, (iv) simplifies the first term by using the facts that (a) the gradient ∇f is 1-Lipschitz and (b) the gradient $\nabla\theta^{-1} \equiv \nabla\psi^{-1}(\mathbf{r}_0)$ is constant, and (iv) also simplifies the second term also using the fact that ∇f is 1-Lipschitz and that θ is L -bi-Lipschitz. Both terms in (iv) can be made to be arbitrarily close to zero by controlling the size of U around $\mathbf{r}_0$. Both terms go to zero as $\mathbf{r}' \rightarrow \mathbf{r}_0$.

F.4.2. PROOF OF LEMMA 13

We begin by making the following observation on comparability of Pareto and Euclidean distances for *linear* Pareto fronts. Let $\mathbf{n} \in \mathbb{R}_{>0}^K$ be a unit normal vector where $n_k > \eta > 0$ is bounded away from 0. Let $\mathfrak{A} = \{\mathbf{r} \in \mathbb{R}^K : \mathbf{r} \cdot \mathbf{n} = 0\}$ be the hyperplane with normal vector $\mathbf{n}$. Then:

$$\frac{\eta}{K(1+\eta)} \|\mathbf{r} - \mathbf{r}'\|_2 \leq d(\mathbf{r}, \mathbf{r}'). \quad (24)$$

Proof of Equation (24). Let $\mathbf{v} = \mathbf{r} - \mathbf{r}'$ for short, and let $I = \{k \in [K] : v_k > 0\}$ be the coordinates where $r'_k < r_k$. Let $J = [K] \setminus I$ be the remaining coordinates. Thus, $d(\mathbf{r}, \mathbf{r}') = \max_{k \in [K]} v_k$. Because $\mathbf{v} \cdot \mathbf{n} = 0$, we have:

$$\sum_{i \in I} n_i v_i = - \sum_{j \in J} n_j v_j.$$

We also have that $\|\mathbf{v}\|_1 = \sum_{i \in I} v_i - \sum_{j \in J} v_j$. The unit vector $\mathbf{n}$ satisfies $1 \geq n_k > \eta > 0$. Together:

$$\sum_{i \in I} v_i \geq \sum_{i \in I} n_i v_i = - \sum_{j \in J} n_j v_j \geq -\eta \cdot \sum_{j \in J} v_j = \eta \cdot \|\mathbf{v}\|_1 - \eta \sum_{i \in I} v_i.$$

Rearranging, we obtain that:

$$\sum_{i \in I} v_i \geq \frac{\eta}{1 + \eta} \|\mathbf{v}\|_1 \geq \frac{\eta}{1 + \eta} \|\mathbf{v}\|_2.$$

Dividing through by K and using the fact that $\max_{k \in [K]} v_k \geq \frac{1}{K} \sum_{i \in I} v_i$ gives the result. $\square$

We can use Equation (24) combined with Lemma 12 to prove Lemma 13.

Proof of Lemma 13. The Pareto distance is bounded above by the ℓ_∞ -distance, which is upper bounded by the ℓ_2 -distance:

$$d(\mathbf{r}, \mathbf{r}') = \max_{k \in [K]} (r_k - r'_k) \leq \|\mathbf{r} - \mathbf{r}'\|_\infty \leq \|\mathbf{r} - \mathbf{r}'\|_2.$$

As for the lower bound, we use the niceness of $\mathfrak{R}$. We claim that for each $\mathbf{r}_0 \in \mathfrak{R}$, there is a sufficiently small open Euclidean ball $B_{\mathbf{r}_0} := B(\mathbf{r}_0, \alpha_{\mathbf{r}_0}) \cap \mathfrak{R}$ such that the following has a positive lower bound:

$$\inf_{\mathbf{r} \in B_{\mathbf{r}_0}} \inf_{\mathbf{r}' \in \mathfrak{R}} \frac{d(\mathbf{r}, \mathbf{r}')}{\|\mathbf{r} - \mathbf{r}'\|_2} > c_{\mathbf{r}_0} > 0. \quad (25)$$

The result follows from compactness of $\mathfrak{R}$. In particular, there is a finite subcover of $\mathfrak{R}$ by balls $B_{\mathbf{r}_1}, \dots, B_{\mathbf{r}_N}$, and we let the constant in the lemma statement be defined as $c = \min_{n \in [N]} c_{\mathbf{r}_n}$.

Fix $\mathbf{r}_0$ and some $\varepsilon > 0$. Lemma 12 constructs a diffeomorphism $\phi : U \rightarrow V$ from a relatively open set in $\mathfrak{R}$ containing $\mathbf{r}_0$ to a locally linearized Pareto front satisfying Equation (18). Choose any ball $B_{\mathbf{r}_0} := B(\mathbf{r}_0, \alpha_{\mathbf{r}_0}) \cap \mathfrak{R}$ such that the ball $B'_{\mathbf{r}_0}$ with twice the radius is contained in U :

$$B'_{\mathbf{r}_0} := B(\mathbf{r}_0, 2\alpha_{\mathbf{r}_0}) \cap \mathfrak{R} \subset U.$$

We show Equation (25) for two cases: (1) $\mathbf{r}' \in B'_{\mathbf{r}_0}$ and (2) $\mathbf{r}' \notin B'_{\mathbf{r}_0}$.

1. Local. Let $\mathbf{r} \in B_{\mathbf{r}_0}$ and $\mathbf{r}' \in B'_{\mathbf{r}_0}$. Let $\hat{\mathbf{r}} = \phi(\mathbf{r})$ and $\hat{\mathbf{r}}' = \phi(\mathbf{r}')$. For each $k \in [K]$, the coordinate map projecting $\mathbf{r}$ down to its k component r_k is 1-Lipschitz smooth. Thus, Lemma 12 shows that:

$$\frac{r_k - r'_k}{\|\mathbf{r} - \mathbf{r}'\|_2} \geq \frac{\hat{r}_k - \hat{r}'_k}{\|\hat{\mathbf{r}} - \hat{\mathbf{r}}'\|_2} - \varepsilon, \quad \forall k \in [K].$$

Taking the max over $k \in [K]$ implies that:

$$\frac{d(\mathbf{r}, \mathbf{r}')}{\|\mathbf{r} - \mathbf{r}'\|_2} \geq \frac{d(\hat{\mathbf{r}}, \hat{\mathbf{r}}')}{\|\hat{\mathbf{r}} - \hat{\mathbf{r}}'\|_2} - \varepsilon \geq \frac{d(\hat{\mathbf{r}}, \hat{\mathbf{r}}')}{\|\hat{\mathbf{r}} - \hat{\mathbf{r}}'\|_2} \frac{\|\hat{\mathbf{r}} - \hat{\mathbf{r}}'\|_2}{\|\mathbf{r} - \mathbf{r}'\|_2} - \varepsilon \geq \frac{\eta}{K(1 + \eta)} \cdot (1 - \varepsilon\sqrt{K}) - \varepsilon,$$

where in the last inequality, we use the facts that (a) Lemma 12 shows that the map ϕ has bounded distortion, and (b) $V \subset T_{\mathbf{r}_0} \mathfrak{R}$ is a linear Pareto front, so that Equation (24) shows that $d(\hat{\mathbf{r}}, \hat{\mathbf{r}}')/\|\mathbf{r} - \mathbf{r}'\| \geq \eta/(K(1 + \eta))$. Once ε is sufficiently small, the lower bound is strictly positive.

2. Global. Consider $\mathbf{r} \in B_{\mathbf{r}_0}$ and $\mathbf{r}' \in \mathfrak{R} \setminus B'_{\mathbf{r}_0}$. The product set $\mathcal{K} = \overline{B_{\mathbf{r}_0}} \times (\mathfrak{R} \setminus B'_{\mathbf{r}_0}) \subset \mathfrak{R}^2$ is compact, while the function $(\mathbf{r}, \mathbf{r}') \mapsto d(\mathbf{r}, \mathbf{r}')/\|\mathbf{r} - \mathbf{r}'\|_2$ is continuous. Thus, it attains its minimum on $\mathcal{K}$. Since $\overline{B_{\mathbf{r}_0}}$ and $(\mathfrak{R} \setminus B'_{\mathbf{r}_0})$ are disjoint, this function is strictly greater than zero. This is because distinct points on $\mathfrak{R}$ are mutually non-dominating; $\mathbf{r}'$ does not dominate $\mathbf{r}$, so there is a coordinate $k \in [K]$ such that $r_k - r'_k > 0$. Thus, the numerator $d(\mathbf{r}, \mathbf{r}') = \max_{k \in [K]} (r_k - r'_k) > 0$ for $(\mathbf{r}, \mathbf{r}') \in \mathcal{K}$. The denominator is upper bounded by the diameter of $\mathfrak{R}$, which is finite as $\mathfrak{R}$ is compact.

That concludes the proof of Lemma 13. $\square$

F.4.3. PROOF OF LEMMA 14

We begin by proving the following fact that inverses of bounded distortions are bounded distortions: Let $\varepsilon \in (0, 1/2)$ and let $\phi : (U, d_U) \rightarrow (V, d_V)$ be an ε -distortion (recalling Definition 6). Then the map $\phi^{-1} : (V, d_V) \rightarrow (U, d_U)$ is a 2ε -distortion.

Indeed, whenever $\mathbf{r}, \mathbf{r}' \in U$, define $\hat{\mathbf{r}} = \phi(\mathbf{r})$ and $\hat{\mathbf{r}}' = \phi(\mathbf{r}')$. By definition, the map ϕ satisfies:

$$(1 - \varepsilon) \cdot d_V(\hat{\mathbf{r}}, \hat{\mathbf{r}}') \leq d_U(\mathbf{r}, \mathbf{r}') \leq (1 + \varepsilon) \cdot d_V(\hat{\mathbf{r}}, \hat{\mathbf{r}}'), \quad \forall \mathbf{r}, \mathbf{r}' \in U.$$

Since $(1 + 2\varepsilon)(1 - \varepsilon) - 1 = \varepsilon(1 - 2\varepsilon) > 0$ whenever $\varepsilon \in (0, 1/2)$, we have:

$$d_V(\hat{\mathbf{r}}, \hat{\mathbf{r}}') \leq \frac{1}{1 - \varepsilon} \cdot d_U(\mathbf{r}, \mathbf{r}') \leq (1 + 2\varepsilon) \cdot d_U(\mathbf{r}, \mathbf{r}').$$

And since $(1 + \varepsilon)(1 - \varepsilon) - 1 = -\varepsilon^2 < 0$, we have that:

$$d_V(\hat{\mathbf{r}}, \hat{\mathbf{r}}') \geq \frac{1}{1 + \varepsilon} \cdot d_U(\mathbf{r}, \mathbf{r}') \geq (1 - \varepsilon) \cdot d_U(\mathbf{r}, \mathbf{r}').$$

Together, these show that ϕ^{-1} is a 2ε -distortion.

We now come to the proof of Lemma 14. We first prove the local result, showing that for each $\mathbf{r}_0 \in \mathfrak{R}$, there is a local linear approximation that preserves the Pareto distance. This is almost an immediate consequence of Lemma 12. The second step applies compactness to cover $\mathfrak{R}$ with a finite collection of well-approximated local regions.

Step 1: Low-distortion for small neighborhoods. Fix any $\mathbf{r}_0 \in \mathfrak{R}$ and without loss of generality, we may fix $\varepsilon \in (0, 1/2)$. Lemma 12 constructs a diffeomorphism $\phi : U \rightarrow V$ satisfying Equation (18). For any $\mathbf{r}, \mathbf{r}' \in U$, let $\hat{\mathbf{r}} = \phi(\mathbf{r})$ and $\hat{\mathbf{r}}' = \phi(\mathbf{r}')$ for short. Applying Equation (18) to the standard basis functions of $\mathbb{R}^K$, where $\pi_k : \mathbb{R}^K \rightarrow \mathbb{R}$ projects to the k th coordinate. Each of these maps are 1-Lipschitz smooth, and so we obtain that for each $k \in [K]$:

$$(r_k - r'_k) = (\hat{r}_k - \hat{r}'_k) + \Delta_k,$$

where $|\Delta_k| \leq \varepsilon \|\mathbf{r} - \mathbf{r}'\|_2$. We obtain the upper bound:

$$d(\mathbf{r}, \mathbf{r}') = \max_{k \in [K]} r_k - r'_k = \max_{k \in [K]} \hat{r}_k - \hat{r}'_k + \Delta_k \leq d(\hat{\mathbf{r}}, \hat{\mathbf{r}}') + \varepsilon \|\mathbf{r} - \mathbf{r}'\|_2.$$

The analogous lower bound holds. Lemma 13 shows that there is a constant $c_{\mathfrak{R}} > 0$ depending on $\mathfrak{R}$ with

$$c_{\mathfrak{R}} \cdot \|\mathbf{r} - \mathbf{r}'\|_2 \leq d(\mathbf{r}, \mathbf{r}') \leq \|\mathbf{r} - \mathbf{r}'\|_2, \quad \forall \mathbf{r}, \mathbf{r}' \in \mathfrak{R}.$$

Let $\varepsilon' = \max\{\varepsilon, \varepsilon/c_{\mathfrak{R}}\}$. It follows that:

$$(1 - \varepsilon') \cdot d(\mathbf{r}, \mathbf{r}') \leq d(\hat{\mathbf{r}}, \hat{\mathbf{r}}') \leq (1 + \varepsilon') \cdot d(\mathbf{r}, \mathbf{r}').$$

Thus, ϕ^{-1} is an ε' -distortion. As shown above, this means ϕ is a $2\varepsilon'$ -distortion. Reparametrizing ε' , we obtain a construction for an ε -distortion ϕ around any point $\mathbf{r}_0$ in $\mathfrak{R}$, where ϕ maps into a linear Pareto front with normal vector $\mathbf{n}(\mathbf{r}_0)$. By construction, $J\phi(\mathbf{r}_0)$ is the identity, so $|J\phi(\mathbf{r}_0)| = 1$. We may restrict the domain of ϕ to an open ball $B(\mathbf{r}_0, \alpha_{\mathbf{r}_0}) \cap \mathfrak{R} \subset U$ so that for all $\mathbf{r}$ in the domain of ϕ ,

$$1 - \varepsilon < |J\phi(\mathbf{r})| < 1 + \varepsilon.$$

Step 2: Partitioning the Pareto front. In the previous step, we saw that for any $\mathbf{r}_0 \in \mathfrak{R}$, there is a triple $(B, \phi, \mathbf{n})$ where B is a relatively open ball in $\mathfrak{R}$ containing $\mathbf{r}_0$, and ϕ is a ε -distortion mapping B into a subset of $\{x \in \mathbb{R}^K : x \cdot \mathbf{n} = 0\}$. We can always shrink the ball so that its radius is less than ε . Since $\mathfrak{R}$ is compact, there must be a finite collection of such triples:

$$(B_1, \phi_1, \mathbf{n}_1), \dots, (B_M, \phi_M, \mathbf{n}_M),$$

where $B_1, \dots, B_M$ cover $\mathfrak{R}$. To obtain the collection $\phi_m : (U_m, d) \rightarrow (V_m, d)$ in the statement, we set:

$$U_m := B_m \setminus \bigcup_{i < m} \overline{B_i} \quad \text{and} \quad V_m = \phi_m(U_m).$$

By construction, the first condition is satisfied. The second condition is also satisfied, since the only points potentially not covered by the family $\{U_m : m \in [M]\}$ are the boundary points of B_m that we removed.

F.4.4. PROOF OF LEMMA 15

We begin with a basic fact about the scaling of ℓ_2 -covering numbers. Let $A \subset \mathbb{R}^K$ and $c, t > 0$. Then:

$$N_2(t, A) \leq (1 + 2c)^K \cdot N_2(t/c, A). \quad (26)$$

Proof of Equation (26). First, recall the following fact (e.g., Corollary 4.2.13 in (Vershynin, 2018)): Let $t > 0$ and let $B \subset \mathbb{R}^K$ be the unit ball in ℓ_2 -norm. Then the covering and packing numbers with respect to the Euclidean distance scale as:

$$N_2(t, B) \leq P_2(t, B) \leq \left(1 + \frac{2}{t}\right)^K. \quad (27)$$

When $c \geq 1$, then $N_2(t/c, A) \geq N_2(t, A)$, and so the inequality holds. Assume in the remainder that $c < 1$. Greedily construct a t -covering S of A as follows: while A is not fully covered by S , choose any point in A that is distance at least t from S , and include it in S . The resulting t -covering is also a t -packing.

Suppose that $S' \subset A$ is a t/c -covering of A . If $s' \in S'$, then by (27), we must have:

$$|B(s', t/c) \cap S| \leq P_2(1/c, B) \leq (1 + 2c)^K.$$

Every point in A is within a distance of t/c from a point in S' , so it follows that:

$$|S| \leq \sum_{s' \in S'} |B(s', t/c) \cap S| \leq (1 + 2c)^K \cdot N_2(t/c, A)$$

which is Equation (26). $\square$

Proof of Lemma 15. This follows almost immediately from Lemma 13. By the upper bound on $d(\mathbf{r}, \mathbf{r}')$ in Equation (19), every t -Euclidean covering of a set A is a t -Pareto covering, implying the first inequality:

$$N_{\text{Par}}(t, A) \leq N_2(t, A).$$

The lower bound also implies that a t -Pareto covering is a $t/c_{\mathfrak{R}}$ -Euclidean covering, so that:

$$\begin{aligned} N_{\text{Par}}(t, A) &\geq N_2(t/c_{\mathfrak{R}}, A) \\ &\geq (1 + 2c_{\mathfrak{R}})^{-K} N_2(t, A), \end{aligned}$$

where the last inequality uses Equation (26).

Now, since $U = B \cap \mathfrak{R}$ is the intersection of a ball with $\mathfrak{R}$, the boundary ∂U is a piecewise smooth, compact $(K - 2)$ -dimensional manifold. It is a standard fact that the Minkowski dimension of such manifolds exists so that we have (Section 3.2 of (Falconer, 2013)):

$$\lim_{t \rightarrow 0} \frac{\log N_2(t, \partial U)}{\log(1/t)} = K - 2.$$

This shows that $N_2(t, \partial U) \leq t^{-(K-2) \cdot (1+o(1))}$ as $t \rightarrow 0$, which implies that

$$\lim_{t \rightarrow 0} N_2(t, \partial U) \cdot t^{K-1} = 0.$$

The second claim then follows immediately from the coarse bound on the covering numbers above, concluding the proof of Lemma 15. $\square$

F.4.5. PROOF OF LEMMA 16

Let $S \subset A$ be a minimal subset such that $\phi(S)$ is a $(t/(1 + \varepsilon))$ -covering of A . Because ϕ is a bijection, the size of S is $N_V(t/(1 + \varepsilon), \phi(A))$. For every $\phi(\mathbf{r}') \in \phi(A)$, there is some $\phi(\mathbf{r}) \in \phi(S)$ such that $d_V(\phi(\mathbf{r}), \phi(\mathbf{r}')) < t/(1 + \varepsilon)$. Since ϕ is an ε -distortion, this implies that $d_U(\mathbf{r}, \mathbf{r}') < t$. In particular, S is a t -covering of A . This implies the upper bound

$$N_U(t, A) \leq N_V(t/(1 + \varepsilon), \phi(A)).$$

The lower bound is similar: if S is a minimal t -covering of A , then $\phi(S)$ is a $(t/(1 - \varepsilon))$ -covering of $\phi(A)$.

F.4.6. PROOF OF LEMMA 19

We claim that there is an affine isomorphism $T : \mathbb{R}^\kappa \rightarrow \mathbb{R}^\kappa$ such that $T(\Omega) = \Omega'$.

Define the simplex $\Omega_0 := \text{conv}(\mathbf{0}, \mathbf{e}_1, \dots, \mathbf{e}_\kappa)$, where $\mathbf{e}_i \in \mathbb{R}^\kappa$ is a standard basis vector. It suffices to show that there is an affine isomorphism from Ω_0 to Ω_1 where $\Omega_1 = \text{conv}(\mathbf{v}_0, \mathbf{v}_1, \dots, \mathbf{v}_\kappa)$. Let $L : \mathbb{R}^\kappa \rightarrow \mathbb{R}^\kappa$ be the following linear map:

$$L\mathbf{e}_i = \mathbf{v}_i - \mathbf{v}_0.$$

It is invertible because the points $\{\mathbf{v}_k : k = 0, 1, \dots, \kappa\}$ are in general linear position. Set $T(\mathbf{u}) = \mathbf{v}_0 + L(\mathbf{u})$. Notice that the determinants of T and V coincide, $\det T = \det L$.

Let $\mathcal{C}(\Omega, \Lambda, S)$ be an Ω -periodic covering of $\mathbb{R}^\kappa$. Then, Fact 2.1 of [Naszódi \(2018\)](#) shows that the following is an Ω -periodic covering of $\mathbb{R}^\kappa$,

$$T(\mathcal{C}) := \mathcal{C}(T(\Omega), L(\Lambda), T(S)).$$

Because any basis V of Λ is mapped to a basis LV of $L(\Lambda)$, the density is preserved:

$$\delta(T(\mathcal{C})) = \frac{|T(S)| \cdot \text{Vol}(T(\Omega))}{\det L(\Lambda)} = \frac{|S| \cdot \det T \cdot \text{Vol}(\Omega)}{\det L \det \Lambda} = \delta(\mathcal{C}),$$

since $\det L(\Lambda) = \det(LV) = \det L \det V = \det L \det \Lambda$, and where the last step used the fact that $\det T = \det L$. Thus, the infimum is also preserved, $\theta(\Omega) = \theta(T(\Omega))$.

F.4.7. PROOF OF LEMMA 17

We first observe that the following set of differences contains only one lattice point, namely $\mathbf{0} \in \Lambda$,

$$\{\mathbf{w} - \mathbf{w}' : \mathbf{w}, \mathbf{w}' \in P(V)\} = \{\lambda_1 \mathbf{v}_1 + \dots + \lambda_\kappa \mathbf{v}_\kappa : \lambda_1, \dots, \lambda_\kappa \in (-1, 1)\}.$$

Set this aside for now. We now construct an injective map $\iota : \mathcal{C}(V) \rightarrow S$, which implies the result.

Let $\mathbf{w} \in \mathcal{C}(V)$. By definition of $\mathcal{C}(V)$, there is some $\mathbf{v} \in \Lambda$ and $\mathbf{s} \in S$ such that $\mathbf{w} = \mathbf{v} + \mathbf{s}$. Choose any such $\mathbf{s}$ and define $\iota(\mathbf{w}) = \mathbf{s}$. We claim that ι is injective. Suppose that there are $\mathbf{w}, \mathbf{w}' \in \mathcal{C}(V)$ such that $\iota(\mathbf{w}) = \iota(\mathbf{w}')$. It follows that

$$\mathbf{w} - \mathbf{w}' = (\mathbf{w} - \iota(\mathbf{w})) - (\mathbf{w}' - \iota(\mathbf{w}')) = \mathbf{v} - \mathbf{v}' \in \Lambda.$$

By the observation at the beginning, we deduce that $\mathbf{w} = \mathbf{w}'$.

F.4.8. PROOF OF LEMMA 18

For any $t > 0$, let A_t denote the scaled set $(1/t)A = [0, 1/t]^\kappa$. To prove the result, we show that the limit supremum and infimum coincide.

Lower bound. For each $t > 0$, let $S_t \subset A$ be a set such that the collection $\{t\Omega + \mathbf{s} : \mathbf{s} \in S_t\}$ forms a minimal covering of A by $t\Omega$. Then, the collection $\{\Omega + (1/t)\mathbf{s} : \mathbf{s} \in S_t\}$ covers A_t . As the set A_t tessellates $\mathbb{R}^\kappa$, we can construct the Ω -periodic covering $\mathcal{C}(\Omega, (1/t)\mathbb{Z}^\kappa, (1/t)S_t)$ of $\mathbb{R}^\kappa$ (recall Definition 9). By definition, the density of this covering $\mathcal{C}$ is at least $\theta(\Omega)$, and so

$$\theta(\Omega) \leq |S_t| \cdot \text{Vol}(\Omega) \cdot t^\kappa,$$

where we use the fact that the set $(1/t)\text{Id} \in \mathbb{R}^{\kappa \times \kappa}$ forms a basis of $(1/t)\mathbb{Z}^\kappa$, so that $\det(1/t)\mathbb{Z}^\kappa = t^{-\kappa}$. Dividing through by $\text{Vol}(\Omega)$ and writing $|S_t| = N(A; t\Omega)$, we obtain the limit infimum:

$$\liminf_{t \rightarrow 0} N(A; t\Omega) \cdot t^\kappa \geq \frac{\theta(\Omega)}{\text{Vol}(\Omega)}.$$

Upper bound. For the upper bound, we will not only need $A_t = [0, 1/t]^\kappa$, but for all $s < 1/t$, we will also let $A_{t,s} \subset A_t$ be the slightly smaller open box:

$$A_{t,s} = \left(s, \frac{1}{t} - s\right)^\kappa.$$

Let $\mathcal{C} = \mathcal{C}(\Omega, \Lambda, S)$ be an Ω -periodic covering of $\mathbb{R}^\kappa$ such that $\delta(\mathcal{C}) < \theta(\Omega) + \varepsilon$. We now show that for sufficiently small $t > 0$, we can use $\mathcal{C}$ to construct an economical covering of A by $t\Omega$, with:

$$\limsup_{t \rightarrow 0} N(A; t\Omega) \cdot t^\kappa < \frac{\theta(\Omega) + \varepsilon}{\text{Vol}(\Omega)}.$$

And since $\varepsilon > 0$ is arbitrary, the result follows for $A = [0, 1]^\kappa$ by taking the limit as ε goes to zero.

Fix any basis V of the lattice Λ , and let $D_V = \text{diam}_{\ell_\infty}(P(V))$ and $D_\Omega = \text{diam}_{\ell_\infty}(\Omega)$ be the diameters of the fundamental parallelepiped and of Ω under the ℓ_∞ -distance. As a high-level roadmap, we will first cover nearly all of A_t by a tessellation of $P(V)$, where the corresponding covering number can be related to the volume. Then, we can transfer to a simplex covering using Lemma 17, which counts the number of elements of $\mathcal{C}$ that are centered in $P(V)$. The main subtlety will be points in A_t that are very close to its boundary, but the covering number of this region grows at a slower rate.

We begin by constructing the *lattice tessellation*. For each $t > 0$, let $\mathcal{L}_t \subset \Lambda$ be the (possibly empty) set of lattice-valued translations $\mathbf{v}$ such that $P(V) + \mathbf{v}$ is fully contained in A_t ,

$$\mathcal{L}_t = \{\mathbf{v} : \mathbf{v} \in \Lambda \text{ and } P(V) + \mathbf{v} \subset A_t\}.$$

Notice that every point $\mathbf{a} \in A_t$ is covered by exactly one translated parallelepiped $P(V) + \mathbf{v}$ where $\mathbf{v} \in \Lambda$, since $P(V)$ tessellates $\mathbb{R}^\kappa$. Moreover, if this point $\mathbf{a}$ is D_V -separated from the boundary ∂A_t under the ℓ_∞ -distance and $\mathbf{a} \in P(V) + \mathbf{v}$, then we also have that $P(V) + \mathbf{v} \subset A_t$. It follows that:

$$A_{t, D_V} \subset \bigcup_{\mathbf{v} \in \mathcal{L}_t} P(V) + \mathbf{v} \subset A_t,$$

where A_{t, D_V} is the slightly smaller box. The volume of each term is:

$$\left(\frac{1}{t} - 2D_V\right)^\kappa \leq |\mathcal{L}_t| \det \Lambda \leq \frac{1}{t^\kappa}.$$

And so, the following limit exists:

$$\lim_{t \rightarrow 0} |\mathcal{L}_t| \cdot t^\kappa = \frac{1}{\det \Lambda}. \quad (28)$$

We now construct the *simplex covering*. Recall that $\mathcal{C}$ is a collection whose elements $\Omega + \mathbf{w}$ are translations of Ω . Let $\mathcal{C}_t \subset \mathcal{C}$ consist of those whose centers $\mathbf{w}$ are covered by some $P(V) + \mathbf{v}$ for $\mathbf{v} \in \mathcal{L}_t$:

$$\mathcal{C}_t = \left\{ \Omega + \mathbf{w} \in \mathcal{C} : \mathbf{w} \in \bigcup_{\mathbf{v} \in \mathcal{L}_t} P(V) + \mathbf{v} \right\}.$$

By Lemma 17, the size of $\mathcal{C}_t$ is at most $|\mathcal{C}_t| \leq |S| \cdot |\mathcal{L}_t|$. This subcollection $\mathcal{C}_t$ covers almost all of A_t , except possibly points near the boundary ∂A_t . By a similar argument as before, if $\mathbf{a} \in A_t$ is $(D_V + D_\Omega)$ -separated from the boundary and if $\mathbf{a} \in \Omega + \mathbf{w}$ is covered by an element of $\mathcal{C}$, then $\mathbf{w}$ must be D_V -separated from the boundary. This further implies that $\mathbf{w} \in P(V) + \mathbf{v}$ for some $\mathbf{v} \in \mathcal{L}_t$. In particular, this argument shows that the smaller box $A_{t, D_V + D_\Omega}$ is covered by $\mathcal{C}_t$:

$$A_{t, D_V + D_\Omega} \subset \bigcup_{\Omega + \mathbf{w} \in \mathcal{C}_t} \Omega + \mathbf{w}. \quad (29)$$

We are almost done, since this shows that the asymptotic covering number of the set on the left satisfies:

$$\limsup_{t \rightarrow 0} N(A_{t, D_V + D_\Omega}; \Omega) \cdot t^\kappa \stackrel{(i)}{\leq} \lim_{t \rightarrow 0} |S| \cdot |\mathcal{L}_t| \cdot t^\kappa \stackrel{(ii)}{=} \frac{|S|}{\det \Lambda} \stackrel{(iii)}{=} \frac{\delta(\mathcal{C})}{\text{Vol}(\Omega)} \stackrel{(iv)}{<} \frac{\theta(\Omega) + \varepsilon}{\text{Vol}(\Omega)},$$

where (i) combines Eq. (29) with the above upper bound $|\mathcal{C}_t| \leq |S| \cdot |\mathcal{L}_t|$, (ii) applies Eq. (28), (iii) uses the definition of $\delta(\mathcal{C})$, and (iv) follows from our choice of $\mathcal{C}$ to be an Ω -periodic covering where $\delta(\mathcal{C}) < \theta(\Omega) + \varepsilon$. To finish the proof, we show that the remaining region $Z_t = A_t \setminus A_{t, D_V + D_\Omega}$ satisfies

$$\limsup_{t \rightarrow 0} N(Z_t; \Omega) \cdot t^\kappa = 0.$$

This follows because Z_t is nearly the $(\kappa - 1)$ -dimensional object ∂A_t . It has just been fattened up by a constant amount $D_V + D_\Omega$, which is independent of the size of the box $1/t$.

Let's formally show that $N(Z_t, \Omega) = O(t^{-(\kappa-1)})$. As $\mathbf{0}$ is in the interior of Ω , for some $r > 0$, this set contains some r -box under the ℓ_∞ -distance,

$$[-r, r]^\kappa \subset \Omega.$$

The region Z_t is a thin shell at the boundary ∂A_t with thickness $D_V + D_\Omega$. The number of $(\kappa - 1)$ -dimensional faces of the box A_t is 2κ ; we can tile Z_t with r -boxes centered in Z_t using no more than

$$2\kappa \times \left\lceil \frac{1}{tr} \right\rceil^{\kappa-1} \times \left\lceil \frac{D_V + D_\Omega}{r} \right\rceil \text{ } r\text{-boxes.}$$

It follows that $N(Z_t, \Omega) = O(t^{-(\kappa-1)})$ grows at a rate dominated by $t^{-\kappa}$. The result follows, since

$$\begin{aligned} \limsup_{t \rightarrow 0} N(A; t\Omega) \cdot t^\kappa &= \limsup_{t \rightarrow 0} N(A_t; \Omega) \cdot t^\kappa \\ &\leq \limsup_{t \rightarrow 0} N(A_{t, D_V + D_\Omega}; \Omega) \cdot t^\kappa + \limsup_{t \rightarrow 0} N(Z_t; \Omega) \cdot t^\kappa < \frac{\theta(\Omega) + \varepsilon}{\text{Vol}(\Omega)}, \end{aligned}$$

using the fact that $N(Z_t, \Omega) \cdot t^\kappa = O(t)$ goes to zero with t .

We can also compute the asymptotic covering number of boxes with any length $\ell A = [0, \ell]^\kappa$, since

$$\lim_{t \rightarrow 0} N(\ell A; t\Omega) \cdot t^\kappa = \lim_{t \rightarrow 0} N(A; (t/\ell)\Omega) \cdot (t/\ell)^\kappa \cdot \ell^\kappa = \frac{\theta(\Omega) \cdot \text{Vol}(\ell A)}{\text{Vol}(\Omega)}.$$

F.4.9. PROOF OF PROPOSITION 5

Let $A \subset \mathbb{R}^\kappa$ be Jordan measurable. Let $D_{\text{in}}, D_{\text{out}} > 0$ satisfy $B_\infty(\mathbf{0}, D_{\text{in}}) \subset \Omega \subset B_\infty(\mathbf{0}, D_{\text{out}})$.

Lower bound. Since A is Jordan measurable (recall Definition 11), there is a countable disjoint union of boxes $B_i \subset A$ covering almost all of A ,

$$\text{Vol}\left(A \setminus \bigcup_{i \in \mathbb{N}} B_i\right) = 0,$$

where each $B_i = B_\infty(\mathbf{v}_i, r_i)$ is an ℓ_∞ -ball. As it is enough to cover the interior of A , this follows from the Whitney covering lemma (Grafakos, 2008). Without loss of generality, we can assume that $r_i \leq 1$. For each $t > 0$, let $S_t \subset A$ be a set such that the collection $\{t\Omega + \mathbf{s} : \mathbf{s} \in S_t\}$ forms a minimal covering of A by translations of $t\Omega$. We now show that there is a constant C depending only on Ω such that for each box B_i , the size of the collection $B_i \cap S_t$ is bounded:

$$|B_i \cap S_t| \geq N(B_i, t\Omega) - Ct^{-(\kappa-1)}.$$

To see this, we augment the centers in $B_i \cap S_t$ by a set T_i so that $\{t\Omega + \mathbf{s} : \mathbf{s} \in (B_i \cap S_t) \cup T_i\}$ covers B_i . The centers in $B_i \cap S_t$ must cover all points in B_i , except possibly those within a distance of D_{out} to the boundary of B_i (since those may have been covered by simplices with centers outside B_i). Thus, to ensure that we cover all of B_i , we just need to cover a thin shell near the boundary of B_i . There are 2κ faces of the ball B_i , and for each face, we need to cover an $r_i \times \cdots \times r_i \times D_{\text{out}}$ box. Recall that we restricted $r_i \leq 1$. Thus, there is a choice of T_i such that:

$$|T_i| \leq 2\kappa \cdot \left\lceil \frac{D_{\text{out}}}{D_{\text{in}}} \right\rceil \cdot \left\lceil \frac{r_i}{tD_{\text{in}}} \right\rceil^{\kappa-1} \leq Ct^{-(\kappa-1)},$$

where C is a constant depending only on Ω . We use this to show that:

$$\liminf_{t \rightarrow 0} N(A; t\Omega) \cdot t^\kappa \geq \frac{\theta(\Omega) \cdot \text{Vol}(A)}{\text{Vol}(\Omega)}. \quad (30)$$

For any $\varepsilon > 0$, choose $M_\varepsilon \in \mathbb{N}$ sufficiently large so that

$$\text{Vol} \left(A \setminus \bigcup_{i=1}^{M_\varepsilon} B_i \right) < \varepsilon.$$

For each $t > 0$, we obtain the chain of inequalities:

$$N(A; t\Omega) = |S_t| \geq \sum_{i=1}^{M_\varepsilon} |B_i \cap S_t| \geq \sum_{i=1}^{M_\varepsilon} N(B_i; t\Omega) - CM_\varepsilon t^{-(\kappa-1)}.$$

Multiplying through by t^κ and taking a limit infimum, we apply Lemma 18 to obtain

$$\liminf_{t \rightarrow 0} N(A; t\Omega) \cdot t^\kappa \geq \sum_{i=1}^{M_\varepsilon} \frac{\theta(\Omega) \cdot \text{Vol}(B_i)}{\text{Vol}(\Omega)} \geq \frac{\theta(\Omega) \cdot (\text{Vol}(A) - \varepsilon)}{\text{Vol}(\Omega)},$$

where the last step uses $\sum_i \text{Vol}(B_i) \geq \text{Vol}(A) - \varepsilon$. Since this holds for all $\varepsilon > 0$, Equation (30) holds.

Upper bound. Fix $r > 0$ and consider a tessellation of $\mathbb{R}^\kappa$ by r -boxes $B_\infty(\mathbf{v}, r)$. Because A is bounded, there is a finite number of such boxes that are fully contained in A . Let us call these $B_1, \dots, B_N$, and let Z denote the remainder

$$Z = A \setminus \bigcup_{i=1}^N B_i.$$

For each $s > 0$, let ∂A^s denote the s -expansion of ∂A , consisting of the points whose distance from ∂A is less than s under the ℓ_∞ -distance. Notice that if $\mathbf{a} \in A \setminus \partial A^{2r}$ has ℓ_∞ -distance more than $2r$ from ∂A , then every r -box containing $\mathbf{a}$ is fully contained in A . Thus, $Z \subset \partial A^{2r}$.

For every $t > 0$, the following holds:

$$N(A; t\Omega) \leq \sum_{i=1}^N N(B_i; t\Omega) + N(Z; t\Omega).$$

Each $N(B_i; t\Omega)$ is bounded by Lemma 18. To bound $N(Z; t\Omega)$, we construct a $t\Omega$ -covering of Z . To this end, let $S_{t;Z} \subset Z$ be a maximal tD_{in} -packing under the ℓ_∞ -norm. Since $S_{t;Z}$ is maximal, this means that every point in Z must be within a distance of tD_{in} of a center in $S_{t;Z}$. As $t\Omega \supset B_\infty(\mathbf{0}, tD_{\text{in}})$, this implies that this set is a $t\Omega$ -covering of Z .

To bound the size of $|S_{t;Z}|$, note that as $S_{t;Z}$ form a tD_{in} -packing, the collection of boxes with ℓ_∞ -radii $tD_{\text{in}}/2$ centered at $S_{t;Z}$ do not overlap. And because the centers of these boxes are contained in $Z \subset \partial A^{2r}$, each of these boxes must be contained in the larger expansion $\partial A^{2r+tD_{\text{in}}}$. It follows that:

$$|S_{t;Z}| \cdot (tD_{\text{in}})^\kappa \leq \text{Vol}(\partial A^{2r+tD_{\text{in}}}).$$

Rearranging, we obtain an upper bound on $|S_{t;Z}| \cdot t^\kappa$. Putting these upper bounds together, we obtain:

$$\begin{aligned} N(A; t\Omega) \cdot t^\kappa &\leq \sum_{i=1}^N N(B_i; t\Omega) \cdot t^\kappa + |S_{t;Z}| \cdot t^\kappa \\ &\leq \sum_{i=1}^N N(B_i; t\Omega) \cdot t^\kappa + \frac{1}{D_{\text{in}}^\kappa} \cdot \text{Vol}(\partial A^{2r+tD_{\text{in}}}). \end{aligned}$$

By the continuity of the Lebesgue measure, the limit of the right-hand side exists as $t \rightarrow 0$, and we obtain:

$$\begin{aligned} \limsup_{t \rightarrow 0} N(A; t\Omega) \cdot t^\kappa &\leq \sum_{i=1}^N \frac{\theta(\Omega) \cdot \text{Vol}(B_i)}{\text{Vol}(\Omega)} + \frac{1}{D_{\text{in}}^\kappa} \cdot \text{Vol}(\partial A^{2r}) \\ &\leq \frac{\theta(\Omega) \cdot \text{Vol}(A)}{\text{Vol}(\Omega)} + \frac{1}{D_{\text{in}}^\kappa} \cdot \text{Vol}(\partial A^{2r}), \end{aligned}$$

where the first inequality uses Lemma 18, and the second inequality uses the fact that the B_i are disjoint subsets of A . Finally, because ∂A is a compact set, we have that $\partial A^{2r} \downarrow \partial A$ as $r \rightarrow 0$. By the continuity of the Lebesgue measure, we also have that $\text{Vol}(\partial A^{2r}) \rightarrow \text{Vol}(\partial A) = 0$, as the boundary has measure zero by assumption. This yields the upper bound:

$$\limsup_{t \rightarrow 0} N(A; t\Omega) \cdot t^\kappa \leq \frac{\theta(\Omega) \cdot \text{Vol}(A)}{\text{Vol}(\Omega)}.$$

This upper bound along with the lower bound Equation (30) gives the result.

F.4.10. PROOF OF LEMMA 20

Let $\mathcal{H} = \{\mathbf{v} \in \mathbb{R}^K : \mathbf{v} \cdot \mathbf{n} = 0\}$ be the hyperplane through the origin. Parameterize $\mathcal{H}$ via the map $\Phi : \mathbb{R}^{K-1} \rightarrow \mathbb{R}^K$ defined as $\Phi(\mathbf{u}) = (u_1, \dots, u_{K-1}, -\frac{1}{n_K} \sum_{k=1}^{K-1} u_k n_k)$. Then the Hausdorff measure $\mathcal{H}^{K-1}$ on $\mathcal{H}$ can be written as $\mathcal{H}^{K-1}(\Phi(A)) = \sqrt{\det((J\Phi)^\top(J\Phi))} \text{Vol}(A)$ (see Theorem 3.2.3 in (Federer, 1996)), where the Jacobian is given by

$$J\Phi = \begin{pmatrix} \mathbf{I}_{K-1} & \\ -n_1/n_K & \dots & -n_{K-1}/n_K \end{pmatrix} \in \mathbb{R}^{K \times (K-1)}.$$

A computation then yields that $\sqrt{\det((J\Phi)^\top(J\Phi))} = 1/n_K$, which implies $\mathcal{H}^{K-1}(\Phi(A)) = \frac{1}{n_K} \text{Vol}(A)$. One can now verify that the preimage of $\Omega_{\mathbf{n}}$ under Φ is given by the set

$$A_t = \left\{ \mathbf{a} \in \mathbb{R}^{K-1} : a_k \geq -t, \sum_{k=1}^{K-1} a_k n_k \leq t n_K \right\} = \left\{ \mathbf{a} \in \mathbb{R}^{K-1} : a_k \geq 0, \sum_{k=1}^{K-1} a_k n_k \leq t \sum_{k=1}^K n_k \right\} - t\mathbf{1}.$$

Another computation using the Jacobian of the map $\mathbf{a} \mapsto \sum_{k=1}^{K-1} n_k a_k$ yields that the volume of this set is given by

$$\mathcal{H}^{K-1}(\Omega_{\mathbf{n}}) = \frac{1}{n_K} \text{Vol}(A_t) = \frac{t^{K-1}}{(K-1)!} \frac{\left(\sum_{k=1}^K n_k\right)^{K-1}}{\prod_{k=1}^K n_k}.$$

That yields the claim.

G. Deferred Proofs

G.1. Proofs for Monotonicity and Pareto Optimality

G.1.1. PROOF OF LEMMA 1

Fix any $f \in \mathcal{F}$ and set $\mathbf{r} = \mathbf{R}(f) \in \mathbf{R}(\mathcal{F})$. Consider the set

$$S = \{\mathbf{u} \in \mathbf{R}(\mathcal{F}) : \mathbf{u} \preceq \mathbf{r}\} = \mathbf{R}(\mathcal{F}) \cap \prod_{k=1}^K (-\infty, r_k],$$

where $\prod_{k=1}^K (-\infty, r_k]$ is the Cartesian product of sets. Clearly $S \neq \emptyset$ since $\mathbf{r} \in S$. Because $\mathbf{R}(\mathcal{F})$ is compact by our standing assumption, and S is a closed subset of $\mathbf{R}(\mathcal{F})$, the set S is also compact. Define the continuous scalarization $s : \mathbb{R}^K \rightarrow \mathbb{R}$ as $s(\mathbf{u}) = \sum_{k=1}^K u_k$. By compactness of S , there exists $\mathbf{u}^* \in S$ such that

$$\mathbf{u}^* \in \arg \min_{\mathbf{u} \in S} s(\mathbf{u}).$$

Since $\mathbf{u}^* \in \mathbf{R}(\mathcal{F})$, there exists $f' \in \mathcal{F}$ with $\mathbf{R}(f') = \mathbf{u}^*$. By construction, $\mathbf{u}^* \preceq \mathbf{r}$, hence $\mathbf{R}(f') \preceq \mathbf{R}(f)$.

It remains to show that f' is Pareto optimal. Suppose otherwise. Then there exists $f'' \in \mathcal{F}$ such that $\mathbf{R}(f'') \prec \mathbf{R}(f') = \mathbf{u}^*$. In particular, $\mathbf{R}(f'') \preceq \mathbf{u}^* \preceq \mathbf{r}$, so $\mathbf{R}(f'') \in S$. Moreover, $s(\mathbf{R}(f'')) < s(\mathbf{u}^*)$ since at least one coordinate is strictly smaller and none is larger, contradicting the minimality of $\mathbf{u}^*$ over S . Therefore $f' \in \text{Par}(\mathcal{F}, \mathbf{R})$ and $\mathbf{R}(f') \preceq \mathbf{R}(f)$. Since f was arbitrary, the claim follows.

Now let $f^* \in \arg \min_{f \in \mathcal{F}} \mathcal{R}_Q(f)$, which exists by our standing assumptions. Then there exists a f with $\mathbf{R}(f) \preceq \mathbf{R}(f^*)$ and $f \in \text{Par}(\mathcal{F}, \mathbf{R})$ by the previous argument. By monotonicity, we hence know that $\mathcal{R}_Q(f) \leq \mathcal{R}_Q(f^*)$, and f is also a minimizer of $\mathcal{R}_Q$. That concludes the proof.

G.1.2. PROOF OF LEMMA 2

For any $\mathbf{r} \in \text{range}(\mathbf{R})$ with some $f \in \mathcal{F}$ satisfying $\mathbf{R}(f) = \mathbf{r}$, define s as

$$s(\mathbf{r}) = \mathcal{R}_Q(f),$$

and let s be arbitrarily monotonic elsewhere. The map s is well-defined: for any f, f' such that $\mathbf{R}(f) = \mathbf{R}(f')$, we have by monotonicity

$$\mathcal{R}_Q(f) \leq \mathcal{R}_Q(f') \quad \text{and} \quad \mathcal{R}_Q(f') \leq \mathcal{R}_Q(f),$$

in which case $\mathcal{R}_Q(f) = \mathcal{R}_Q(f')$. By construction, we have that $\mathcal{R}_Q = s \circ \mathbf{R}$.

G.2. Proof of Theorem 2

We split the proof into the bound for Pareto ERM and the bound for Pareto EW.

G.2.1. PROOF FOR PARETO ERM

Fix $t > 0$, let $N = N_{\text{Par}}(t, \mathcal{F}, \mathbf{R})$ and let $f_1, \dots, f_N$ be the minimal t -Pareto set in $\mathcal{F}$ on which we compute the Pareto ERM $\hat{f}_{\text{Par}(t)}$. By Definition 5, this means that for every $f \in \mathcal{F}$ there is a f_j with $\mathcal{R}_k(f_j) \leq \mathcal{R}_k(f) + t$ for all k . By Definition 4, we know that hence $\mathcal{R}_Q(f_j) \leq \mathcal{R}_Q(f) + \bar{\omega}_Q(t)$. Now consider the event

$$\mathcal{E} = \left\{ \max_{j \in [N]} \left| \hat{\mathcal{R}}_Q(f_j) - \mathcal{R}_Q(f_j) \right| \leq r \right\} \quad \text{with} \quad r = \sqrt{\frac{\log(2N/\delta)}{2n}}.$$

Recalling that $\hat{\mathcal{R}}_Q(f) = \frac{1}{n} \sum_{i=1}^n \ell(f, z_i)$, we may apply Hoeffding's bound to these i.i.d. $[0, 1]$ -valued random variables $\ell(f, z_i)$ with expectation $\mathcal{R}_Q(f)$. In particular, we get that

$$\mathbb{P}(\mathcal{E}^c) \leq \sum_{j \in [N]} \mathbb{P} \left(\left| \sum_{i=1}^n \ell(f_j, z_i) - n\mathcal{R}_Q(f_j) \right| > nr \right) \leq 2N \exp \left(-\frac{2n^2 r^2}{n} \right) = \delta$$

and so, $\mathbb{P}(\mathcal{E}) \geq 1 - \delta$. Let $f^* \in \arg \min_{f \in \mathcal{F}} \mathcal{R}_Q(f)$. On $\mathcal{E}$, for $i = i(f^*)$ being the index in the covering that corresponds to f^* (meaning that $\mathcal{R}_Q(f_i) \leq \mathcal{R}_Q(f^*) + \bar{\omega}_Q(t)$), we get that

$$\begin{aligned} & \mathcal{R}_Q(\hat{f}_{\text{Par}(t)}) - \mathcal{R}_Q(f^*) \\ & \leq \underbrace{\mathcal{R}_Q(\hat{f}_{\text{Par}(t)}) - \hat{\mathcal{R}}_Q(\hat{f}_{\text{Par}(t)})}_{\leq r} + \underbrace{\hat{\mathcal{R}}_Q(\hat{f}_{\text{Par}(t)}) - \hat{\mathcal{R}}_Q(f_i)}_{\leq 0} + \underbrace{\hat{\mathcal{R}}_Q(f_i) - \mathcal{R}_Q(f_i)}_{\leq r} + \underbrace{\mathcal{R}_Q(f_i) - \mathcal{R}_Q(f^*)}_{\leq \bar{\omega}_Q(t)}, \end{aligned}$$

where we used that $\hat{f}_{\text{Par}(t)}$ is the empirical risk minimizer on the Pareto covering. And so, $\mathcal{R}_Q(\hat{f}_{\text{Par}(t)}) - \mathcal{R}_Q(f^*) \leq 2r + \bar{\omega}_Q(t)$, concluding the proof.

G.2.2. PROOF FOR PARETO EW

Throughout the proof, we assume that $\mathfrak{R}$ is a nice Pareto front (Assumption 1), so that the limiting distribution μ computed in Theorem 1 is well-defined. Recall μ induces the prior π over the Pareto set, obtained by pulling back and normalizing,¹¹

$$\pi = \frac{1}{\mu(\mathfrak{R})} \cdot \mathbf{R}^* \mu,$$

so that $\pi(A) \propto \mu(\mathbf{R}(A))$ whenever $A \subset \text{Par}(\mathcal{F}, \mathbf{R})$ is a measurable set.

We begin with an observation, which relates the Pareto covering number and the limiting distribution. First, we lower bound the mass of near-optimal models close to $f^* \in \arg \min_{f \in \mathcal{F}} \mathcal{R}_Q(f)$, where recall that we assumed $f^* \in \text{Par}(\mathcal{F}, \mathbf{R})$. In particular, for any $t > 0$, define the following subset:

$$\mathcal{N}_t := \{f \in \text{Par}(\mathcal{F}, \mathbf{R}) : \mathbf{R}(f) \leq \mathbf{R}(f^*) + t\mathbf{1}\}. \quad (31)$$

¹¹Technically, the distribution is defined on the quotient space $\text{Par}(\mathcal{F}, \mathbf{R})/\mathbf{R}$. But, without loss of generality, we can assign a canonical representative to each class, and assume that $\mathbf{R}$ is injective.

Lemma 21 (Mass on near-optimal models). *Let $\mathfrak{R}$ be a nice Pareto front and let $\mathbf{R}(f^*)$ be in the relative interior of $\mathfrak{R}$. For every $\gamma > 0$, there is some $t_0 > 0$ such that for all $t \in (0, t_0]$,*

$$\pi(\mathcal{N}_t) \geq \frac{1 - \gamma}{N_{\text{Par}}(t, \mathcal{F}, \mathbf{R})}. \quad (32)$$

Proof. The set $\mathbf{R}(\mathcal{N}_t)$ is a Pareto ball in $\mathfrak{R}$. Let $\mathbf{r}^* = \mathbf{R}(f^*)$. Then $\mathbf{R}(\mathcal{N}_t) = B_t$, where:

$$B_t := \{\mathbf{r} \in \mathfrak{R} : d(\mathbf{r}^*, \mathbf{r}) \leq t\} \quad \text{and} \quad \pi(\mathcal{N}_t) = \frac{\mu(B_t)}{\mu(\mathfrak{R})}.$$

We hence proceed by lower bounding the volume of Pareto balls centered at $\mathbf{r}^*$ in $\mathfrak{R}$ by linearizing the manifold at $\mathbf{r}^*$. Let $\mathbf{n}^* = \mathbf{n}(\mathbf{r}^*)$, where $\mathbf{n}(\mathbf{r})$ is the normal vector of $\mathfrak{R}$ at $\mathbf{r}$. Niceness imposes a lower bound $n_k(\mathbf{r}) > \eta > 0$ for all $k \in [K]$. Also recall that $\beta(\mathbf{n})$ is defined in Equation (21) as:

$$\beta(\mathbf{n}) = \theta_K \cdot (K - 1)! \cdot \frac{\prod_{k \in [K]} n_k}{(\mathbf{1} \cdot \mathbf{n})^{K-1}},$$

where θ_K is the covering density of $(K - 1)$ -dimensional simplices (see Lemma 19). Since the covering density is lower bounded $\theta_K \geq 1$ (Rogers, 1957), by niceness we have that $\beta(\mathbf{n}^*) > 0$ is bounded away from zero.

For any $\varepsilon > 0$, Lemma 14 shows that there exists a smooth diffeomorphism $\phi : U \rightarrow V$ centered at $\mathbf{r}^*$, where $U \subset \mathfrak{R}$ is relatively open in $\mathfrak{R}$ containing $\mathbf{r}^*$ and $V \subset \{\mathbf{v} \in \mathbb{R}^K : \mathbf{v} \cdot \mathbf{n}^* = 0\}$ is in the linear subspace orthogonal to $\mathbf{n}^*$. The diffeomorphism can be constructed so that $\phi(\mathbf{r}^*) = \mathbf{r}^*$, the map ϕ is an ε -distortion, and it is locally volume-preserving at $\mathbf{r}^*$, where the determinant of the Jacobian is $|J\phi(\mathbf{r})| = 1$. Define the ball $\tilde{B}_{t,\varepsilon}$ by:

$$\tilde{B}_{t,\varepsilon} := \{\mathbf{r} \in V : d(\mathbf{r}^*, \mathbf{r}) < t/(1 + \varepsilon)\}.$$

Because $\mathbf{r}^*$ is in the relative interior of $\mathfrak{R}$ and ϕ is an ε -distortion, we have that $\tilde{B}_{t,\varepsilon} \subset \phi(B_t)$.

We now lower bound $\mu(B_t)$ using the definition of μ from Theorem 1, given by $\mu(B_t) = \int_{B_t} \beta(\mathbf{n}(\mathbf{r})) \, d\mathcal{H}(\mathbf{r})$. The functions $\phi(\mathbf{r})$ and $\beta(\mathbf{n}(\mathbf{r}))$ are smooth, so for any $\gamma > 0$, by shrinking U to a sufficiently small relatively open set around $\mathbf{r}^*$, we can ensure that

$$|J\phi(\mathbf{r})| > 1 - \gamma \quad \text{and} \quad \beta(\mathbf{n}(\mathbf{r})) > (1 - \gamma) \cdot \beta(\mathbf{n}^*), \quad (33)$$

where the second inequality is possible because $\beta(\mathbf{n}^*) > 0$. We have:

$$\begin{aligned} \mu(B_t) &\geq (1 - \gamma) \cdot \beta(\mathbf{n}^*) \cdot \int_{\phi(B_t)} |J\phi(\mathbf{r})| \, d\mathcal{H}(\mathbf{r}) \\ &\geq (1 - 2\gamma) \cdot \beta(\mathbf{n}^*) \cdot \mathcal{H}(\phi(B_t)) \\ &\geq (1 - 2\gamma) \cdot \beta(\mathbf{n}^*) \cdot \mathcal{H}(\tilde{B}_{t,\varepsilon}), \end{aligned}$$

where the first two inequalities follow from Eq. (33) and the fact that $(1 - \gamma)(1 - \gamma') \geq 1 - (\gamma + \gamma')$ whenever $\gamma, \gamma' \geq 0$, and the last inequality uses the fact that $\phi(B_t)$ contains $\tilde{B}_{t,\varepsilon}$. Now Lemma 20 implies that

$$\mathcal{H}(\tilde{B}_{t,\varepsilon}) = \frac{1}{(K - 1)!} \frac{(\mathbf{1} \cdot \mathbf{n}^*)^{K-1}}{\prod_{k \in [K]} n_k^*} \cdot \left(\frac{t}{1 + \varepsilon} \right)^{K-1} \geq (1 - \gamma) \cdot \frac{1}{(K - 1)!} \frac{(\mathbf{1} \cdot \mathbf{n}^*)^{K-1}}{\prod_{k \in [K]} n_k^*} \cdot t^{K-1},$$

where the inequality holds for $\varepsilon < \log(1/\gamma)/K$, since $1/(1 + \varepsilon)^{K-1} \geq 1/e^{\varepsilon \cdot (K-1)}$. Continuing from the above inequality, we obtain:

$$\begin{aligned} \mu(B_t) &\geq (1 - 2\gamma) \cdot \beta(\mathbf{n}^*) \cdot \mathcal{H}(\tilde{B}_{t,\varepsilon}) \\ &= (1 - 2\gamma) \cdot \beta(\mathbf{n}^*) \cdot (1 - \gamma) \cdot \frac{1}{(K - 1)!} \frac{(\mathbf{1} \cdot \mathbf{n}^*)^{K-1}}{\prod_{k \in [K]} n_k^*} \cdot t^{K-1} \\ &\geq (1 - 3\gamma) \cdot \theta_K \cdot t^{K-1}, \end{aligned}$$

where the last inequality again uses the fact that $(1 - \gamma)(1 - \gamma') \geq 1 - (\gamma + \gamma')$. Finally, Theorem 1 also shows that:

$$\lim_{t \rightarrow 0} N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}) \cdot t^{K-1} = \mu(\mathfrak{R}),$$

so that for sufficiently small t , we have that $\mu(\mathfrak{R}) \leq (1 + \gamma) \cdot N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}) \cdot t^{K-1}$. We obtain that:

$$\frac{\mu(B_t)}{\mu(\mathfrak{R})} \geq \frac{(1 - 3\gamma) \cdot \theta_K \cdot t^{K-1}}{(1 + \gamma) \cdot N_{\text{Par}}(t, \mathcal{F}, \mathfrak{R}) \cdot t^{K-1}} \geq (1 - 4\gamma) \cdot \frac{\theta_K}{N_{\text{Par}}(t, \mathcal{F}, \mathfrak{R})},$$

where we use the fact that for $\gamma < 1$, that $1/(1 + \gamma) \geq 1 - \gamma$.

The result then follows from the lower bound $\theta_K \geq 1$, and reparametrizing γ . $\square$

We now prove the bound for Pareto EW from Theorem 2. Fix $t \in (0, 1]$ and recall $\mathcal{N}_t$ from Equation (31). We just established by Lemma 21 that the mass $\pi(\mathcal{N}_t)$ is bounded away from zero. Recall the definition of Pareto EW from Section 4.1 as the aggregated model $f_{\hat{\rho}} = \mathbb{E}_{f \sim \rho}[f]$ using the Gibbs posterior $\hat{\rho}$ from Eq. (2). We now show that for any $\delta \in (0, 1)$ and $\lambda > 0$, with probability at least $1 - \delta$

$$\mathcal{R}_Q(f_{\hat{\rho}}) - \inf_{f \in \mathcal{F}} \mathcal{R}_Q(f) \leq \bar{\omega}_Q(t) + \frac{\lambda}{4n} + \frac{2 \log(2/(\delta \pi(\mathcal{N}_t)))}{\lambda}. \quad (34)$$

Indeed, by Theorem 4.2 in Alquier (2024), the exponential weights estimator satisfies with probability at least $1 - \delta$ and for every $\lambda > 0$,

$$\mathbb{E}_{f \sim \hat{\rho}}[\mathcal{R}_Q(f)] \leq \inf_{\rho \ll \pi} \left\{ \mathbb{E}_{f \sim \rho}[\mathcal{R}_Q(f)] + \frac{\lambda}{4n} + \frac{2(\text{KL}(\rho, \pi) + \log(\frac{2}{\delta}))}{\lambda} \right\}.$$

Now, define the localized distribution

$$\rho_t(A) := \pi(A \mid \mathcal{N}_t) = \frac{\pi(A \cap \mathcal{N}_t)}{\pi(\mathcal{N}_t)}.$$

Then ρ is absolutely continuous with respect to π and $\text{KL}(\rho_t, \pi) = \int \log\left(\frac{d\rho_t}{d\pi}\right) d\rho_t = \log\left(\frac{1}{\pi(\mathcal{N}_t)}\right)$. Then the infimum is upper bounded by the evaluation at this specific ρ_t so that we obtain

$$\mathbb{E}_{f \sim \hat{\rho}}[\mathcal{R}_Q(f)] \leq \mathbb{E}_{f \sim \rho_t}[\mathcal{R}_Q(f)] + \frac{\lambda}{4n} + \frac{2 \log(2/(\delta \pi(\mathcal{N}_t)))}{\lambda}.$$

Next, since ρ_t is supported on $\mathcal{N}_t$, by the definition of the upper modulus (Definition 4, applied with comparator f^*) we have for all $f \in \mathcal{N}_t$, $\mathcal{R}_Q(f) \leq \mathcal{R}_Q(f^*) + \bar{\omega}_Q(t)$, and therefore

$$\mathbb{E}_{f \sim \rho_t}[\mathcal{R}_Q(f)] \leq \mathcal{R}_Q(f^*) + \bar{\omega}_Q(t).$$

Combining the last two displays yields

$$\mathbb{E}_{f \sim \hat{\rho}}[\mathcal{R}_Q(f)] \leq \mathcal{R}_Q(f^*) + \bar{\omega}_Q(t) + \frac{\lambda}{4n} + \frac{2 \log(2/(\delta \pi(\mathcal{N}_t)))}{\lambda},$$

for all $\lambda > 0$ and with probability at least $1 - \delta$. Finally, since the loss defining $\mathcal{R}_Q$ is convex in the prediction, Jensen's inequality implies that the aggregated predictor satisfies $\mathcal{R}_Q(f_{\hat{\rho}}) = \mathcal{R}_Q(\mathbb{E}_{f \sim \hat{\rho}}[f]) \leq \mathbb{E}_{f \sim \hat{\rho}}[\mathcal{R}_Q(f)]$. Therefore the same bound holds with $\mathcal{R}_Q(f_{\hat{\rho}})$ on the left-hand side, and recalling that $\inf_{f \in \mathcal{F}} \mathcal{R}_Q(f) = \mathcal{R}_Q(f^*)$ concludes the proof of Eq. (34).

Finally, recall that by Lemma 21 we have that for a sufficiently small $t_0 > 0$

$$\pi(\mathcal{N}_t) \geq \frac{1}{2N_{\text{Par}}(t, \mathcal{F}, \mathfrak{R})}, \quad \forall t \in (0, t_0].$$

We can hence upper bound the log term $\log\left(\frac{2}{\delta \pi(\mathcal{N}_t)}\right) \leq \log\left(\frac{4N_{\text{Par}}(t, \mathcal{F}, \mathbf{R})}{\delta}\right)$. Finally, optimizing the right-hand side of Eq. (34) (where we upper bounded as above the log term) over $\lambda > 0$ yields the choice

$$\lambda = \sqrt{8n \log\left(\frac{4N_{\text{Par}}(t, \mathcal{F}, \mathbf{R})}{\delta}\right)},$$

which gives the bound for the Pareto EW stated in Theorem 2.

G.3. Proof of Corollary 1

The first part of the proof is a direct consequence of Theorem 2. Indeed, recall that t_n is defined in Corollary 1 as

$$t_n = \inf \left\{ t > 0 : t^2 \geq \frac{2 \log(2N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}))}{L^2 n} \right\}.$$

By definition of the Pareto covering number, the function $t \mapsto 2 \log(2N_{\text{Par}}(t, \mathcal{F}, \mathbf{R})) / (L^2 n)$ is non-increasing, and so for any $t > t_n$, it holds that

$$t^2 \geq \frac{2 \log(2N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}))}{L^2 n}.$$

Combining this with Theorem 2, we obtain that, for any $t > t_n$,

$$\mathcal{R}_Q(\hat{f}_{\text{Par}(t)}) - \inf_{f \in \mathcal{F}} \mathcal{R}_Q(f) \leq \bar{\omega}_Q(t) + \sqrt{\frac{2 \log(2N_{\text{Par}}(t, \mathcal{F}, \mathbf{R})/\delta)}{n}} \leq Lt + Lt + \sqrt{\frac{2 \log(1/\delta)}{n}}.$$

As this is valid for all $t > t_n$, we can conclude the first part of the proof.

For the second part, note that the assumption $L^2 n > 2 \log 2$ together with the fact that $\mathbf{R}(\mathcal{F}) \subset [0, 1]^K$ implies that

$$\frac{2 \log(2N_{\text{Par}}(1, \mathcal{F}, \mathbf{R}))}{L^2 n} = \frac{2 \log 2}{L^2 n} \leq 1.$$

As a consequence, $t_n \leq 1$ and we can make the following case distinction based on the value of $\theta := \frac{K-1}{L^2 n} \log \left(1 + \frac{16L^2 n}{K-1} \right)$.

- If $\theta \geq 1$ the claimed bound from Corollary 1 becomes $t_n^2 \leq 1 \wedge \theta = 1$, which we already verified.
- If $\theta < 1$, Lemma 9 implies that $\log N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}) \leq (K-1) \log(1 + \frac{1}{t})$. Thus, any solution to

$$t^2 \geq \frac{2 \log(2(1 + 1/t)^{K-1})}{L^2 n}$$

also upper bounds t_n . Now, write $a = 2(K-1)/(L^2 n)$ and $b = 2^{K/(K-1)}$. We claim that $t^2 = \frac{a}{2} W\left(\frac{2b^2}{a}\right)$ satisfies the inequality above, where W denotes the Lambert W -function. Indeed,

$$t^2 = \frac{a}{2} W\left(\frac{2b^2}{a}\right) \leq \frac{a}{2} \log\left(1 + \frac{2b^2}{a}\right) = \frac{K-1}{L^2 n} \log\left(1 + \frac{L^2 n 2^{2\frac{K}{K-1}}}{K-1}\right) \leq \theta, \quad (35)$$

where we used $W(x) \leq \log(1+x)$ for $x > 0$. Furthermore, $\theta < 1$ implies $t < 1$ and, consequently,

$$\frac{2 \log(2(1 + 1/t)^{K-1})}{L^2 n} \leq \frac{2 \log(2^K/t^{K-1})}{L^2 n} = \frac{2(K-1)}{L^2 n} \log\left(\frac{2^{K/(K-1)}}{t}\right) = a \log\left(\frac{b}{t}\right).$$

Finally, by definition of the Lambert W -function, we know that our choice of t^2 satisfies $t^2 = a \log(b/t)$, and the claim follows. Hence, by Eq. (35) we have

$$t_n^2 \leq t^2 \leq \theta = \frac{K-1}{L^2 n} \log\left(1 + \frac{16L^2 n}{K-1}\right).$$

Putting both cases together yields the bound $t_n^2 \leq 1 \wedge \theta$, concluding the proof. Note that, by going through the same calculations, one can derive an analogous bound for the Pareto EW.

G.4. Proof of Theorem 3

The high-level road map for the proof of the lower bound is the following. We construct a set of N points $\mathbf{r} \in [0, 1]^K$, which allows us to choose a function f such that $\mathbf{R}(f) = \mathbf{r}$ for each point $\mathbf{r}$. We choose the points so that for any pair $\mathbf{r}, \mathbf{r}'$, neither point lies in the t -Pareto ball centered at the other, that is, their pairwise Pareto distance exceeds t . Further, we add two dominated points separated by exactly $t\mathbf{1}$ along the diagonal. These points do not change the t -Pareto covering number, but they force the modulus at scale t to be exactly t . Finally, we construct a loss and a family of distributions $\mathcal{Q}$ in a way that for each point $\mathbf{r}$, there is a corresponding distribution $Q \in \mathcal{Q}$ under which the associated hypothesis is the unique risk minimizer and every other point has excess risk at least Ct . Estimating a nearly optimal model therefore induces a multiple testing problem, which is lower bounded by a standard application of Fano's inequality.

We begin by noting that, by the assumptions on n and t , we have $t \leq 1/60$. Let $m \in [8]$ and define the values $v_m = \frac{1}{2} + 8t(m - \frac{9}{2})$. Then, $v_m \in [0, 1]$ since $t \leq 1/56$ and $|v_m - \frac{1}{2}| \leq 8t\frac{7}{2} = 28t \leq \frac{1}{2}$. For $\mathbf{a} \in [8]^{K-1}$, define the points in risk space

$$\mathbf{r}(\mathbf{a}) := (v_{a_1}, \dots, v_{a_{K-1}}, \frac{1}{2}) \in [0, 1]^K.$$

For the sum $S(\mathbf{a}) = \sum_{i=1}^{K-1} a_i$, let $A_s = \{\mathbf{a} \in [8]^{K-1} : S(\mathbf{a}) = s\}$; if $\mathbf{a} \prec \mathbf{a}'$ then clearly $S(\mathbf{a}) < S(\mathbf{a}')$. Thus, for all $\mathbf{a}, \mathbf{a}' \in A_s$ it cannot be that $\mathbf{a} \neq \mathbf{a}'$ and $\mathbf{a} \preceq \mathbf{a}'$. Since there are exactly 8^{K-1} different $\mathbf{a} \in [8]^{K-1}$ and exactly $7(K-1) + 1$ different values that S can take (which are $K-1, K, \dots, 8(K-1)$), by the *pigeonhole principle* there must be at least one s such that

$$|A_s| \geq \frac{8^{K-1}}{7(K-1) + 1} \geq \exp(K-1) \geq N,$$

where in the second inequality we used that $K \geq 4$ and in the third the assumption on N . Let $A \subset A_s$ be any subset of size exactly N . Then, for any $\mathbf{a} \neq \mathbf{a}' \in A$, it holds that $\mathbf{a} \not\preceq \mathbf{a}'$, and there is one coordinate $j \leq K-1$ with $a_j > a'_j$. As a consequence

$$r_j(\mathbf{a}) - r_j(\mathbf{a}') = v_{a_j} - v_{a'_j} = 8t(a_j - a'_j) \geq 8t.$$

Now, by definition of Pareto distance (cf. Definition 3), it follows that

$$d(\mathbf{r}(\mathbf{a}), \mathbf{r}(\mathbf{a}')) = \min \{t' \geq 0 : \mathbf{r}(\mathbf{a}) - t'\mathbf{1} \preceq \mathbf{r}(\mathbf{a}')\} \geq 8t$$

and hence no t -Pareto ball can cover any two of the points $\{\mathbf{r}(\mathbf{a}) : \mathbf{a} \in A\}$. In particular, we have constructed a set that has t -Pareto covering number $N_{\text{Par}}(t, \{\mathbf{r}(\mathbf{a}) : \mathbf{a} \in A\}) = N$.

Let us introduce the hypothesis space $\mathcal{F} = \{f_{\mathbf{a}} : \mathbf{a} \in A\} \cup \{g_0, g_1\}$ so that $|\mathcal{F}| = N + 2$. Define $\mathbf{R} : \mathcal{F} \rightarrow [0, 1]^K$ as

$$\mathbf{R}(f_{\mathbf{a}}) = \mathbf{r}(\mathbf{a}), \quad \mathbf{R}(g_0) = \left(\frac{1}{2} + 29t\right)\mathbf{1}, \quad \text{and} \quad \mathbf{R}(g_1) = \left(\frac{1}{2} + 30t\right)\mathbf{1}.$$

By construction, every $f_{\mathbf{a}}$ is Pareto optimal, and both g_0 and g_1 are dominated by all $f_{\mathbf{a}}$, implying that we retain $N_{\text{Par}}(t, \mathcal{F}, \mathbf{R}) = N$.

Let us now introduce the coordinate-wise independent distributions $\mathcal{Q} = \{Q_{\mathbf{q}} : \mathbf{q} \in A\}$ via $Z \in \mathcal{Z} := \{0, 1\}^{A \cup \{0, 1\}}$ that, for all $\mathbf{a} \in A$,

$$Z_{\mathbf{a}} \sim \text{Ber}(\theta_{\mathbf{q}, \mathbf{a}}) \quad \text{with} \quad \theta_{\mathbf{q}, \mathbf{a}} = \begin{cases} \frac{1}{2} - 2t & \mathbf{a} = \mathbf{q}, \\ \frac{1}{2} & \mathbf{a} \neq \mathbf{q}. \end{cases} \quad \text{and} \quad Z_0 \sim \text{Ber}\left(\frac{1}{2} + 29t\right), \quad Z_1 \sim \text{Ber}\left(\frac{1}{2} + 30t\right).$$

Moreover, let us denote the risks $\mathcal{R}_{\mathbf{q}} := \mathcal{R}_{Q_{\mathbf{q}}}$ through the loss on $\mathcal{F} \times \mathcal{Z}$

$$\ell(f, z) = \begin{cases} z_{\mathbf{a}} & f = f_{\mathbf{a}}, \\ z_0 & f = g_0, \\ z_1 & f = g_1. \end{cases} \implies \mathcal{R}_{\mathbf{q}}(f) = \begin{cases} 1/2 - 2t & f = f_{\mathbf{q}} \\ 1/2 & f = f_{\mathbf{a}}, \mathbf{a} \neq \mathbf{q} \\ 1/2 + 29t & f = g_0 \\ 1/2 + 30t & f = g_1 \end{cases}$$

Importantly, the loss is independent of the distribution Q , and between 0 and 1. By comparing each ordered pair in $\mathcal{F}$ we see that the only one within margin t is (g_1, g_0) ; $\mathbf{R}(g_1) = \mathbf{R}(g_0) + t\mathbf{1}$. Since $\mathcal{R}_Q(g_1) - \mathcal{R}_Q(g_0) = t$, this verifies that $\bar{\omega}_Q(t) = t$ for all Q .

In the following lower bound we consider proper estimators $\hat{f} \in \mathcal{F}$, but notice that we could set the loss to 1 outside of $\mathcal{F}$, so that any improper estimator must have excess risk at least $1/2$.

Let $\hat{f} \in \mathcal{F}$ be any estimator and define the test

$$\hat{\mathbf{q}} = \begin{cases} \mathbf{a} & \hat{f} = f_{\mathbf{a}} \\ \mathbf{a} \sim \text{Uniform}(A) & \hat{f} \in \{g_0, g_1\} \end{cases}.$$

Then, by design, on the event that $\hat{\mathbf{q}} \neq \mathbf{q}$

$$\mathcal{R}_{\mathbf{q}}(\hat{f}) - \min_{f \in \mathcal{F}} \mathcal{R}_{\mathbf{q}}(f) \geq \frac{1}{2} - \left(\frac{1}{2} - 2t\right) = 2t.$$

Therefore,

$$\sup_{\mathbf{q}} \mathbb{P} \left(\mathcal{R}_{\mathbf{q}}(\hat{f}) - \min_{f \in \mathcal{F}} \mathcal{R}_{\mathbf{q}}(f) \geq 2t \right) \geq \sup_{\mathbf{q}} \mathbb{P}(\hat{\mathbf{q}} \neq \mathbf{q}).$$

We are now ready to apply Fano's inequality.

Lemma 22 (Fano's inequality (Yu, 1997)). *Let $J \sim \text{Uniform}([N])$, and let $(Z|J=j) \sim Q_j$. Assume that $\text{KL}(Q_i, Q_j) \leq \alpha$ for all i, j . Then, for any estimator $\psi(Z)$ computed from Z ,*

$$\mathbb{P}(\psi(Z) \neq J) \geq 1 - \frac{\alpha + \log 2}{\log N}.$$

In order to apply Lemma 22, we bound the KL-divergence between the $Q_{\mathbf{q}}$ we constructed. By Wegel et al. (2025, Lemma D.2), we have

$$\begin{aligned} \text{KL}(Q_{\mathbf{q}}^{\otimes n}, Q_{\mathbf{q}'}^{\otimes n}) &= n \left(\text{KL}(\text{Ber}(\tfrac{1}{2} - 2t), \text{Ber}(\tfrac{1}{2})) + \text{KL}(\text{Ber}(\tfrac{1}{2}), \text{Ber}(\tfrac{1}{2} - 2t)) \right) \\ &\leq n (4(2t)^2 + 4(2t)^2) = 32nt^2. \end{aligned}$$

As the supremum over $\mathbf{q}$ is lower bounded by the average over $\mathbf{q}$, Lemma 22 implies that

$$\sup_{\mathbf{q}} \mathbb{P}(\hat{\mathbf{q}} \neq \mathbf{q}) \geq \frac{1}{N} \sum_{\mathbf{q} \in A} \mathbb{P}(\hat{\mathbf{q}} \neq \mathbf{q}) \geq 1 - \frac{32nt^2 + \log 2}{\log N} \geq 1/2,$$

where we used that $t \leq \sqrt{\log(N/4)/(64n)}$ and $N \geq 4$. Finally, it follows that

$$\sup_{\mathbf{q} \in A} \mathbb{E} \left[\mathcal{R}_{\mathbf{q}}(\hat{f}) - \min_{f \in \mathcal{F}} \mathcal{R}_{\mathbf{q}}(f) \right] \geq 2t \sup_{\mathbf{q}} \mathbb{P} \left(\mathcal{R}_{\mathbf{q}}(\hat{f}) - \min_{f \in \mathcal{F}} \mathcal{R}_{\mathbf{q}}(f) \geq 2t \right) \geq t,$$

which yields the result and concludes the proof of Theorem 3.

G.5. Proof of Proposition 1

Recall that $\hat{f}_{\text{adapt}} = \hat{f}_{\gamma_n}$. Therefore, to prove Proposition 1, we may apply the same decomposition presented in the proof of Proposition 3:

$$\mathcal{R}_Q(\hat{f}_{\gamma_n}) - \mathcal{R}_Q(f^*) \leq 4\mathcal{R}_n(\ell \circ \mathcal{F}_{\gamma_n}) + 2\sqrt{\frac{2\log(4/\delta)}{n}} + \underbrace{\inf_{f \in \mathcal{F}_{\gamma_n}} \mathcal{R}_Q(f) - \mathcal{R}_Q(f^*)}_{=:\varepsilon_n},$$

where, as opposed to the proof of Proposition 3, we do not have a guarantee that $\varepsilon_n = 0$ for all n since it is not guaranteed that $\gamma_n > \underline{\gamma}_Q$ for all n . We can further bound the Rademacher complexity of a finite function class using Massart's Lemma (Shalev-Shwartz and Ben-David, 2014, Lemma 26.8) as

$$4\mathcal{R}_n(\ell \circ \mathcal{F}_{\gamma_n}) + 2\sqrt{\frac{2\log(4/\delta)}{n}} \leq 4\sqrt{\frac{2\log(2|\mathcal{F}_{\gamma_n}|)}{n}} + 2\sqrt{\frac{2\log(4/\delta)}{n}} \leq 40\sqrt{\frac{\log(8|\mathcal{F}_{\gamma_n}|/\delta)}{n}}.$$

It remains to argue that $\varepsilon_n = 0$ if $n \geq n_0 := \min_{\gamma > \underline{\gamma}_Q} |\mathcal{F}_\gamma|$. As demonstrated in the proof of Proposition 3, for every $\gamma > \underline{\gamma}_Q$, we know that $\inf_{f \in \mathcal{F}_\gamma} \mathcal{R}_Q(f) = \inf_{f \in \mathcal{F}} \mathcal{R}_Q(f)$. Thus, it is enough if the adaptive set $\mathcal{F}_{\gamma_n}$ contains $\mathcal{F}_\gamma$ for some $\gamma > \underline{\gamma}_Q$.

By construction, $\gamma_n = \gamma_{(m_n)}$, where recall that

$$m_n = \min \{|\mathcal{F}|, \max \{n, |\text{Par}(\mathcal{F}, \mathbf{R})|\}\}.$$

Hence either $\gamma_n = 0$ and $\mathcal{F}_{\gamma_n} = \text{Par}(\mathcal{F}, \mathbf{R})$, or $\gamma_n > 0$ and $\mathcal{F}_{\gamma_n} = \{f \in \mathcal{F} : \Gamma(f) \leq \gamma_{(m_n)}\}$ keeps all models up to the m_n -th smallest margin (including ties). Since $\gamma \mapsto |\mathcal{F}_\gamma|$ is an increasing step function of γ , we know that $n_0 = \min_{\gamma > \underline{\gamma}_Q} |\mathcal{F}_\gamma|$ is well-defined and attained by some γ^* . Moreover, because $\text{Par}(\mathcal{F}, \mathbf{R}) \subseteq \mathcal{F}_{\gamma^*} \subseteq \mathcal{F}$ for any $\gamma^* > \underline{\gamma}_Q$, we know that $n \geq |\mathcal{F}_{\gamma^*}|$ implies the second inequality in $|\mathcal{F}_{\gamma_{(m_n)}}| \geq m_n \geq |\mathcal{F}_{\gamma^*}|$, and therefore $\gamma_{(m_n)} \geq \gamma^*$ which implies $\mathcal{F}_{\gamma^*} \subseteq \mathcal{F}_{\gamma_{(m_n)}} = \mathcal{F}_{\gamma_n}$ by nestedness. As argued above, this yields the claim.

G.6. Proof of Theorem 4

We demonstrate that the proof of Theorem 4 follows almost verbatim from the proof of the Q -aggregation estimator from Lecué and Rigollet (2014). There are, however, some important differences that we highlight along the way.

In this section we use the shorthand notation $\mathcal{R} = \mathcal{R}_Q$, $\widehat{\mathcal{R}} = \widehat{\mathcal{R}}_Q$ and $\|f\|_2 = \|f\|_{L^2(Q_X)}$, where Q remains the target distribution. Moreover, we associate the simplex $\triangle^{K-1}$ with the set of all distributions $\mathcal{P}([K])$ through the canonical map, and switch between both notations without explicitly stating it.

Define the function $\widetilde{\mathcal{R}}(\rho) = \frac{1}{2}\mathcal{R}(\phi_\rho) + \frac{1}{2}\mathbb{E}_{k \sim \rho} \mathcal{R}(f_k)$ and denote $\widetilde{\ell}_\rho(x, y) = \frac{1}{2}\ell(\phi_\rho(x), y) + \frac{1}{2}\mathbb{E}_{k \sim \rho} \ell(f_k(x), y)$. Note that $\widetilde{\mathcal{R}}(\rho) = Q\widetilde{\ell}_\rho$ where we use the standard empirical process theory notation $Qf = \int f dQ$. We actually prove the result for a generalized version of the estimator using a prior $\pi \in \mathcal{P}([K])$ defined as

$$\widehat{\rho}_{\text{Par}} \in \arg \min_{\rho \in \mathcal{P}([K])} \widehat{F}(\rho) \quad \text{where} \quad \widehat{F}(\rho) = \frac{1}{2}\widehat{\mathcal{R}}_Q(\phi_\rho) + \frac{1}{2}\mathbb{E}_{k \sim \rho} \widehat{\mathcal{R}}_Q(f_k) + \frac{K(\rho, \pi)}{\lambda n},$$

where $K(\rho, \pi) = \sum_{k=1}^K \rho_k \log(1/\pi_k)$. It reduces to the estimator in Section 5.1 by taking the uniform prior $\pi_k = 1/K$, since then $K(\rho, \pi) = \log(K) \sum_{k=1}^K \rho_k = \log K$ is independent of ρ and does not affect the optimization problem. We use the shorthand $\widehat{\rho} = \widehat{\rho}_{\text{Par}}$ throughout this section. Define the population version of $\widehat{F}$ and the comparator ρ^* as

$$\rho^* \in \arg \min_{\rho \in \mathcal{P}([K])} \{F(\rho) + \mu V(\rho)\} \quad \text{where} \quad F(\rho) = \frac{1}{2}\mathcal{R}(\phi_\rho) + \frac{1}{2}\mathbb{E}_{k \sim \rho} \mathcal{R}(f_k) + \frac{K(\rho, \pi)}{\lambda n}$$

so that $F(\rho) = \widetilde{\mathcal{R}}(\rho) + \frac{K(\rho, \pi)}{\lambda n}$ and where we denote the variance $V(\rho) := \mathbb{E}_{X \sim Q_X} \text{Var}_{k \sim \rho} f_k(X) = \mathbb{E}_{k \sim \rho} \|f_k - f_\rho\|_2^2$.

G.6.1. PRELIMINARY LEMMATA

To start the proof, we need some lemmata that essentially demonstrate why the proof of the Q -aggregation estimator applies almost verbatim to the aggregator on the Pareto set.

Lemma 23. *If $\underline{\omega}_Q(t) \geq t$ and $\overline{\omega}_Q(t) \leq t$, then $\Phi(\rho) = \mathcal{R}(f_\rho) - \mathcal{R}(\phi_\rho)$.*

Proof. Follows from the following sandwich inequalities: Since $\mathbf{R}(\phi_\rho) = \mathbf{R}(f_\rho) - \Phi(\rho)\mathbf{1}$, we have by definition of the moduli (Definition 4) that

$$\Phi(\rho) \leq \underline{\omega}_Q(\Phi(\rho)) \leq \mathcal{R}(f_\rho) - \mathcal{R}(\phi_\rho) \leq \overline{\omega}_Q(\Phi(\rho)) \leq \Phi(\rho),$$

implying the equality. $\square$

Lemma 23 implies that we know exactly the improvement an estimator ϕ_ρ has over the estimator with f_ρ . Recall now that the strong concavity of the loss implies the following Jensen's gap.

Lemma 24 (Proposition 2 in [Lecué and Rigollet \(2014\)](#)). *If the loss $\ell(\cdot, y)$ is κ -strongly convex, then*

$$\mathcal{R}(f_\rho) \leq \mathbb{E}_{k \sim \rho} \mathcal{R}(f_k) - \frac{\kappa}{2} V(\rho).$$

Importantly, this is also true for $\kappa = 0$, when the loss is merely convex.

Next, we show that the strong concavity of Φ , ensured by Assumption 2, implies that we can effectively control the variance of the estimator.

Lemma 25. *If Φ is η -strongly concave in $\|f_\rho - f_\gamma\|_2$, as defined in Eq. (5), then $\Phi(\rho) \geq \frac{\eta}{2} V(\rho)$.*

Proof. To prove this, define the difference $D(\rho) = \Phi(\rho) - \frac{\eta}{2} V(\rho)$. Since Φ is η -strongly concave and V is 2-strongly concave, that is,

$$V(\alpha\rho + (1-\alpha)\gamma) = \alpha V(\rho) + (1-\alpha)V(\gamma) + \alpha(1-\alpha)\|f_\rho - f_\gamma\|_2^2,$$

we know that D must be concave (e.g., by adding the two equations / inequalities). Let δ_j now denote the dirac delta on $j \in [K]$. Then, because $V(\delta_j) = \Phi(\delta_j) = 0$ for all j , we know that $D(\delta_j) = 0$ for all j . And so, by Jensen's inequality, for all $\rho \in \mathcal{P}([K])$

$$\Phi(\rho) - \frac{\eta}{2} V(\rho) = D(\rho) = D\left(\sum_{k \in [K]} \rho_k \delta_k\right) \geq \sum_{k \in [K]} \rho_k D(\delta_k) = 0.$$

This yields the claim. $\square$

Combining Lemmas 23 to 25 yields the proof of Eq. (7):

$$\mathcal{R}(\phi_\rho) = \mathcal{R}(f_\rho) - \Phi(\rho) \leq \mathbb{E}_{k \sim \rho} \mathcal{R}(f_k) - \frac{\kappa}{2} V(\rho) - \Phi(\rho) \leq \mathbb{E}_{k \sim \rho} \mathcal{R}(f_k) - \frac{\kappa + \eta}{2} V(\rho).$$

This provides some intuition why the proof works; even when the loss is not strongly convex ($\kappa = 0$), the estimator can penalize variance. Next, we prove the key property gained from the strongly convex objective. This is analogous to Proposition 4 in [Lecué and Rigollet \(2014\)](#).

Lemma 26. *Let $\mu < (\eta + \kappa)/4$. It holds for any ρ that*

$$\tilde{\mathcal{R}}(\rho) - \tilde{\mathcal{R}}(\rho^*) \geq -\frac{1}{\lambda n} (K(\rho, \pi) - K(\rho^*, \pi)) - \mu(V(\rho) - V(\rho^*)) + \left(\frac{\eta + \kappa}{4} - \mu\right) \|f_\rho - f_{\rho^*}\|_2^2$$

Proof. First note that from the definition,

$$F(\rho) = \tilde{\mathcal{R}}(\rho) + \frac{K(\rho, \pi)}{\lambda n} = \frac{1}{2} \mathcal{R}(\phi_\rho) + \frac{1}{2} \mathbb{E}_{k \sim \rho} \mathcal{R}(f_k) + \frac{K(\rho, \pi)}{\lambda n}$$

is $(\eta + \kappa)/2$ -strongly convex with respect to $\|f_\rho - f_\gamma\|_2$: Since $\rho \mapsto \Phi(\rho)$ is η -strongly concave and $\rho \mapsto \mathcal{R}(f_\rho)$ is κ -strongly convex (Proposition 3 in [Lecué and Rigollet, 2014](#)) we have that $\rho \mapsto \mathcal{R}(\phi_\rho) = \mathcal{R}(f_\rho) - \Phi(\rho)$ is $\eta + \kappa$ strongly convex in $\|f_\rho - f_\gamma\|_2^2$, where we used Lemma 23 for the equality. Hence, $\rho \mapsto \tilde{\mathcal{R}}(\rho)$ is $(\eta + \kappa)/2$ -strongly convex. Finally $\rho \mapsto K(\rho, \pi)$ is convex.

Moreover, since V is 2-strongly concave, that is,

$$V(\alpha\rho + (1-\alpha)\gamma) = \alpha V(\rho) + (1-\alpha)V(\gamma) + \alpha(1-\alpha)\|f_\rho - f_\gamma\|_2^2,$$

we have that $F_\mu(\rho) := F(\rho) + \mu V(\rho)$ is $(\eta + \kappa)/2 - 2\mu$ -strongly convex. This strong convexity implies that

$$F_\mu(\alpha\rho + (1-\alpha)\rho^*) \leq \alpha F_\mu(\rho) + (1-\alpha)F_\mu(\rho^*) - \left(\frac{\eta + \kappa}{4} - \mu\right) \alpha(1-\alpha) \|f_\rho - f_{\rho^*}\|_2^2.$$

Now, bounding $F_\mu(\rho^*) \leq F_\mu(\alpha\rho + (1-\alpha)\rho^*)$, subtracting $(1-\alpha)F_\mu(\rho^*)$ from both sides and dividing by α yields

$$F_\mu(\rho^*) \leq F_\mu(\rho) - \left(\frac{\eta + \kappa}{4} - \mu\right) (1-\alpha) \|f_\rho - f_{\rho^*}\|_2^2.$$

Taking $\alpha \rightarrow 0$ yields that $F_\mu(\rho) - F_\mu(\rho^*) \geq \left(\frac{\eta+\kappa}{4} - \mu\right) \|f_\rho - f_{\rho^*}\|_2^2$. Therefore, we get that

$$\begin{aligned} F_\mu(\rho) - F_\mu(\rho^*) &= \mu(V(\rho) - V(\rho^*)) + (\tilde{\mathcal{R}}(\rho) - \tilde{\mathcal{R}}(\rho^*)) + \frac{1}{\lambda n}(K(\rho, \pi) - K(\rho^*, \pi)) \\ &\geq \left(\frac{\eta+\kappa}{4} - \mu\right) \|f_\rho - f_{\rho^*}\|_2^2 \end{aligned}$$

Rearranging yields the result. $\square$

G.6.2. MAIN PROOF

We now come to the main proof. We begin by extracting the main empirical error term deterministically. This is analogous to Proposition 5 in [Lecué and Rigollet \(2014\)](#). Let Q_n denote the empirical measure, and for some parameter $s > 0$, define the random error term

$$Z_n = (Q - Q_n)(\tilde{\ell}_{\hat{\rho}} - \tilde{\ell}_{\rho^*}) - \mu \left(2V(\hat{\rho}) + V(\rho^*) + 2\|f_{\hat{\rho}} - f_{\rho^*}\|_2^2 \right) - \frac{2}{s}K(\hat{\rho}, \pi).$$

Notably, recall that the loss $\tilde{\ell}_\rho$ was defined via our map ϕ_ρ . Hence, Z_n differs from the error term in [Lecué and Rigollet \(2014\)](#) via the loss functions and via the choice of $\hat{\rho}$.

Proposition 6. Assume that $\mu \leq \frac{1}{20}(\kappa + \eta)$, $\frac{1}{\lambda n} \geq \frac{4}{s}$. Then it holds that

$$\mathcal{R}(\phi_{\hat{\rho}}) \leq \min_{k \in [K]} \left\{ \mathcal{R}(f_k) + \frac{\log(1/\pi_k)}{\lambda n} \right\} + 2Z_n$$

Proof. From the definition of $\hat{\rho}$ we know that $\hat{F}(\hat{\rho}) \leq \hat{F}(\rho^*)$, spelled out being

$$Q_n \tilde{\ell}_{\hat{\rho}} + \frac{K(\hat{\rho}, \pi)}{\lambda n} \leq Q_n \tilde{\ell}_{\rho^*} + \frac{K(\rho^*, \pi)}{\lambda n}$$

which we can rearrange and add $\tilde{\mathcal{R}}(\hat{\rho}) - \tilde{\mathcal{R}}(\rho^*) = Q(\tilde{\ell}_{\hat{\rho}} - \tilde{\ell}_{\rho^*})$ on both sides to obtain

$$\begin{aligned} \tilde{\mathcal{R}}(\hat{\rho}) - \tilde{\mathcal{R}}(\rho^*) &\leq (Q - Q_n)(\tilde{\ell}_{\hat{\rho}} - \tilde{\ell}_{\rho^*}) + \frac{1}{\lambda n}(K(\rho^*, \pi) - K(\hat{\rho}, \pi)) \\ &= Z_n + \mu \left(2V(\hat{\rho}) + V(\rho^*) + 2\|f_{\hat{\rho}} - f_{\rho^*}\|_2^2 \right) + \frac{2}{s}K(\hat{\rho}, \pi) + \frac{1}{\lambda n}(K(\rho^*, \pi) - K(\hat{\rho}, \pi)) \end{aligned}$$

where the second equality is simply the definition of Z_n . Combining this with the lower bound from Lemma 26 and cancelling $\frac{1}{\lambda n}(K(\rho^*, \pi) - K(\hat{\rho}, \pi)) + \mu V(\rho^*)$ on both sides yields that

$$\left(\frac{\eta+\kappa}{4} - 3\mu \right) \|f_{\hat{\rho}} - f_{\rho^*}\|_2^2 \leq Z_n + 3\mu V(\hat{\rho}) + \frac{2}{s}K(\hat{\rho}, \pi)$$

which we can plug back in using $\left(\frac{\eta+\kappa}{4} - 3\mu\right)^{-1} = \frac{4}{\eta+\kappa-12\mu}$ to get

$$\begin{aligned} \tilde{\mathcal{R}}(\hat{\rho}) - \tilde{\mathcal{R}}(\rho^*) &\leq Z_n + \mu(2V(\hat{\rho}) + V(\rho^*)) + \frac{2}{s}K(\hat{\rho}, \pi) + \frac{8\mu}{\eta+\kappa-12\mu} \left(Z_n + \mu 3V(\hat{\rho}) + \frac{2}{s}K(\hat{\rho}, \pi) \right) \\ &\quad + \frac{1}{\lambda n}(K(\rho^*, \pi) - K(\hat{\rho}, \pi)) \\ &= \left(1 + \frac{8\mu}{\eta+\kappa-12\mu} \right) \left(Z_n + \frac{2}{s}K(\hat{\rho}, \pi) \right) + \left(2\mu + \frac{24\mu^2}{\eta+\kappa-12\mu} \right) V(\hat{\rho}) \\ &\quad + \frac{1}{\lambda n}(K(\rho^*, \pi) - K(\hat{\rho}, \pi)) + \mu V(\rho^*). \end{aligned}$$

Now notice that by definition of $\tilde{\mathcal{R}}$, Assumption 2 and Lemmas 24 and 25 we have that

$$\begin{aligned}\tilde{\mathcal{R}}(\rho) &= \frac{1}{2}\mathcal{R}(\phi_\rho) + \frac{1}{2} \mathbb{E}_{k \sim \rho} \mathcal{R}(f_k) \\ &= \mathcal{R}(\phi_\rho) - \frac{1}{2}(\mathcal{R}(f_\rho) - \Phi(\rho)) + \frac{1}{2} \mathbb{E}_{k \sim \rho} \mathcal{R}(f_k) \\ &\geq \mathcal{R}(\phi_\rho) + \frac{\eta}{4}V(\rho) + \frac{\kappa}{4}V(\rho) = \mathcal{R}(\phi_\rho) + \frac{\eta + \kappa}{4}V(\rho),\end{aligned}$$

cf. Eq. (7). It follows that

$$\begin{aligned}\mathcal{R}(\phi_{\hat{\rho}}) &\leq \tilde{\mathcal{R}}(\rho^*) + \mu V(\rho^*) + \frac{1}{\lambda n} K(\rho^*, \pi) + \left(1 + \frac{8\mu}{\eta + \kappa - 12\mu}\right) Z_n \\ &\quad + \left(2\mu + \frac{24\mu^2}{\eta + \kappa - 12\mu} - \frac{\eta + \kappa}{4}\right) V(\hat{\rho}) \\ &\quad + \left(\left(1 + \frac{8\mu}{\eta + \kappa - 12\mu}\right) \frac{2}{s} - \frac{1}{\lambda n}\right) K(\hat{\rho}, \pi)\end{aligned}$$

Now, if the choice of parameters satisfies

$$20\mu \leq \eta + \kappa, \quad \text{and} \quad \frac{1}{\lambda n} \geq \frac{4}{s}.$$

we can treat each term: $1 + \frac{8\mu}{\eta + \kappa - 12\mu} \leq 2$ since $\eta + \kappa - 12\mu \geq 8\mu$. The second line is non-positive since $2\mu + \frac{24\mu^2}{\eta + \kappa - 12\mu} - \frac{\eta + \kappa}{4} \leq 5\mu - \frac{\eta + \kappa}{4} \leq 0$. The third line is non-positive since $\left(1 + \frac{8\mu}{\eta + \kappa - 12\mu}\right) \frac{2}{s} - \frac{1}{\lambda n} \leq \frac{4}{s} - \frac{1}{\lambda n} \leq 0$.

It follows by definition of $\rho^* \in \arg \min_\rho F(\rho) + \mu V(\rho) = \tilde{\mathcal{R}}(\rho) + \frac{1}{\lambda n} K(\rho, \pi) + \mu V(\rho)$ that

$$\begin{aligned}\mathcal{R}(\phi_{\hat{\rho}}) &\leq \tilde{\mathcal{R}}(\rho^*) + \mu V(\rho^*) + \frac{1}{\lambda n} K(\rho^*, \pi) + 2Z_n \\ &= \inf_{\rho} \left\{ \tilde{\mathcal{R}}(\rho) + \mu V(\rho) + \frac{1}{\lambda n} K(\rho, \pi) \right\} + 2Z_n \\ &\leq \min_{k \in [K]} \left\{ \underbrace{\tilde{\mathcal{R}}(\delta_k)}_{=\mathcal{R}(f_k)} + \underbrace{\mu V(\delta_k)}_{=0} + \underbrace{\frac{1}{\lambda n} K(\delta_k, \pi)}_{=\log(1/\pi_k)/(\lambda n)} \right\} + 2Z_n.\end{aligned}$$

This yields the result of the proposition. $\square$

Finally, we now show that a similar bound to that proved in [Lecué and Rigollet \(2014\)](#) on “their Z_n ” also applies to our Z_n . Again, note that while they look the same, they differ both in the loss function and the $\hat{\rho}$. Luckily, this does not change the proof too much given the following assumptions. For completeness, we state the whole proof here.

Remark 1. In the work by [Lecué and Rigollet \(2014\)](#), there is a mistake in this step. When deriving their Equations (3.7) and (3.8), a factor $1/2$ is missing in front of $K(\hat{\theta})$. This can be fixed by using $-\frac{2}{s}K(\hat{\rho}, \pi)$ instead of $-\frac{1}{s}K(\hat{\rho}, \pi)$ in the definition of Z_n , assuming $\beta \geq 4n/s$ in their Proposition 5 and later adjusting the assumption on β in the main result accordingly. We already applied this fix in our Proposition 6.

Proposition 7. Assume that all functions and Y are in $[0, 1]$, and that the loss and ϕ are Lipschitz in this sense:

$$\forall \hat{y}, \hat{y}' \in [0, 1] : |\ell(\hat{y}, y) - \ell(\hat{y}', y)| \leq L_\ell |\hat{y} - \hat{y}'| \quad \text{and} \quad \forall x \in \mathcal{X} : |\phi_\rho(x) - \phi_\gamma(x)| \leq L_\phi |f_\rho(x) - f_\gamma(x)|.$$

Assume that $\lambda n \leq s/4$ and $s < \min \{n/\sqrt{3}L_\ell L_\phi, 8\mu n/(L_\ell(L_\ell + 8\mu))\}$. Then it holds that

$$\mathbb{P}\left(Z_n \geq \frac{\log(1/\delta)}{\lambda n}\right) \leq \delta \quad \text{and} \quad \mathbb{E}[Z_n] \leq 0.$$

Proof. Thanks to Jensen's inequality and Chernoff's bound, it is sufficient to prove that $\mathbb{E} \exp(\lambda n Z_n) \leq 1$, because then

$$\mathbb{P}(\lambda n Z_n > x) \leq \frac{\mathbb{E} \exp(\lambda n Z_n)}{\exp x} \leq \exp(-x) \quad \text{and} \quad \mathbb{E}[Z_n] \leq \frac{1}{\lambda n} \log \mathbb{E} \exp(\lambda n Z_n) \leq 0.$$

Denote $\ell_\rho(x, y) = \ell(\phi_\rho(x), y)$, $\ell^{(k)}(x, y) = \ell(f_k(x), y)$ and write Z_n as

$$\begin{aligned} Z_n &= (Q - Q_n)(\tilde{\ell}_{\hat{\rho}} - \tilde{\ell}_{\rho^*}) - \mu \left(2V(\hat{\rho}) + V(\rho^*) + 2\|f_{\hat{\rho}} - f_{\rho^*}\|_2^2 \right) - \frac{2}{s} K(\hat{\rho}, \pi) \\ &= (Q - Q_n) \left(\frac{1}{2}(\ell_{\hat{\rho}} - \ell_{\rho^*}) + \frac{1}{2} \sum_{k=1}^K (\hat{\rho}_k - \rho_k^*) \ell^{(k)} \right) - \mu \left(\sum_{k \in [K]} \hat{\rho}_k \|f_k - f_{\rho^*}\|_2^2 + \hat{\rho} \mathbf{H} \rho^* \right) - \frac{2}{s} K(\hat{\rho}, \pi) \\ &= \underbrace{\frac{1}{2}(Q - Q_n)(\ell_{\hat{\rho}} - \ell_{\rho^*}) - \mu \sum_{k \in [K]} \hat{\rho}_k \|f_k - f_{\rho^*}\|_2^2 - \frac{1}{s} K(\hat{\rho}, \pi)}_{=: A_n} \\ &\quad + \underbrace{\frac{1}{2}(Q - Q_n) \sum_{k=1}^K (\hat{\rho}_k - \rho_k^*) \ell^{(k)} - \mu \hat{\rho} \mathbf{H} \rho^* - \frac{1}{s} K(\hat{\rho}, \pi)}_{=: B_n} \end{aligned}$$

where we used the $K \times K$ matrix $\mathbf{H}_{j,k} = \|f_j - f_k\|_2^2$ and the identities

$$\begin{aligned} V(\hat{\rho}) + \|f_{\hat{\rho}} - f_{\rho^*}\|_2^2 &= \sum_{k \in [K]} \hat{\rho}_k \|f_k - f_{\rho^*}\|_2^2, \\ V(\hat{\rho}) + V(\rho^*) + \|f_{\rho^*} - f_{\hat{\rho}}\|_2^2 &= \hat{\rho} \mathbf{H} \rho^*, \end{aligned}$$

see Equations (3.2) and (3.3) from [Lecué and Rigollet \(2014\)](#).

Apply Cauchy-Schwarz $\mathbb{E} \exp \lambda n Z_n = \mathbb{E} \exp \lambda n A_n \exp \lambda n B_n \leq \sqrt{\mathbb{E} \exp 2\lambda n A_n} \sqrt{\mathbb{E} \exp 2\lambda n B_n}$, and prove $\mathbb{E} \exp 2\lambda n A_n \leq 1$ and $\mathbb{E} \exp 2\lambda n B_n \leq 1$, respectively.

Our assumptions ensure that $|\ell_{\hat{\rho}} - \ell_{\rho^*}| \leq L_\ell |\phi_{\hat{\rho}} - \phi_{\rho^*}| \leq L_\ell L_\phi |f_{\hat{\rho}} - f_{\rho^*}|$, and so symmetrization and contraction yields

$$\begin{aligned} \mathbb{E} \exp 2\lambda n A_n &\leq \mathbb{E} \exp \left(2\lambda n \max_{\rho \in \mathcal{P}([K])} \left\{ \frac{1}{2}(Q - Q_n)(\ell_\rho - \ell_{\rho^*}) - \mu \sum_{k \in [K]} \rho_k \|f_k - f_{\rho^*}\|_2^2 - \frac{1}{s} K(\rho, \pi) \right\} \right) \\ &\leq \mathbb{E} \exp \left(s \max_{\rho \in \mathcal{P}([K])} \left\{ L_\ell L_\phi Q_{\sigma, n}(f_\rho - f_{\rho^*}) - \mu \sum_{k \in [K]} \rho_k \|f_k - f_{\rho^*}\|_2^2 - \frac{1}{s} K(\rho, \pi) \right\} \right) \end{aligned}$$

where we used that $2\lambda n \leq s$ and denote $Q_{\sigma, n}$ as the symmetrized measure with i.i.d. Rademacher variables. Note that the choice of ρ^* is irrelevant for the rest of the proof; hence from here on, the proof is identical to [Lecué and Rigollet \(2014\)](#) and we get that $\mathbb{E} \exp 2\lambda n A_n \leq 1$ if $s < n/\sqrt{3} L_\ell L_\phi$.

Moreover, by similar calculations as in Equation (3.15) in [Lecué and Rigollet \(2014\)](#), we have that

$$\begin{aligned}
 \mathbb{E} \exp 2\lambda n B_n &\leq \mathbb{E} \exp \left(\sum_{k=1}^K \hat{\rho}_k \sum_{j=1}^K \rho_j^* s \left(\frac{1}{2} (Q - Q_n)(\ell^{(j)} - \ell^{(k)}) - \mu \|f_j - f_k\|_2^2 \right) - K(\hat{\rho}, \pi) \right) \\
 &\leq \mathbb{E} \exp \left(\max_{\rho \in \mathcal{P}([K])} \left\{ \sum_{k=1}^K \rho_k \sum_{j=1}^K \rho_j^* s \left(\frac{1}{2} (Q - Q_n)(\ell^{(j)} - \ell^{(k)}) - \mu \|f_j - f_k\|_2^2 \right) - K(\rho, \pi) \right\} \right) \\
 &\leq \sum_{k=1}^K \pi_k \mathbb{E} \exp \left(\sum_{j=1}^K \rho_j^* s \left(\frac{1}{2} (Q - Q_n)(\ell^{(j)} - \ell^{(k)}) - \mu \|f_j - f_k\|_2^2 \right) \right) \\
 &\leq \sum_{k \in [K]} \pi_k \sum_{j \in [K]} \rho_j^* \mathbb{E} \exp \left(s \left(\frac{1}{2} (Q - Q_n)(\ell^{(j)} - \ell^{(k)}) - \mu \|f_j - f_k\|_2^2 \right) \right)
 \end{aligned}$$

where the first inequality follows from expanding the quadratic term $\hat{\rho} \mathbf{H} \rho^*$ and $2\lambda n \leq s$, the third one follows from

$$\max_{\rho \in \mathcal{P}([K])} \left\{ \sum_{k=1}^K \rho_k a_k - K(\rho, \pi) \right\} = \max_{k \in [K]} \{a_k + \log \pi_k\},$$

and the last one is Jensen's inequality. Since $f_{\delta_k} = f_k$, the Lipschitz assumption gives

$$Q(\ell^{(j)} - \ell^{(k)})^2 \leq L_\ell^2 Q(f_j - f_k)^2 = L_\ell^2 \|f_k - f_j\|_2^2 \implies -\mu \|f_k - f_j\|_2^2 \leq -\frac{\mu}{L_\ell^2} Q(\ell^{(j)} - \ell^{(k)})^2.$$

Plugging this in and using Proposition 1 in [Lecué and Rigollet \(2014\)](#) yields $\mathbb{E} \exp 2\lambda n B_n \leq 1$ whenever $s < 8\mu n / (L_\ell(L_\ell + 8\mu))$. That concludes the proof of this proposition. $\square$

Combining Proposition 6 and Proposition 7 concludes the proof of Theorem 4 by noting that all conditions from Lemma 26 and Propositions 6 and 7 are satisfied whenever we have that

$$\mu \leq \frac{1}{20}(\eta + \kappa), \quad 1 \leq \lambda n \leq \frac{s}{4} < \frac{1}{4} \min \left\{ \frac{n}{\sqrt{3}L_\ell L_\phi}, \frac{8\mu n}{L_\ell(L_\ell + 8\mu)} \right\}.$$

One set of parameters that satisfies these constraints is

$$\mu = \frac{\eta + \kappa}{20}, \quad \text{and} \quad \lambda = \frac{s}{4n} = \frac{1}{11L_\ell} \min \left\{ \frac{1}{L_\phi}, \frac{\eta + \kappa}{L_\ell + \eta + \kappa} \right\}.$$

Further, this not only yields the high-probability bound, but also a bound in expectation:

$$\mathbb{E} \left[\mathcal{E}_Q(\phi_{\hat{\rho}}, \{f_k\}_{k=1}^K) \right] \leq \frac{\log(1/\pi_k)}{\lambda n}. \quad (36)$$

G.7. Proof of Theorem 6

The proof idea of Theorem 6 is to extend the setting of Example 7 by creating $d := \lfloor \log_2(K) \rfloor$ copies of it. To this end, choose $\mathcal{X} = \{(i, j) : i \in [d], j \in \{-1, 1\}\}$ and, for some $\varepsilon \in (0, 1/4]$ to be determined later, let us define the family of distributions $\mathcal{Q} = \{Q_\sigma : \sigma \in \{-1, +1\}^d\}$ via the random variables $(X, Y) \sim Q_\sigma$ as

$$\mathbb{P}(X = (i, -1)) = \frac{1}{d} \left(\frac{1}{2} + \sigma_i \varepsilon \right), \quad \mathbb{P}(X = (i, 1)) = \frac{1}{d} \left(\frac{1}{2} - \sigma_i \varepsilon \right), \quad (Y|X = (i, j)) = \mathbf{1}\{j = +1\}.$$

We consider the absolute loss $\ell(f, (x, y)) = |y - f(x)|$ and the dictionary $\{f_\tau : \tau \in \{-1, +1\}^d\}$ with $f_\tau(i, j) = \mathbf{1}\{\tau_i = +1\}$. $\tau \in \{\pm 1\}^d$ is the index of the dictionary and yields a dictionary of size $2^d = 2^{\lfloor \log_2(K) \rfloor} \in [K/2, K]$. As the hypothesis space, we extend the dictionary to the following set

$$\mathcal{F} = \text{conv} \left(\{f_\tau : \tau \in \{-1, +1\}^d\} \right) \cup \{g_u : u \in [0, 1]^d\},$$

with the functions $g_{\mathbf{u}}$ defined as $g_{\mathbf{u}}(i, -1) = u_i^2$ and $g_{\mathbf{u}}(i, +1) = 2u_i - u_i^2$. Finally, note that $\mathcal{R}_{Q_{\sigma}} = \sum_{\tau \in \{\pm\}^d} q_{\tau}(\sigma) \mathcal{R}_{\tau}$. Here, $\mathcal{R}_{\tau}(f) = \mathbb{E}_{(X,Y) \sim P_{\tau}} |f(X) - Y|$, where P_{τ} is defined by

$$\mathbb{P}(X = (i, -1)) = \frac{1}{d} \mathbf{1}\{\tau_i = -1\}, \quad \mathbb{P}(X = (i, +1)) = \frac{1}{d} \mathbf{1}\{\tau_i = +1\}, \quad (Y|X = (i, j)) = \mathbf{1}\{j = +1\}.$$

Indeed, for the weights defined as

$$q_{\tau}(\sigma) = \prod_{i=1}^d \left(\frac{1}{2} - \sigma_i \varepsilon \right)^{\mathbf{1}\{\tau_i = +1\}} \left(\frac{1}{2} + \sigma_i \varepsilon \right)^{\mathbf{1}\{\tau_i = -1\}}$$

a direct calculation verifies that $\sum_{\tau \in \{\pm\}^d} q_{\tau}(\sigma) = 1$ and $Q_{\sigma} = \sum_{\tau \in \{\pm\}^d} q_{\tau}(\sigma) P_{\tau}$. The claim then follows from Lemma 5.

G.7.1. LOWER BOUND

We begin by noticing that we can write the convex combination of dictionary elements as

$$f_{\rho}(i) \equiv f_{\rho}(i, j) = \sum_{\tau \in \{-1, +1\}^d} \rho_{\tau} \mathbf{1}\{\tau_i = +1\}.$$

For any expert f_{τ} , the loss reads as $|f_{\tau}(i, j) - \mathbf{1}\{j = +1\}| = |\mathbf{1}\{\tau_i = +1\} - \mathbf{1}\{j = +1\}| = \mathbf{1}\{\tau_i \neq j\}$ for any (i, j) . Therefore, we can write the risk $\mathcal{R}_{\sigma} = \mathcal{R}_{Q_{\sigma}}$ of f_{τ} as

$$\mathcal{R}_{\sigma}(f_{\tau}) = \frac{1}{d} \sum_{i=1}^d \left(\frac{1}{2} - \sigma_i \varepsilon \right) \mathbf{1}\{\tau_i = -1\} + \left(\frac{1}{2} + \sigma_i \varepsilon \right) \mathbf{1}\{\tau_i = +1\} = \frac{1}{2} + \frac{\varepsilon}{d} \sum_{i=1}^d \sigma_i \tau_i.$$

Thus, the best dictionary element is the one corresponding to $\tau = -\sigma$ which achieves risk $1/2 - \varepsilon$, and each coordinate that differs in τ increases the excess risk of f_{τ} by $2\varepsilon/d$. Moreover, for the aggregate f_{ρ} we have

$$\mathcal{R}_{\sigma}(f_{\rho}) = \frac{1}{d} \sum_{i=1}^d \left(\frac{1}{2} - \sigma_i \varepsilon \right) (1 - f_{\rho}(i)) + \left(\frac{1}{2} + \sigma_i \varepsilon \right) f_{\rho}(i) = \frac{1}{2} + \frac{\varepsilon}{d} \sum_{i=1}^d \sigma_i (2f_{\rho}(i) - 1),$$

and the excess risk is given by

$$\mathcal{R}_{\sigma}(f_{\rho}) - \min_{\tau} \mathcal{R}_{\sigma}(f_{\tau}) = \varepsilon + \frac{\varepsilon}{d} \sum_{i=1}^d \sigma_i (2f_{\rho}(i) - 1).$$

For any estimator $\hat{\rho} \in \Delta^{2^d-1} \cong \mathcal{P}(\{\tau : \tau \in \{\pm 1\}^d\})$ that we may compute on n i.i.d. samples from Q_{σ} , that is, $\hat{\rho} \equiv \hat{\rho}(Z)$ where $Z \sim Q_{\sigma}^{\otimes n}$, define the estimator $\hat{\sigma}$ as $\hat{\sigma}_i = -\text{sign}(2\hat{f}_{\rho}(i) - 1) \in \{-1, +1\}$ (with arbitrary tie breaks when it is zero). Then,

$$\mathcal{R}_{\sigma}(f_{\hat{\rho}}) - \min_{\tau} \mathcal{R}_{\sigma}(f_{\tau}) \geq \frac{\varepsilon}{2d} \|\hat{\sigma} - \sigma\|_1.$$

We can now apply Assouad's Lemma to lower bound the excess risk.

Lemma 27 (Assouad's lower bound (Yu, 1997)). *Let $\mathcal{Q}_n = \{Q_{\sigma}^n : \sigma \in \{-1, 1\}^d\}$ be a family of probability measures. Write $\sigma \sim \sigma'$ if σ and σ' differ in only one coordinate. Then*

$$\max_{\sigma} \mathbb{E}_{Z \sim Q_{\sigma}^n} [\|\hat{\sigma}(Z) - \sigma\|_1] \geq d \min \left\{ 1 - \sqrt{\frac{1}{2} \text{KL}(Q_{\sigma}^n, Q_{\sigma'}^n)} : \sigma \sim \sigma' \right\}.$$

In particular, we need to bound the Kullback-Leibler divergence between the two distributions. A calculation shows that for $\sigma \sim \sigma'$. Then, for $\varepsilon \leq 1/4$ and by [Tsybakov \(2009, Lemma 2.7\)](#)

$$\begin{aligned} \text{KL}(Q_{\sigma}^{\otimes n}, Q_{\sigma'}^{\otimes n}) &= \frac{n}{d} \text{KL}(\text{Ber}(\tfrac{1}{2} \pm \varepsilon), \text{Ber}(\tfrac{1}{2} \mp \varepsilon)) \\ &\leq \frac{n}{d} \chi^2(\text{Ber}(\tfrac{1}{2} \pm \varepsilon), \text{Ber}(\tfrac{1}{2} \mp \varepsilon)) \\ &= \frac{n}{d} \frac{4\varepsilon^2}{(1/2 + \varepsilon)(1/2 - \varepsilon)} \leq \frac{22n\varepsilon^2}{d}. \end{aligned}$$

We get that the minimax expected excess risk is lower bounded by

$$\max_{\sigma} \mathbb{E}_{Q_{\sigma}^{\otimes n}} [\mathcal{R}_{\sigma}(f_{\hat{\rho}}) - \min_{\tau} \mathcal{R}_{\sigma}(f_{\tau})] \geq \frac{\varepsilon}{2} \left(1 - \sqrt{\frac{11n\varepsilon^2}{d}}\right) \geq \frac{\varepsilon}{4} = \frac{1}{8} \sqrt{\frac{d}{11n}} = \frac{1}{8} \sqrt{\frac{\lfloor \log_2(K) \rfloor}{11n}} \geq \frac{1}{38} \sqrt{\frac{\log(K)}{n}}.$$

where the second and third (in-)equalities hold for $\varepsilon = \frac{1}{2} \sqrt{\frac{d}{11n}}$ which we now show to be smaller than $1/4$. Also note that we used that for any $K \geq 4$,

$$\lfloor \log_2(K) \rfloor \geq \log_2(K) - 1 \geq \frac{1}{2} \log_2(K).$$

Indeed, since we assumed that $\log(K)/n \leq 1$, we have that

$$\varepsilon = \frac{1}{2} \sqrt{\frac{\lfloor \log_2(K) \rfloor}{11n}} \leq \frac{1}{2} \sqrt{\frac{\log K}{22 \log(2)n}} < \frac{1}{7} \sqrt{\frac{\log(K)}{n}} \leq \frac{1}{4}.$$

That concludes the proof of the lower bound.

G.7.2. UPPER BOUND

Finally, we check that Theorem 4 applies. In particular, we claim that we may choose (ϕ, Φ) as (ψ, Ψ) from Eq. (4). To begin, we calculate that for any τ and any f ,

$$\mathcal{R}_{\tau}(f) = \frac{1}{d} \sum_{i=1}^d (\mathbf{1}\{\tau_i = -1\} f(i, -1) + \mathbf{1}\{\tau_i = +1\} (1 - f(i, +1))).$$

Recall the definition of ψ_{ρ} from Eq. (4), which in the notation of this setting is

$$\psi_{\rho} \in \arg \min_{f \in \mathcal{F}} \max_{\tau \in \{\pm\}^d} \{\mathcal{R}_{\tau}(f) - \mathcal{R}_{\tau}(f_{\rho})\}.$$

Furthermore, the objective is non-negative on the convex hull; thus, the minimizer will be one of the $g_{\mathbf{u}}$. For $g_{\mathbf{u}}$ we can calculate the risk compared to f_{ρ}

$$\mathcal{R}_{\tau}(g_{\mathbf{u}}) - \mathcal{R}_{\tau}(f_{\rho}) = \frac{1}{d} \sum_{i=1}^d (\mathbf{1}\{\tau_i = -1\} (u_i^2 - f_{\rho}(i)) + \mathbf{1}\{\tau_i = +1\} ((1 - u_i)^2 - (1 - f_{\rho}(i)))).$$

Maximizing over τ , which can be done coordinate-wise, yields

$$\max_{\tau \in \{\pm\}^d} \{\mathcal{R}_{\tau}(g_{\mathbf{u}}) - \mathcal{R}_{\tau}(f_{\rho})\} = \frac{1}{d} \sum_{i=1}^d \max \{u_i^2 - f_{\rho}(i), (1 - u_i)^2 - (1 - f_{\rho}(i))\}.$$

On the other hand, minimizing over $\mathbf{u}$ (also coordinate-wise), yields that the minimum is attained for $u_i = f_{\rho}(i)$. Hence, $\psi_{\rho} = g_{\mathbf{u}}$ with

$$-\Psi(\rho) = \min_{\mathbf{u} \in [0,1]^d} \max_{\tau \in \{\pm\}^d} \{\mathcal{R}_{\tau}(g_{\mathbf{u}}) - \mathcal{R}_{\tau}(f_{\rho})\} = \frac{1}{d} \sum_{i=1}^d f_{\rho}^2(i) - f_{\rho}(i) = -\frac{1}{d} \sum_{i=1}^d f_{\rho}(i)(1 - f_{\rho}(i)) \leq 0.$$

We can now verify that (ψ, Ψ) satisfies Assumption 2.

1. Clearly, when we plug in any Dirac delta δ_τ , it holds that $\Psi(\delta_\tau) = \frac{1}{d} \sum_{i=1}^d f_\tau(i)(1 - f_\tau(i)) = 0$ because $f_\tau(i)$ is either 0 or 1.
2. Ψ is $\eta = 2$ -strongly concave in $\|f_\rho - f_\gamma\|_2^2 = \frac{1}{d} \sum_{i=1}^d (f_\rho(i) - f_\gamma(i))^2$: indeed, for $h(u) = u(1 - u)$, it holds $h''(u) = -2$. As a consequence, for $\mathbf{u} = (f_\rho(1), \dots, f_\rho(d))$ we have

$$\nabla_{\mathbf{u}}^2 \Psi(\rho) = \nabla_{\mathbf{u}}^2 \frac{1}{d} \sum_{i=1}^d h(u_i) = -\frac{2}{d} \mathbf{I}_d.$$

3. Finally, our previous calculations verify that $\mathcal{R}_\tau(f_\rho) - \mathcal{R}_\tau(\psi_\rho) = \Psi(\rho)$ for all τ , and thus $\mathbf{R}(\psi_\rho) = \mathbf{R}(f_\rho) - \Psi(\rho)\mathbf{1}$.

Moreover, all the other assumptions of Theorem 4 are also satisfied.

1. Boundedness is obvious as all functions and Y take values in $[0, 1]$.
2. The absolute loss is 1-Lipschitz, that is, $L_\ell = 1$.
3. ψ_ρ is 2-Lipschitz in f_ρ : if $\psi_\rho = g_{\mathbf{u}}$ and $\psi_\gamma = g_{\mathbf{v}}$, then in each coordinate we have that $|\psi_\rho(i, -1) - \psi_\gamma(i, -1)| = |u_i^2 - v_i^2| \leq 2|u_i - v_i| = 2|f_\rho(i) - f_\gamma(i)|$ and $|\psi_\rho(i, +1) - \psi_\gamma(i, +1)| = |(u_i - v_i)(2 - (u_i + v_i))| \leq 2|u_i - v_i| = 2|f_\rho(i) - f_\gamma(i)|$. Hence $L_\phi = 2$.
4. Finally, since $\mathcal{R}_Q$ is a linear scalarization of the source risks, Lemma 4 implies that $\underline{\omega}_Q(t) \geq t$ and $\bar{\omega}_Q(t) \leq t$.

Hence, we apply Theorem 4 (through Eq. (36)) to get

$$\lambda = \frac{1}{11L_\ell} \min \left\{ \frac{1}{L_\phi}, \frac{\eta + \kappa}{L_\ell + \eta + \kappa} \right\} = \frac{1}{11} \min \left\{ \frac{1}{2}, \frac{2 + 0}{1 + 2 + 0} \right\} = \frac{1}{22},$$

by plugging in all the constants. That concludes the proof of Theorem 6.